Quantum Consistency and Renormalization of the Axis Model Effective Field Theory

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Abstract

We present the quantum–field–theoretic formulation of the Axis Model as a predictive effective field theory (EFT) on the interval $E_{\rm IR} < \mu < \min(\Lambda_\Phi, \Lambda_q)$, with a scalar–coherence cutoff $\Lambda_\Phi \sim 10^5$ GeV and a morton–dissolution scale $\Lambda_q \sim 10^{16}$ GeV. We construct a BRST-invariant gauge fixing, prove anomaly cancellation via scalar-bundle triviality, and derive the one- and two-loop renormalization group equations (RGEs) for the renormalizable subsector. The minimal Abelian completion exhibits no perturbative non-Gaussian UV fixed point in (λ, y) and a positive one-loop Abelian coefficient, so the EFT is not asymptotically safe and is not intended as a UV-complete theory above its thresholds. Within its window, the quartic remains positive and flows are perturbative; dispersion/positivity bounds control leading higher-dimensional operators. We also provide the electroweak mass matrix as an emergent effect of a composite stiffness Z_χ multiplying the kinetic term of a collective $SU(2)$ orientation coordinate χ, clarifying how $v_{\rm eff}^2 \equiv Z_\chi v^2$ enters the standard mass matrix. This paper establishes the quantum consistency, RG control, and domain of validity of the Axis EFT, and implements a UV–consistent Abelian–sector extension (Scenario A) via threshold matching at $\mu = \Lambda_q$ and basis termination above Λ_q. Other Standard Model factors $(SU(2)_L, SU(3)_C)$ are treated as external inputs. In this framework, the low-energy electromagnetic coupling $\alpha(0)$ is obtained as a consistency relation between Axis $U(1)_Z$ running and the empirical weak input $g_L(\mu_{\rm EW})$, not as a de-novo prediction. All results are obtained within the Axis–EFT window $E_{\rm IR} < \mu < \min(\Lambda_\Phi, \Lambda_q)$ with $\Lambda_\Phi \sim 10^5$ GeV and $\Lambda_q \sim 10^{16}$ GeV; above Λ_q the Abelian basis does not exist and the theory matches onto its non-Abelian/pre-geometric parent. The $SU(2)$ parent in the UV extension is a pre-geometric/non-electroweak factor; $SU(2)_L$ is treated as external input (Scenario A) when comparing to data.

1 Introduction

Why ultraviolet extension matters. Effective field theories are predictive only up to the scale where new physics enters. A consistent ultraviolet (UV) extension is therefore essential both conceptually and phenomenologically: it ensures that couplings remain finite, vacuum stability is maintained, and unitarity is preserved at arbitrarily high energies.

Well–known examples include the embedding of the Fermi four–fermion interaction into the renormalizable electroweak theory [1, 2, 3], and the UV safety of non–Abelian gauge theories due to asymptotic freedom [4, 5]. Conversely, Abelian theories such as QED are only effective descriptions, with the Landau–pole pathology implying the need for a more fundamental parent theory [6].

*This work was conducted independently and does not represent the views of Indiana University.

Axis model context. In this paper we ask whether the Axis EFT, which successfully organizes low–energy matter and geometry, can be consistently extended into the UV. The challenge is twofold: (i) the $U(1)_Z$ sector has the same Landau–pole problem as QED, and (ii) the scalar–Yukawa subsystem admits no perturbative non–Gaussian fixed point. The purpose of our UV extension is therefore to replace the ill–posed Abelian extrapolation with a metric–independent, pre–geometric parent theory that condenses to generate spacetime and the Axis EFT at scale Λ_q.

1.1 Success and limitations of the Axis Model EFT

EFT achievements. The Axis framework realizes observed low-energy physics in an effective field theory (EFT) built from: (i) a complex scalar $\Phi = \rho\, e^{i\theta}$ whose coherent domains gate projection into spacetime; (ii) composite matter excitations (*mortons*) stabilized by scalar coherence and organized by an orthogonal tri-vector frame; and (iii) gauge sectors that, in the coherent phase, reproduce the Standard Model (SM) gauge structure and predict fermion hierarchies and mixings with good accuracy. Gravitational dynamics emerge via coherence-gated projection and a vierbein/metric reconstructed from internal data, providing a unified EFT treatment of matter and geometry.

Conceptual picture. At a high level, the scale Λ_q marks a phase transition between two descriptions: *below* Λ_q the world is geometric—the tri-vector frame exists, mortons are well-defined composites, and an Abelian Axis basis $U(1)_Z$ appears as a Cartan projection; *above* Λ_q the description is pre-geometric—the scalar condensate is melted, mortons dissolve, and the relevant variables are topological and undifferentiated.

Limitations. As an EFT, the Axis model is predictive on

$$E_{\mathrm{IR}} < \mu < \min(\Lambda_\Phi, \Lambda_q),$$

but exhibits two familiar UV issues if naively extrapolated: (i) the Abelian $U(1)_Z$ coupling has a positive one-loop coefficient, leading to a Landau-pole artifact in a purely Abelian continuation; and (ii) the (λ, y) subsystem (scalar quartic and Yukawa) lacks a perturbative non-Gaussian UV fixed point. These features signal the need for new degrees of freedom and a non-Abelian (or pre-geometric) parent above Λ_q. A complementary paper develops the flavor-sector organization and Standard-Model parameter predictions [7]; the present work is confined to EFT consistency, renormalization, and matching.

1.2 The ultraviolet imperative

Our goal is to replace the Abelian extrapolation by an explicit pre-geometric theory that: (i) has no background metric in the deep UV; (ii) condenses at $\rho = \rho_0$ to generate geometry and the Axis–EFT below Λ_q; and (iii) delivers matched two-loop evolution of EFT couplings, avoiding the spurious Abelian Landau-pole artifact because the $U(1)_Z$ basis terminates at $\mu = \Lambda_q$ and matches to the non-Abelian parent, and maintaining absolute vacuum stability.

In keeping with the scope of this paper (Scenario A), we implement a UV–consistent Abelian–sector extension via threshold matching at $\mu = \Lambda_q$ and basis termination above Λ_q; $SU(2)_L$ and $SU(3)_C$ remain external inputs for phenomenology in Sec. 5.[1]

[1] A more ambitious "Scenario B" would extend the pre-geometric construction to derive $\mathrm{SU}(2)_L$ (and possibly $\mathrm{SU}(3)_C$), converting the α consistency relation into a fully ab initio electroweak prediction. That program is orthogonal to the Abelian pathology and beyond our present scope.

1.3 A pre-geometric proposal

We postulate a metric-independent UV action of BF type with compact gauge group $G = \mathrm{SU}(2)$, an adjoint two-form B, and an order parameter $\Sigma = \rho\, e^{i\vartheta}$: simplicity constraints enforce that, in the coherent phase, B becomes a simple self-dual bivector from which the Urbantke metric is reconstructed. At $\mu \simeq \Lambda_q$ a condensation mechanism (driven by the Σ sector and a compact three-form C with $H = dC$) selects a coherent branch, *geometry emerges*, and the Axis–EFT appears as the low-energy description. Instanton-like configurations aligned by a triad interaction seed the internal frame; after Axis projection these map to morton tri-vectors in the EFT. Matching at $\mu = \Lambda_q$ relates EFT couplings $\{\lambda,\ y \equiv g_{\psi\Phi},\ g_{\Phi V},\ g_Z\}$ to UV parameters, providing initial conditions for two-loop running below Λ_q and eliminating the spurious Abelian UV behavior by basis termination.

1.4 Outline of the paper

Section 2 reviews the Axis EFT: its core ingredients (mortons, scalar coherence, projection), the one-loop renormalization structure, and the ultraviolet breakdowns that motivate a parent theory. Section 3 introduces the pre-geometric UV theory: field content and symmetries, the action, the simplicity/Urbantke mechanism for emergent geometry, the condensation process, and matching onto the Axis EFT. Appendices A (variational identities), B (spectra), C (threshold derivations), and D (projector normalizations) provide detailed derivations. Section 4 presents the complete two-loop RGEs for the EFT, applies threshold matching at Λ_q, and evolves from the Planck scale to the electroweak scale, demonstrating that the spurious Abelian Landau-pole artifact is avoided because the $U(1)_Z$ basis terminates at Λ_q and matches to the non-Abelian parent, and showing vacuum stability. Section 5 propagates these results to low-energy observables, including the running of $\alpha(0)$, two-loop fermion mass evolution, and signatures of the pre-geometric phase. Section 6 concludes with scope, limitations, and future directions.

Scope. This paper treats only the Axis $U(1)_Z$ sector. The construction is UV-consistent in the sense that $U(1)_Z$ above Λ_q is replaced by a pre-geometric non-Abelian parent, with two-loop matched evolution below Λ_q. Other gauge factors (e.g. $SU(2)_L$, $SU(3)_C$) are taken as external inputs (Scenario A). Accordingly, the fine-structure constant result is a consistency relation between the Axis $U(1)_Z$ dynamics and the empirical weak sector, not an ab initio electroweak prediction. The EFT is valid on $E_{\mathrm{IR}} < \mu < \min(\Lambda_\Phi, \Lambda_q)$, with $\Lambda_\Phi \sim 10^5$ GeV and $\Lambda_q \sim 10^{16}$ GeV, within which BRST consistency, anomaly cancellation, perturbative running, and vacuum stability are maintained [8, 7, 9]. Outside this window the EFT is not predictive. A more ambitious Scenario B would extend the pre-geometric construction to derive $SU(2)_L$ (and possibly $SU(3)_C$) above Λ_q, converting the present α relation into a parameter-free prediction.

Position relative to prior attempts. Across frameworks, the electroweak group is typically *embedded* and then broken in GUTs—e.g. SU(5) [10], SO(10) [11], and E_6 [12]—*engineered* via string compactifications and Wilson lines (heterotic $E_8 \times E_8/\mathrm{SO}(32)$ and Calabi–Yau compactifications) [13, 14, 15, 16, 17], or *reconstructed* from axioms in spectral/noncommutative geometry [18, 19, 20]. None of these yields a unique, assumption-free ab initio derivation of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$; in practice, hypercharge embeddings, matter content, and couplings depend on the chosen group, compactification, or internal algebra [21, 22]. Within that landscape, our contribution is orthogonal and deliberately narrower: we UV-extend the *Abelian* Axis sector by replacing $U(1)_Z$

above Λ_q with a pre-geometric non-Abelian parent and performing explicit single-threshold matching with full two-loop running below Λ_q (§§3–4), then propagate to α and fermion observables (§5). In Scenario A we treat $g_L(\mu_{\rm EW})$ as empirical input, so the α statement is a *consistency relation* rather than an ab initio electroweak prediction. Methodologically this is more explicit and reproducible on thresholds/RG bookkeeping than typical sector-only treatments (single threshold, two-loop coefficients, and positivity/stability checks are all spelled out), but conceptually less ambitious than GUT/string programs. The path to a genuine ab initio electroweak claim is Scenario B: extend the same pre-geometric construction so that $\mathrm{SU}(2)_L$ (and possibly $\mathrm{SU}(3)_C$) emerges above Λ_q, converting the present consistency relation for α into a parameter-free prediction.

2 Review of the Axis Model as an Effective Field Theory

This section summarizes the Axis Model in its EFT incarnation: field content and projection (Sec. 2.1), one-loop renormalization structure (Sec. 2.2), and the specific UV breakdowns that motivate the UV extension of Sec. 3 (Sec. 2.3).

2.1 Core components

Mortons and scalar coherence. Matter is realized as scalar-stabilized composites (*mortons*) built from quantized internal vector displacements along orthogonal axes, with the simplest "ordinary matter" morton carrying the tri-vector content $(1_z, 2_x)$.[2] Coherent binding—and therefore projection into spacetime observables—exists only in domains where the complex scalar $\Phi = \rho\, e^{i\theta}$ is phase-aligned. Outside such domains mortons dissolve and geometric projection fails. This picture is formalized by (i) a scalar alignment operator $\widehat{\Pi}_\Phi$ that filters to the coherent subspace, and (ii) a local projection map $P^\mu_\Phi[\cdot]$ that embeds internal displacements into the tangent bundle only when $|\nabla\theta|^2 \ll \Lambda_\Phi^2$. These constructions underlie emergent vierbein/metric definitions. See Refs. [23, 24] for operator definitions and examples.

Master Lagrangian (EFT window). The gauge-consistent EFT used for quantum calculations organizes the renormalizable pieces into scalar, vector (two Abelian basis fields X_μ, Z_μ[3]), and fermion sectors, plus leading EFT operators. A canonical presentation is (indices and signs as in Ref. [8], App. M):

$$\begin{aligned}\mathcal{L}_{\rm Axis} = {} & \tfrac{1}{2}(\partial_\mu\Phi)(\partial^\mu\Phi) \;-\; V(\Phi) \\ & -\; \tfrac{1}{4}\, F_X^{\mu\nu} F^X_{\mu\nu} \;+\; \tfrac{1}{2}\, g^{(X)}_{\Phi V}\, (\Phi^\dagger\Phi)\, X_\mu X^\mu \\ & -\; \tfrac{1}{4}\, F_Z^{\mu\nu} F^Z_{\mu\nu} \;+\; \tfrac{1}{2}\, g_{\Phi V}\, (\Phi^\dagger\Phi)\, Z_\mu Z^\mu \\ & +\; i\,\bar\psi\,\partial\!\!\!/\psi \;-\; y\,\Phi\,\bar\psi\psi \;+\; g_Z\, Q_\psi\, \bar\psi\gamma^\mu Z_\mu\psi \\ & +\; \frac{g_A}{M}\,\Phi_A\, F_X^{\mu\nu}\tilde F^X_{\mu\nu} \;+\; \kappa_{\alpha Z}\,(\partial_\mu\alpha)\, Z^\mu \;+\; \cdots \end{aligned} \tag{1}$$

Here $y \equiv g_{\psi\Phi}$ is the Yukawa coupling; $g_{\Phi V}$ is the scalar–vector mass coupling with $m_Z^2 = g_{\Phi V}\,\langle\Phi^\dagger\Phi\rangle$ (so $m_Z^2 = g_{\Phi V}\, v_\Phi^2$ in the EFT vacuum); $g_Z \equiv g_{\psi Z}$ is the fermion–Z gauge coupling; $\kappa_{\alpha Z}$ is the derivative portal of the EFT phase α. The X-sector uses $g^{(X)}_{\Phi V}$ analogously.

[2] We use the internal labels (x, z) to match the EFT vector sectors below.

[3] Here X_μ is the axis-x gauge potential whose *coherent* excitations realize the composite photon. In the foundational Axis notes, the photon is a six–morton composite with internal period $6r$ (observable wavelength $12r$) built from x–axis displacements; see the composite–photon discussion there for details and consistency checks

Projection and EFT domain. The EFT is predictive on the interval

$$E_{\rm IR} < \mu < \min(\Lambda_\Phi, \Lambda_q),$$

where Λ_Φ is the scalar-coherence cutoff and Λ_q the morton-dissolution scale; below $E_{\rm IR}$ one matches to the usual low-energy limits (SMEFT, chiral Lagrangian).[4]

Two ultraviolet scales and their roles. The Axis EFT is predictive on

$$E_{\rm IR} < \mu < \min(\Lambda_\Phi, \Lambda_q)$$

with two physically distinct upper thresholds:

Λ_Φ (scalar coherence scale). This is the energy at which phase fluctuations of the complex scalar $\Phi = \rho e^{i\theta}$ destroy scalar coherence. Above Λ_Φ, the projection map $P^\mu_\Phi[\cdot]$ ceases to be reliable, and the identification of the non-Abelian projected connection from the internal vector basis is no longer well-defined. The Abelian *basis* fields (X_μ, Z_μ) may still be written, but their coherent projection into the emergent SM-like gauge structure fails.

Λ_q (morton-dissolution scale). This higher scale marks the melting of the scalar condensate that stabilizes mortons. Above Λ_q morton composites dissolve, the tri-vector frame has no meaning, and the Abelian Axis basis does not exist; the correct description is the pre-geometric (topological) parent sector. This is the true end of the Axis EFT.

Operationally, Λ_Φ is where the *projection mechanism* breaks, while Λ_q is where the *composite degrees of freedom* themselves disappear and the theory enters the pre-geometric phase.

2.2 One-loop renormalization group structure

With the coupling basis $\{\,\lambda,\ y \equiv g_{\psi\Phi},\ g_{Z\Phi},\ g_Z \equiv g_{\psi Z}\,\}$ fixed by (1), one-loop running in the minimal Abelian Axis–EFT takes the form (normalization as in Ref. [9]):

$$16\pi^2\,\beta^{(1)}_\lambda = +18\,\lambda^2 \;-\; 24\,y^4 \;+\; 8\,\lambda\,y^2, \qquad 16\pi^2\,\beta^{(1)}_y = +5\,y^3, \tag{1}$$

$$16\pi^2\,\beta^{(1)}_{g_{\Phi V}} = a\,g^3_{\Phi V} \quad (a>0), \qquad 16\pi^2\,\beta^{(1)}_{g_Z} = b^{(1)}_Z\,g^3_Z, \qquad b^{(1)}_Z > 0 \;\;\text{(Abelian)}. \tag{2}$$

The explicit two-loop structures for (λ, y) used later in §4 match the signs above and are collected in App. E of this paper / Appendix X of Ref. [9]. Within the EFT window the quartic remains positive for representative inputs (see the stability scans and figures in Ref. [25]).

Vacuum stability band. At one loop the competing effects in β_λ (negative y^4 vs. positive λ^2 and gauge-induced pieces) define a broad stability band in which $\lambda(\mu) > 0$ across $m_Z \lesssim \mu \lesssim \Lambda_\Phi$; two-loop corrections preserve this qualitative picture in scans.

2.3 Identifying the UV breakdowns

Two distinct issues arise when one *formally* extrapolates the Abelian Axis–EFT beyond its intended domain:

[4] Within this window the non-Abelian projected connection governs physical amplitudes; the Abelian X, Z basis is a computational device tied to the projection map.

(i) Abelian Landau pole. Because $b_Z^{(1)} > 0$, the Abelian $U(1)_Z$ coupling $g_Z(\mu)$ grows and would encounter a Landau pole at exponentially high scales in a purely Abelian continuation. In the Axis program this is an *artifact* of using the Abelian basis outside its validity: above Λ_q the Abelian description is not defined, and the appropriate variables are the non-Abelian projected connection or (in this paper) the pre-geometric UV fields.

(ii) No perturbative UV fixed point in the (λ, y) plane. The one-loop Yukawa equation is UV-repulsive ($\beta_y^{(1)} \propto +y^3$), and while two-loop terms slow the growth at very large coupling, there is no small-coupling non-Gaussian UV fixed point for (λ, y). Thus a perturbative UV extension cannot live *within* the renormalizable Abelian EFT. This is consistent with interpreting the Axis Lagrangian as an EFT valid only up to $\min(\Lambda_\Phi, \Lambda_q)$.

Coherence breakdown scale Λ_Φ. We estimate the end of scalar coherence by the integral criterion $\int_{\mu_0}^{\Lambda_\Phi} d\ln\mu\, \gamma_\Pi(\mu) \simeq \varepsilon_{\rm coh}$ with $\varepsilon_{\rm coh} = 0.3$–$0.5$. Using the background-field one-loop form of γ_Π and the EW-anchored inputs in App. E, the Abelian-dominated estimate yields $\Lambda_\Phi \simeq (0.8\text{–}1.2) \times 10^5$ GeV; Yukawa screening only increases this value. A self-contained derivation and recipe to reproduce the number are given in App. H.

Implication. The EFT is sound and predictive on $E_{\rm IR} < \mu < \min(\Lambda_\Phi, \Lambda_q)$ but cannot itself address the UV. Sections 3–4 construct a pre-geometric, non-Abelian parent above Λ_q and perform matched two-loop evolution below it, removing the Abelian pole by basis termination and providing a positive quartic throughout the flow.

2.4 Conventions

Conventions for the scalar, phases, and mixing (used throughout).

Symbol	**Domain**	**Meaning / Usage**
$\Phi(x)$	UV/EFT	Complex scalar field. Two equivalent parameterizations are used: *(polar)* $\Phi = \rho\, e^{i\theta}$ and *(VEV–shift)* $\Phi = (v_\Phi + h)\, e^{i\alpha/f_\alpha}$.
$\rho(x)$, $\theta(x)$	UV	Polar variables. ρ is radial modulus; θ is a dimensionless phase coordinate.
v_Φ	EFT	Vacuum modulus: $v_\Phi \equiv \rho_0$.
$h(x)$	EFT	Canonical radial fluctuation about v_Φ.
$\vartheta(x)$	UV	Axion–like UV phase variable (dimensionless, 2π-periodic). Appears with flux/3-form data via $S = q + f\,\vartheta$.
$\alpha(x)$	EFT	Canonical EFT phase (dimensionful). Enter exponents as $e^{i\alpha/f_\alpha}$; transforms $\alpha \to \alpha + f_\alpha \epsilon$.
f_α, f_ϑ	EFT/UV	Decay constants for α (EFT) and ϑ (UV), respectively. Locally $\theta \equiv \vartheta/f \approx \alpha/f_\alpha$ up to mixing/renormalization.
χ	Matching	*Single mixing angle* between radial and phase directions that defines the light mass eigenstate at $\mu = \Lambda_q$. Defined once (Eq. (3)).

When to use which: Use $\{\rho, \vartheta\}$ (polar/UV) for writing the UV action and for matching formulae expressed in terms of $\tau(\rho), \kappa(\rho), \eta(\rho)$. Use $\{h, \alpha\}$ (VEV–shift/EFT) for all renormalized couplings and running $\{\lambda, y, g_{\Phi V}, g_Z\}$ below Λ_q. The angle χ appears only in matching relations that project the UV polar basis onto EFT mass eigenstates (e.g. $g_{\Phi V}$).

Mixing convention (defined once). Near the vacuum, write UV fluctuations in the polar basis $(\delta\rho,\, \delta\vartheta)$ and define a single rotation by χ to the light/heavy mass eigenstates $(\Phi_{\text{light}},\, \Phi_{\text{heavy}})$:

$$\begin{pmatrix} \delta\rho \\ f_\vartheta\, \delta\vartheta \end{pmatrix} = \begin{pmatrix} \cos\chi & -\sin\chi \\ \sin\chi & \cos\chi \end{pmatrix} \begin{pmatrix} \Phi_{\text{light}} \\ \Phi_{\text{heavy}} \end{pmatrix}. \tag{3}$$

By this convention, the light scalar couples to vectors via the $\delta\rho$ component ($\propto \cos\chi$), which is the origin of the $\cos\chi$ factor in $g_{\Phi V}$ at matching.

Conventions. We denote the Abelian Axis factor by $U(1)_Z$, with gauge field Z_μ and coupling g_Z. Below Λ_q the EFT coherence phase is α; the UV Pontryagin phase is ϑ; they are related by a single mixing angle χ at threshold (Sec. 3.5). The dimension-4 scalar–vector operator is written

$$\frac{1}{2}\, g_{\Phi V}\, (\Phi^\dagger \Phi)\, Z_\mu Z^\mu,$$

and we use $g_{\Phi V}$ exclusively for its coefficient. Throughout we take $\mathrm{Tr}(T^i T^j) = \delta^{ij}$ so the Cartan projector has $\omega_{\text{proj}} = 1$ at matching.

3 The UV Extension: Dynamics Above the Morton Scale

3.1 Dissolution and the Pre-Geometric Phase ($\mu > \Lambda_q$)

Clarifying note (pre-geometric). "pre-geometric" here means the fundamental field is the adjoint-valued two-form B; no background metric is assumed in the UV action. The spacetime *manifold* (with its differential-form structure) is taken as given; what *emerges* in the coherent phase is the metric/vierbein via the Urbantke construction from simple B on the simplicity manifold (see App. A.3; cf. [26, 27, 28, 29]); and see Sec. 3.2 for how integrating out B around a simple background induces the Yang–Mills kinetic term.

Above the morton scale Λ_q, the theory undergoes a phase transition to a *pre-geometric* regime. The energy density is sufficient to melt the scalar condensate that stabilizes the low-energy vacuum; as a result, mortons dissolve and the entire geometric axis structure (the orthogonal tri-vector frame) ceases to exist as a meaningful set of variables. What remains is a topological BF sector (plus the compact three-form) with no background metric and no Abelian Axis basis. The action reduces to a topological BF sector weakly perturbed by the order parameter $\Sigma = \rho\, e^{i\vartheta}$ and by the three-form sector:

$$S_{\text{UV}} \xrightarrow[\rho\to 0]{} \int \mathrm{Tr}(B\wedge F) \;+\; \frac{\xi}{8\pi^2}\,\vartheta\,\mathrm{Tr}(F\wedge F) \;+\; \int H\wedge q\,. \tag{4}$$

There is no background metric and thus no Hodge operator; local geometric excitations are absent. In this regime: (i) the Axis Abelian basis does not exist (only the compact non-Abelian connection A is meaningful), (ii) mortons dissolve (their IR definition requires scalar-coherent projection and simple bivectors), and (iii) vacuum sectors are labeled by topological invariants $\int \mathrm{Tr}(F\wedge F) \in \mathbb{Z}$ and the four-form flux $\int H$.

Boundary remark. On manifolds with boundary, the Pontryagin density is a total derivative, $\mathrm{Tr}(F\wedge F) = d\,\mathrm{Tr}(A\wedge dA + \frac{2}{3}A\wedge A\wedge A)$, so the $\xi\,\vartheta\,\mathrm{Tr}(F\wedge F)$ term induces a boundary Chern–Simons action. The three-form sector similarly yields $\int_{\partial\mathcal{M}} C\wedge(q + f\vartheta)$. All variational statements below assume compact spacetime or boundary conditions consistent with these surface terms.

Scope of the UV SU(2). The SU(2) gauge symmetry in $S_{\rm UV}$ is the ultraviolet parent of $U(1)_Z$ only (see Sec. 2 for scope).

Notation. We denote the emergent Cartan Abelian factor by $U(1)_Z$; this is the same object previously labeled $U(1)_n$, i.e. $U(1)_Z \equiv U(1)_n$. We use $U(1)_Z$ henceforth for consistency with the EFT couplings g_Z and $g_{\Phi V}$.

Scenario A: $\mathrm{SU}(2)_{\rm UV} \longrightarrow U(1)_Z$ (Cartan), primarily UV–extending the EFT $U(1)_Z$ sector.

This makes explicit why the EFT Abelian Landau–pole artifact disappears above Λ_q: the Abelian basis is simply not defined in the pre-geometric phase.[5]

3.2 The Pre-Geometric Lagrangian and Emergent Geometry

This section defines the ultraviolet (UV) pre-geometric theory whose condensed phase yields the Axis–EFT. Above the morton–dissolution scale Λ_q there is no background metric; dynamics are governed by a gauge–topological sector, an order parameter $\Sigma = \rho\, e^{i\vartheta}$, and a compact three-form. Geometry, the Abelian Axis basis, and mortons themselves arise only after condensation.

Field content and symmetries. We take a compact gauge group $G = \mathrm{SU}(2)$ with connection $A = A^i T^i$ and curvature $F = dA + A \wedge A$. The variable $B = B^i T^i$ is an adjoint-valued two-form. The order parameter $\Sigma = \rho\, e^{i\vartheta}$ carries a global $\mathrm{U}(1)_{\rm coh}$ "coherence" phase. A compact three-form C with field strength $H = dC$ implements metric-independent locking of ϑ. We also use a symmetric traceless multiplier $\Psi_{ij} = \Psi_{ji}$ with $\delta^{ij}\Psi_{ij} = 0$ to impose simplicity. Discrete $B \mapsto -B$ parity can be imposed to forbid odd-B terms if desired.[6]

Action and UV metric independence. We postulate the metric-independent action

$$S_{\rm UV} = \int \mathrm{Tr}\big[B \wedge F(A)\big] + \frac{\kappa(\rho)}{2}\int \Psi_{ij}(\Sigma)\Big(B^i \wedge B^j - \tfrac{1}{3}\delta^{ij} B^k \wedge B^k\Big) + \frac{\tau(\rho)}{2}\int \mathrm{Tr}\big(B \wedge \star_B B\big)$$
$$+ \frac{\xi}{8\pi^2}\int \vartheta\, \mathrm{Tr}\big(F \wedge F\big) + \int \Big[\tfrac{\eta(\rho)}{2}\, H \wedge \star_B H + H \wedge (q + f\,\vartheta)\Big] + \zeta \int \epsilon_{ijk}\, B^i \wedge F^j\, n^k(\Sigma)\,. \quad (5)$$

Here $\star_B$ is the Hodge operator built from the Urbantke metric of B (defined below). The scalar-dependent couplings obey

$$\kappa(0) = \tau(0) = \eta(0) = 0, \qquad \kappa(\rho_0),\ \tau(\rho_0),\ \eta(\rho_0) > 0 \quad (\rho = \rho_0 \text{ coherent phase}).$$

Consequently, in the strict UV limit $\rho \to 0$ all $\star_B$-dependent terms decouple and (5) reduces to the topological sector $\int \mathrm{Tr}(B \wedge F) + \frac{\xi}{8\pi^2}\int \vartheta\, \mathrm{Tr}(F \wedge F) + \int H \wedge q$, with no background metric.

On the Hodge $\star_B$. Because $\star_B$ is defined by the Urbantke metric of B, kinetic terms carrying $\star_B$ are physically meaningful only *after* condensation, where a stable simple background B_0 exists. All quadratic spectra and couplings used below are computed as an expansion around B_0; deep in the UV ($\rho \to 0$) the theory is purely topological and metric-free.

[5] A more ambitious "Scenario B" would extend the pre-geometric construction to derive $\mathrm{SU}(2)_L$ itself (and possibly $\mathrm{SU}(3)_C$), yielding a fully *ab initio* α prediction. That program requires additional model choices and lies beyond our present scope.

[6] We normalize $\mathrm{Tr}(T^i T^j) = \delta^{ij}$. With this choice, $\frac{1}{8\pi^2}\int \mathrm{Tr}(F \wedge F) \in \mathbb{Z}$. If one prefers $\mathrm{Tr}(T^i T^j) = \frac{1}{2}\delta^{ij}$, rescale integer factors accordingly.

Quantization of the ξ coupling. Since $\frac{1}{8\pi^2}\int \mathrm{Tr}(F\wedge F)\in\mathbb{Z}$, invariance of the path integral under the 2π-periodic shift $\vartheta\to\vartheta+2\pi$ requires $\xi\in\mathbb{Z}$. This also lets one treat ϑ as an angular variable on the vacuum branches selected by the three-form sector.

Simplicity constraint and Urbantke metric. Integrating out Ψ_{ij} imposes

$$B^i\wedge B^j-\tfrac{1}{3}\delta^{ij}B^k\wedge B^k=0, \tag{6}$$

whose solutions are simple bivectors. On such configurations the (densitized) Urbantke metric

$$\tilde g_{\mu\nu}(B)=\frac{1}{12}\,\epsilon^{\alpha\beta\gamma\delta}\,\epsilon_{ijk}\,B^i{}_{\mu\alpha}\,B^j{}_{\beta\gamma}\,B^k{}_{\delta\nu},\qquad g_{\mu\nu}=\tilde g_{\mu\nu}\,(\det\tilde g)^{-1/4} \tag{7}$$

is non-degenerate; the Hodge $\star_B$ is then defined with respect to $g_{\mu\nu}(B)$.

Variational motivation. See Appendix K for a simple variational motivation that shows the simplicity tensor

$$C_{ij}\;\equiv\;B_i\wedge B_j\;-\;\tfrac{1}{3}\,\delta_{ij}\,B_k\wedge B^k$$

is dynamically selected—i.e. it minimizes a gauge-invariant "free-energy" functional—and that this condition connects directly to the Plebanski/GR simplicity constraint.

Gauge/BRST structure and $n(\Sigma)$. Gauge invariance under G is manifest. The triad interaction $\epsilon_{ijk}B^i\wedge F^j\,n^k(\Sigma)$ is gauge-invariant provided $n(\Sigma)$ transforms in the adjoint; one explicit construction is $n(\Sigma)=\mathrm{Ad}_{U[\Sigma]}\hat z$ with $U[\Sigma]\in\mathrm{SU}(2)$ and a fixed Cartan direction $\hat z$.[7] BRST quantization proceeds as in BF theory with simplicity constraints; the multiplier sector does not break gauge invariance.

Limiting regimes and emergent YM kinetic. In the deep UV ($\rho\to 0$) the theory reduces to the topological BF sector with no metric. In the coherent phase ($\rho=\rho_0$) the constraint (6) and $\tau(\rho_0)>0$ define $g_{\mu\nu}[B]$ and induce a Yang–Mills kinetic term for A. Integrating out δB at quadratic order—using the linearized Urbantke variation $\delta(\star_B)$ and the definition of K_B given in Appendix A.3—yields (App. B and App. A)

$$S_{\rm eff}[A]\supset-\frac{1}{2\,\tau(\rho_0)}\int \mathrm{Tr}\big(F\wedge\star_{B_0}F\big),\qquad \frac{1}{g^2_{\rm NA}}=\frac{\omega_{\rm YM}}{\tau(\rho_0)}\quad(\omega_{\rm YM}=1\text{ in our conventions})\,.$$

Below Λ_q the Abelian Axis field Z appears as the Cartan projection along the coherence frame, with projector weight $\omega_{\rm proj}=1$ on the simple background of App. B.

Role of the three-form (pointer). The three-form sector yields $\eta(\rho)\star_B H+(q+f\vartheta)=0$ and $dH=0$; eliminating H produces the effective potential $V_{\rm 4f}=(q+f\vartheta)^2/\big[2\,\eta(\rho)\big]\times\mathrm{vol}_B$, which locks ϑ and biases ρ to condense. The condensation mechanism and the definition of Λ_q are developed in Sec. 3.3.

[7] If Σ is taken as an SU(2) doublet, another covariant choice is $n^i(\Sigma)=\Sigma^\dagger T^i\Sigma/(\Sigma^\dagger\Sigma)$.

Clarification on "pre-geometric." In this paper, the term *pre-geometric* means that the ultraviolet action (5) is written without a background metric: the two-forms B, the curvature F, and the three-form H are treated as algebraic variables on a four-manifold, independent of any metric structure. A metric first appears only in the coherent phase, when the simplicity constraint enforces that B becomes a simple bivector and the Urbantke construction yields $g_{\mu\nu}(B)$. Accordingly, the Hodge operator $\star_B$ has meaning only *after* condensation, once $g_{\mu\nu}(B)$ is defined. Throughout, "pre-geometric" denotes this metric-independent regime, not an assumption of hidden background geometry.

Conceptual summary. *Below* Λ_q the scalar is condensed, giving a coherent background that defines the Urbantke metric and a tri-vector frame; mortons are stable composites, and an Abelian Axis basis $U(1)_Z$ exists as a Cartan projection. *Above* Λ_q the condensate is melted: the metric and tri-vector structure are not defined, the Abelian basis is absent, and the dynamics are topological. The UV extension we construct shows how these two regimes are matched consistently by threshold corrections at Λ_q.

3.3 The Condensation Mechanism at Λ_q

Varying the three-form sector yields $\eta(\rho) \star_B H + (q + f\vartheta) = 0$ and $dH = 0$. Eliminating H generates a metric-independent but ρ-dependent potential,

$$V_{4\text{f}}(\vartheta, \rho) = \frac{(q + f\,\vartheta)^2}{2\,\eta(\rho)}\,\text{vol}_B, \tag{8}$$

with vol_B the emergent volume density. Minimization locks the phase, $q + f\vartheta = 2\pi N$, selecting a discrete vacuum branch $N \in \mathbb{Z}$, and biases toward $\rho \neq 0$ if $\eta'(\rho) > 0$.

Including local UV terms and the B-sector,

$$V_{\text{eff}}(\rho, \vartheta; B) = \frac{m_0^2}{2}\rho^2 + \frac{\lambda_\Sigma}{4}\rho^4 + \frac{\kappa(\rho)}{2}\,\mathcal{S}[B;\Sigma] + \frac{\tau(\rho)}{2}\,\text{Tr}(B \wedge \star_B B) + \frac{(q + f\vartheta)^2}{2\,\eta(\rho)}\,\text{vol}_B\,. \tag{9}$$

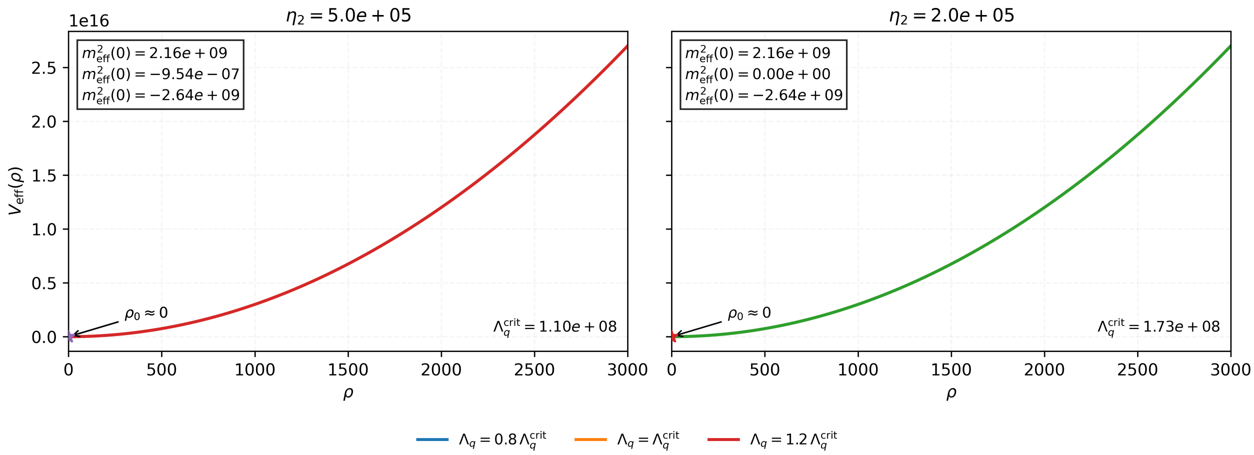

Figure 1: **Phase structure of $V_{\rm eff}(\rho)$ across Λ_q.** Two panels show $V_{\rm eff}(\rho)$ for $\eta(\rho) = \eta_0 + \eta_2\rho^2$ and $\tau(\rho) = \tau_2\rho^2$ at two choices of η_2. In each panel we sweep $\Lambda_q \in \{0.8, 1.0, 1.2\}\Lambda_q^{\rm crit}$; the middle curve satisfies $m^2_{\rm eff}(0) = V''_{\rm eff}(0) = 0$, and for $\Lambda_q > \Lambda_q^{\rm crit}$ an off–origin minimum ρ_0 emerges. Here $\Lambda_q^{\rm crit} = \eta_0\sqrt{(m_0^2 + 6v^2\tau_2)/(\eta_2\,{\rm vol}_B)}$.

Reverse of dissolution. The condensation we analyze here is the time-reverse of the high-energy dissolution: as the scale runs *down* through Λ_q, the pre-geometric substrate freezes into a structured, geometric phase, dynamically selecting an orthogonal tri-vector frame and re-instating the Axis projection rules.

The condensation scale Λ_q is defined by the RG scale where the curvature at $\rho = 0$ changes sign:

$$m^2_{\rm eff}(\mu) = \partial_\rho^2 V_{\rm eff}\big|_{\rho=0} = m_0^2(\mu) - \frac{(q + f\vartheta_*)^2}{2}\,\frac{\eta_2}{\eta_0^2} + \cdots, \qquad \Lambda_q:\ m^2_{\rm eff}(\Lambda_q) = 0, \tag{10}$$

with $\eta(\rho) = \eta_0 + \eta_2\rho^2 + \cdots$ near the origin and ϑ_* the locked phase.

Definition of Λ_q. We define the condensation scale Λ_q as the RG scale at which the curvature of the effective potential at $\rho = 0$ changes sign, i.e. $m^2_{\rm eff}(\Lambda_q) = 0$. This marks the onset of the transition to the coherent, geometric phase. As derived in App. I, this condition depends on the running radial mass $m^2_\rho(\mu)$ together with contributions from the three-form sector. For a representative benchmark, this yields the quantitative estimate $\Lambda_q \simeq 1.1 \times 10^{16}$ GeV.

Finite-temperature potential and nucleation. We approximate the thermal effective potential for the radial mode by the standard high-T form (with daisy resummation implicit in the coefficients):

$$V_{\rm eff}(\rho; T) \;\simeq\; D\,(T^2 - T_0^2)\,\rho^2 \;-\; E\,T\,\rho^3 \;+\; \frac{\lambda_T}{4}\,\rho^4, \tag{11}$$

where $D > 0$, $E > 0$ receive bosonic contributions from Z and heavy modes with ρ-dependent masses, while $\lambda_T > 0$ is the quartic at temperature T.

The phase transition is first order when $E \neq 0$. Degenerate minima occur at the critical temperature T_c:

$$\frac{\rho_c}{T_c} = \frac{2E}{\lambda_T}, \qquad T_c^2 = \frac{T_0^2}{1 - \frac{E^2}{D\,\lambda_T}}, \tag{12}$$

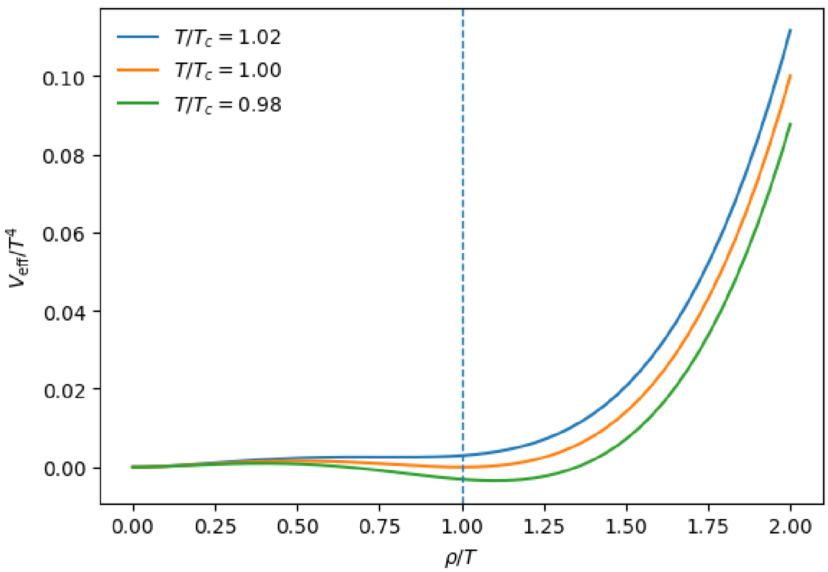

Figure 2: **Finite-temperature effective potential** $V_{\rm eff}(\rho;T)$ of Eq. 11. A barrier opens near T_c for $E \neq 0$ (degenerate minima), consistent with a weakly first-order transition; for smaller E (or larger screening) the barrier disappears and the transition becomes a smooth crossover. Benchmarks shown: $D = 0.10$, $E = 0.05$, $\lambda_T = 0.10$, giving $T_0/T_c = \sqrt{1 - E^2/(D\lambda_T)} \simeq 0.866$ and $\rho_c/T_c = 2E/\lambda_T = 1$. See App. J.

and the surface tension and latent heat at T_c are

$$\sigma(T_c) = \int_0^{\rho_c} d\rho \sqrt{2V_{\rm eff}(\rho;T_c)} = \frac{4\,E^3\,T_c^3}{3\sqrt{2}\,\lambda_T^{5/2}}, \qquad L \equiv -T_c\Big[\partial_T \Delta V\Big]_{T_c} = \frac{8\,E^2 T_c^4}{\lambda_T^3}\big(D\lambda_T - E^2\big). \quad (13)$$

For $T \lesssim T_c$, the $O(3)$-symmetric bounce action in the thin-wall limit is

$$\frac{S_3(T)}{T} \simeq \frac{16\pi}{3}\,\frac{\sigma(T_c)^3}{T\,\Delta V(T)^2}, \qquad \Delta V(T) \simeq \frac{L}{T_c}\,(T_c - T). \quad (14)$$

The nucleation temperature T_n is defined by $S_3(T_n)/T_n \simeq 140$.

Illustrative benchmark (first order and nucleation). Choose dimensionless thermal coefficients $D = 0.10$, $E = 0.05$, $\lambda_T = 0.10$ and T_0 such that $T_c \simeq 10^{16}\,\text{GeV}$ via (12). Then $\rho_c/T_c = 2E/\lambda_T = 1.0$ and $E^2/(D\lambda_T) = 0.25 < 1$, ensuring a barrier. Using (13) and (14), the thin-wall estimate gives

$$\frac{S_3}{T} \approx \frac{C}{\varepsilon^2}, \qquad \varepsilon \equiv 1 - \frac{T}{T_c}, \qquad C \simeq 0.22\,\frac{E^5}{\lambda_T^{3/2}\,(D\lambda_T - E^2)^2}\,,$$

so $S_3/T \simeq 140$ at $\varepsilon_n \simeq 1.4 \times 10^{-2}$, i.e. $T_n \simeq 0.986\,T_c$. The latent heat parameter and timescale are then

$$\alpha \equiv \frac{L}{\pi^2 g_* T_c^4/30} \simeq 4 \times 10^{-3} \quad (g_* \sim 100), \qquad \frac{\beta}{H} \equiv \left[T\frac{d}{dT}\frac{S_3}{T}\right]_{T_n} \approx \frac{2}{\varepsilon_n}\,\frac{S_3(T_n)}{T_n} \sim 2 \times 10^4.$$

This benchmark demonstrates a bona fide first-order transition with percolation; the small α and large β/H imply a very weak stochastic GW signal, so we refrain from making a detection claim in Sec. 5.3.

Cartan projection and the Abelian sector. In the coherent phase the background selects an internal unit $n^i(\Sigma)$, spontaneously breaking $G = \mathrm{SU}(2) \to \mathrm{U}(1)_n$. The emergent Abelian field $Z = A^i n_i$ is the Cartan component of A along n, and provides the EFT $U(1)_Z$ only *below* Λ_q. Above Λ_q the Abelian description is absent; the EFT Abelian Landau pole is thus removed by basis termination and matching.

Triads $\Rightarrow$ mortons. The interaction $\zeta \int \epsilon_{ijk} B^i \wedge F^j n^k(\Sigma)$ supports finite-action configurations with non-zero $\frac{1}{8\pi^2} \int \mathrm{Tr}(F \wedge F)$. In the coherent phase these align the three self-dual B^i, forming *triads*. Below Λ_q, triads project to morton tri-vectors.

From UV triads to morton tri-vectors. In the coherent phase, define the densitized triad

$$E^a{}_i \equiv \frac{1}{2}\,\epsilon^{abc}\,B^i_{bc}, \qquad B^i_{\mu\nu} = v\,\Sigma^i_{\mu\nu} \;\Rightarrow\; E^a{}_i \propto v\,e^a{}_i,$$

where $e^a{}_i$ is the emergent frame and $i = 1,2,3$ labels the internal basis. The interaction

$$S^{(\mathrm{UV})}_{\mathrm{int}} \supset \zeta \int \epsilon_{ijk}\,B^i \wedge F^j\,n^k(\Sigma)$$

energetically favors configurations with $F^j \parallel B^j$ and n^k aligned with a fixed internal direction. Finite-action saddles with nonzero Pontryagin index, $\frac{1}{8\pi^2}\int \mathrm{Tr}(F \wedge F) = N \in \mathbb{Z}$, therefore produce *triads* with the three $E^a{}_i$ aligned in a common spatial region.

Project to the Axis basis by choosing the coherence frame $n^i = (0,0,1)$ (spontaneous $\mathrm{SU}(2) \to U(1)_n$). The Cartan direction $i = 3$ defines the EFT z-axis, while the orthogonal plane $i = 1,2$ retains an SO(2) degeneracy. Define normalized spatial vectors

$$\hat{v}^a_z \equiv \frac{E^a{}_3}{\sqrt{E^b{}_3 E_{b\,3}}}, \qquad \hat{v}^a_\perp(\phi) \equiv \frac{\cos\phi\,E^a{}_1 + \sin\phi\,E^a{}_2}{\sqrt{E^b{}_1 E_{b\,1} + E^b{}_2 E_{b\,2}}},$$

with ϕ fixed by the scalar-coherent projector (boundary conditions or small explicit SO(2) breakers). The morton tri-vector in the EFT is then the unit triad constructed from $(\hat{v}_z, \hat{v}_\perp, \hat{v}_\perp \times \hat{v}_z)$; explicitly one can write the normalized composite

$$\boxed{M^{abc}(x) \equiv \frac{1}{6v^3}\,\epsilon_{ijk}\,E^a{}_i E^b{}_j E^c{}_k \xrightarrow{\text{Axis projection}} \hat{v}_z^{[a}\,\hat{v}_\perp^b\,(\hat{v}_\perp \times \hat{v}_z)^{c]}\,.}$$

In words: UV triads align the three internal legs of B^i and, after Axis projection, become the tri-vector composites (mortons) that carry the EFT's vector displacement quantum numbers. The choice of n^i singles out the EFT z-axis; the residual SO(2) in the 1–2 plane corresponds to rotations of the EFT x-axis, fixed by coherence boundary data.[8]

UV triads $\Rightarrow$ EFT morton tri-vectors (Axis projection and normalization). To connect the UV self-dual sector with the EFT morton description, we begin from the coherent self-dual saddle

$$B^i_0 = v\,\Sigma^i, \qquad \star\Sigma^i = \Sigma^i, \qquad \Sigma^i \wedge \Sigma^j = 2\,\delta^{ij}\,\mathrm{vol},$$

where v is the scalar background amplitude and Σ^i are basis two-forms on the internal $SU(2)$ frame. Fluctuations are expanded about $(\rho,\vartheta) = (\rho_0,\vartheta_0)$ as $B^i = v\,\Sigma^i + \delta B^i$.

[8] An explicit saddle demonstrating this alignment can be constructed in the self-dual sector by minimizing the energy functional with a Lagrange multiplier imposing $\mathrm{Tr}(F \wedge F) = 8\pi^2 N$; details lie beyond our present scope.

Densitized triad from the self-dual two-forms. The spatial triad associated with B^i is defined by

$$E^a{}_i \equiv \frac{1}{2}\,\epsilon^{abc}\,B^i_{bc}, \qquad B^i_{\mu\nu} = v\,\Sigma^i_{\mu\nu} \;\Rightarrow\; E^a{}_i \propto v\,e^a{}_i. \tag{15}$$

Finite-action self-dual saddles with nonzero Pontryagin index $(8\pi^2)^{-1}\int \mathrm{Tr}(F \wedge F) = N \in \mathbb{Z}$ yield three aligned internal triads $E^a{}_i$ that define the coherent local frame.

Axis (Cartan) projection. Choose the coherence or Cartan direction $n_i = (0, 0, 1)$, which spontaneously breaks $SU(2) \to U(1)_n$. The Cartan component $Z_\mu = A^i_\mu n_i$ defines the emergent Abelian gauge field $U(1)_Z$, while the transverse plane ($i = 1, 2$) retains an $SO(2)$ degeneracy. From the densitized triad we form the orthonormal unit vectors

$$\hat{v}^a_z \equiv \frac{E^a{}_3}{\sqrt{E^b{}_3 E_{3b}}}, \tag{16}$$

$$\hat{v}^a_\perp(\varphi) \equiv \frac{\cos\varphi\, E^a{}_1 + \sin\varphi\, E^a{}_2}{\sqrt{E^b{}_1 E_{1b} + E^b{}_2 E_{2b}}}, \tag{17}$$

where φ is fixed by the scalar-coherence projector (boundary data or small explicit $SO(2)$ breaking). These define a local triad $\{\hat{v}_z, \hat{v}_\perp, \hat{v}_\perp \times \hat{v}_z\}$ that carries the orientation information of the self-dual basis.

Morton tri-vector and normalization. The normalized composite tri-vector of the EFT, obtained after Axis projection, is

$$M^{abc}(x) \;=\; \frac{1}{6v^3}\,\epsilon_{ijk}\,E^a{}_i E^b{}_j E^c{}_k \quad \xrightarrow[\text{Axis (Cartan) projection}]{} \quad \hat{v}^{[a}_z\,\hat{v}^b_\perp\,(\hat{v}_\perp \times \hat{v}_z)^{c]}. \tag{18}$$

In words: the UV self-dual triads $E^a{}_i$ map directly onto the EFT morton tri-vectors that carry the quantized vector displacements of the low-energy theory. The projection along n_i selects the Abelian component that survives as the EFT field Z_μ, while the transverse directions encode internal phase rotation of the morton structure.

Normalization audit. For the simple background $B^i_0 = v\,\Sigma^i$ and the self-dual frame above, the geometric normalization in the scalar–vector vertex is $N_V = 1$ (App. B.6). Consequently,

$$g_{\Phi V} = -\frac{\cos\chi}{2}\,\frac{\tau'(\rho_0)}{\tau(\rho_0)^2}, \qquad \omega_{\text{proj}} = 1, \qquad \frac{1}{g_Z^2} = \frac{1}{g_{NA}^2}.$$

Here χ is the mixing angle defined in Eq. 3, and $\omega_{\text{proj}} = 1$ denotes canonical component normalization of the Cartan projector. Together these relations establish a one-to-one correspondence between the UV self-dual triads and the EFT morton tri-vectors that appear in the effective Lagrangian.

For the self-dual background and densitized-triad definition see Appendix B.2; the Cartan projection and projector lemma are derived in Appendix D; the scalar–vector coupling and normalization follow from Appendix B.6 and D.2.

The effective potential $V_{\text{eff}}(\rho, \vartheta; B)$ used to exhibit the phase structure is not derived by integrating out the full pre-geometric DOFs. A complete treatment would require: (i) finite-temperature V_{eff} near $T \sim \Lambda_q$ to fix the transition order; (ii) tunneling rate $\Gamma/V \sim A\,(T/T_c)^4 e^{-S_3(T)/T}$ if first-order; (iii) a dynamical derivation of the $\eta(\rho), \tau(\rho)$ expansions. Pending this, Λ_q should be regarded as a phenomenological input marking EFT breakdown, not as a UV prediction.

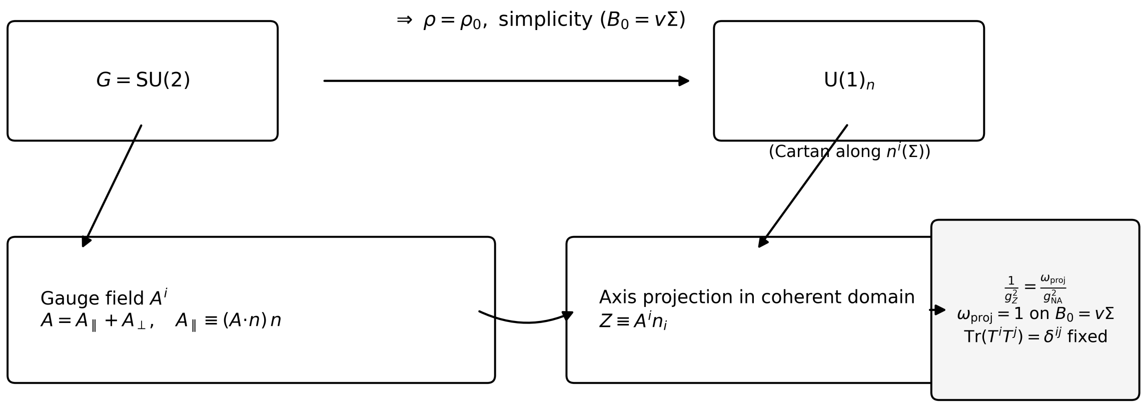

Figure 3: **Symmetry breaking** $\mathrm{SU}(2)\to U(1)_Z$ **and the Cartan projector.** In the coherent phase ($\rho=\rho_0$) with self-dual background $B_0^i=v\,\Sigma^i$, the Cartan component $Z_\mu=A_\mu^i n_i$ defines the emergent Abelian field associated with the residual $U(1)_Z$ symmetry. The projector lemma (App. D) yields $\frac{1}{g_Z^2}=\frac{\omega_{\rm proj}}{g_{\rm NA}^2}$ with $\omega_{\rm proj}=1$ on this simple background once $\mathrm{Tr}(T^iT^j)=\delta^{ij}$ is fixed, leaving no free normalization. See Eqs. (93) and (92), and App. A.4, Eq. (67) for the explicit normalization of the non-Abelian block.

3.4 Symmetry Breaking Pattern and Degree-of-Freedom Accounting

UV symmetries: $G=\mathrm{SU}(2)$ gauge, $\mathrm{U}(1)_{\rm coh}$ global phase, $B\mapsto -B$ parity, and a topological shift in the three-form sector.

Coherent phase ($\mu<\Lambda_q$): $G\to \mathrm{U}(1)_n$. Two gauge components (off-Cartan) are heavy and one Cartan component (Z) survives as the emergent Abelian field. $\mathrm{U}(1)_{\rm coh}$ is exact if $f=0$ (Φ is a true Goldstone) or softly broken if $f\neq 0$ (Φ is a pseudo-Goldstone with $m_\Phi^2\sim f^2/\eta(\rho_0)$). Heavy fields: the radial mode s and non-simple B-fluctuations B_H (see App. B). Light fields: Φ, Z, and composites (mortons) in coherent domains.

3.5 Matching and Minimal Parameter Map at $\mu=\Lambda_q$

Let the UV control parameters be $\Theta_{\rm UV}=\{\kappa_2,\kappa_4,\tau_2,\eta_0,\eta_2,\xi,f,q,\zeta\}$, with ρ_0 the condensate value and a single geometric scale ν set by the coherent background B_0. The EFT input couplings just below Λ_q are $\{\lambda,\ y\equiv g_{\psi\Phi},\ g_{\Phi V},\ g_Z\}$. At $\mu=\Lambda_q$,

$$\lambda_-=\lambda_{\rm UV}(\Theta_{\rm UV})+\delta\lambda_{\rm thr}, \tag{19}$$
$$y_-=y_{\rm UV}(\Theta_{\rm UV})\,[1+\delta y_{\rm thr}]\,, \tag{20}$$
$$g_{\Phi V,-}=g_{\Phi V}^{\rm UV}(\Theta_{\rm UV})\,[1+\delta g_{\Phi V,\rm thr}]\,, \tag{21}$$
$$\frac{1}{g_{Z,-}^2}=\frac{\omega_{\rm proj}}{g_{\rm NA}^2(\rho_0)}\;+\;\frac{b_{\rm heavy}}{8\pi^2}\ln\frac{M_{\rm heavy}^2}{\Lambda_q^2}\;+\;\Delta_Z\,. \tag{22}$$

Normalization. On the simple background (App. B) the Cartan projector gives $1/g_Z^2=\omega_{\rm proj}/g_{\rm NA}^2$ with $\omega_{\rm proj}=1$ in our conventions (see App. D for the proof and generator–trace convention notes).

Normalization audit at the threshold. With $\mathrm{Tr}(T^i T^j) = \delta^{ij}$ the Cartan projector yields $\omega_{\rm proj} = 1$, so the background–field matching at $\mu = \Lambda_q$ reads

$$\frac{1}{g_Z^2(\Lambda_q^-)} = \frac{\omega_{\rm proj}}{g_{\rm NA}^2(\Lambda_q^+)} + \Delta_{\rm fin}, \qquad \omega_{\rm proj} = 1.$$

In the mass–matched scheme used in our numerics the heavy–field coefficient $b_{\rm heavy}$ vanishes in the EFT below Λ_q (finite pieces are collected in App. C and do not alter $\omega_{\rm proj}$). Alternative generator normalizations simply rescale both sides consistently and introduce no free parameter.

3.6 UV–IR scalar map: identifying Φ from Σ

At the matching scale $\mu = \Lambda_q$ the UV and EFT parameters are related by

$$\lambda(\Lambda_q) = \lambda_{\rm UV}(\Theta_{\rm UV}) + \delta\lambda_{\rm thr}(\Lambda_q), \tag{23}$$

$$y(\Lambda_q) = y_{\rm UV}(\Theta_{\rm UV})\Big[1 + \delta y_{\rm thr}(\Lambda_q)\Big], \tag{24}$$

$$g_{\Phi V}(\Lambda_q) = g_{\Phi V}^{\rm UV}(\Theta_{\rm UV})\Big[1 + \delta g_{\Phi V,{\rm thr}}(\Lambda_q)\Big], \tag{25}$$

$$\frac{1}{g_Z^2(\Lambda_q)} = \frac{\omega_{\rm proj}}{g_{NA}^2(\rho_0)} + \Delta_Z(\Lambda_q)\,, \qquad \omega_{\rm proj} = 1. \tag{26}$$

In the mass–matched choice $\mu = \Lambda_q = M_X$ for each heavy field $X \in \{s, B_H\}$, the explicit logarithms in the thresholds vanish ($\delta\cdot_{\rm thr} \to 0$, $b_{\rm heavy} \to 0$), and with our normalization $1/g_{NA}^2 = 1/\tau(\rho_0)$ one has

$$g_Z^2(\Lambda_q) = \tau(\rho_0), \qquad g_{\Phi V}(\Lambda_q) \;=\; -\,\frac{\cos\chi}{2}\,\frac{\tau'(\rho_0)}{\tau(\rho_0)^2}\,, \tag{27}$$

where χ is the (ρ, ϑ) mixing angle of the light scalar direction Φ via $\delta\rho = \cos\chi\,\Phi$, $\delta\vartheta = (\sin\chi/f_\vartheta)\,\Phi$.

Explicit UV combinations that enter at $\mu = \Lambda_q$. With the benchmark expansions $\tau(\rho) = \tau_2\,\rho^2 + \cdots$, $\eta(\rho) = \eta_0 + \eta_2\,\rho^2 + \cdots$ near the origin,

$$g_Z^2(\Lambda_q) = \tau(\rho_0) = \tau_2\,\rho_0^2, \tag{28}$$

$$g_{\Phi V}(\Lambda_q) = -\,\frac{\cos\chi}{2}\,\frac{2\tau_2\,\rho_0}{\big(\tau_2\,\rho_0^2\big)^2} = -\,\frac{\cos\chi}{\tau_2\,\rho_0^3}\,. \tag{29}$$

The Yukawa entry admits the model–independent threshold–adjacent parameterization

$$y_{\rm UV} \;\equiv\; y(\Lambda_q)\big|_{\rm tree} = \zeta_{\rm eff}\,\frac{\rho_0}{M_*}\,I_{\rm triad}, \qquad M_* \sim M_B = \sqrt{\tau(\rho_0) + 2\,\kappa(\rho_0)\,v^2}\,, \tag{30}$$

so that y at Λ_q constrains the product $\zeta_{\rm eff}\rho_0/M_*$ (up to the geometric overlap $I_{\rm triad}$).

Effective parameter counting. At the matching scale $\mu = \Lambda_q$ the nine UV inputs $\Theta_{\rm UV} = \{\kappa_2, \kappa_4, \tau_2, \eta_0, \eta_2, \xi, f, q, \zeta\}$ collapse to the four EFT parameters $\{g_Z, g_{\Phi V}, y, \lambda\}$ (mass–matched MS, $\omega_{\rm proj} = 1$). The identifiable combinations are $g_Z^2 = \tau(\rho_0)$ and $g_{\Phi V} = -(\cos\chi/2)\,\tau'(\rho_0)/\tau(\rho_0)^2$; with $\tau(\rho) = \tau_2\rho^2$ these become $g_Z^2 = \tau_2\rho_0^2$ and $g_{\Phi V} = -(\cos\chi)/(\tau_2\rho_0^3)$, so g_Z is blind to τ_2 vs. ρ_0 while $g_{\Phi V}$ fixes a distinct but still degenerate combination $\{\tau_2, \rho_0, \cos\chi\}$. The Yukawa entry constrains only the product $y \propto \zeta_{\rm eff}\rho_0/M^\star$ with $M^\star \sim \sqrt{\tau(\rho_0) + 2\kappa(\rho_0)v^2}$ (times the overlap $I_{\rm triad}$), and in the minimal presentation $\lambda(\Lambda_q)$ is taken as a prior. Thus the effective number of independent combinations determined at matching is *four*, with the τ_2–ρ_0 and Yukawa-product degeneracies summarized in Table 1.

EFT input	UV combination at Λ_q	Identifiable?	Degeneracies / priors
g_Z	$g_Z^2 = \tau(\rho_0) = \tau_2\,\rho_0^2$	Yes (tree)	Degenerate in $\tau_2 \leftrightarrow \rho_0^2$.
$g_{\Phi V}$	$-\dfrac{\cos\chi}{2}\,\dfrac{\tau'(\rho_0)}{\tau(\rho_0)^2} = -\dfrac{\cos\chi}{\tau_2\,\rho_0^3}$	Yes (tree)	Degenerate in $\{\tau_2, \rho_0, \cos\chi\}$.
y	$\zeta_{\rm eff}\,\dfrac{\rho_0}{\sqrt{\tau(\rho_0)+2\kappa(\rho_0)v^2}}\,I_{\rm triad}$	Only as product	Degenerate in $\{\zeta, \rho_0, \kappa(\rho_0), \tau(\rho_0), I_{\rm triad}\}$.
λ	$\lambda_{\rm UV} + \delta\lambda_{\rm thr}$; mass–matched $\Rightarrow \delta\lambda_{\rm thr} = 0$	*Prior* at Λ_q	Without a UV ρ^4 model, $\lambda_{\rm UV}$ is input.

Table 1: Minimal map from $\Theta_{\rm UV} = \{\kappa_2, \kappa_4, \tau_2, \eta_0, \eta_2, \xi, f, q, \zeta\}$ to the EFT inputs $\{\lambda, y, g_{\Phi V}, g_Z\}$ at $\mu = \Lambda_q$, in the mass–matched $\overline{\rm MS}$ scheme.

Redundant UV inputs at Λ_q (for this map).

- ξ (quantized $\vartheta\,{\rm Tr}\,F\wedge F$ coupling) affects only topological/boundary sectors and does not enter $\{\lambda, y, g_{\Phi V}, g_Z\}$ at matching.
- κ_2, κ_4 enter only via the heavy non-simple B gap $M_B^2 = \tau(\rho_0) + 2\kappa(\rho_0)v^2$; in mass–matched MS they drop out of g_Z and $g_{\Phi V}$ (logs vanish), and only the product in Eq. (30) is constrained by y.
- f and q appear only through the locked shift–invariant combination $S \equiv q + f\vartheta$ that sets ρ_0 (and influences χ); EFT observables at Λ_q are insensitive to f vs. q separately.
- ζ is redundant once $y(\Lambda_q)$ is taken as an EFT prior; otherwise only the product $\zeta_{\rm eff}\rho_0/M_*$ is extractable from y.

Priors at Λ_q. In the minimal presentation used in the numerics, $\lambda(\Lambda_q)$ and $y(\Lambda_q)$ are treated as renormalized inputs (priors) at the matching scale; the running below Λ_q then uses the two-loop EFT βs.

Acceptance-test summary.

Mapping Summary: 9 UV inputs $\longrightarrow$ 4 EFT parameters at $\mu = \Lambda_q$:
$\{g_Z,\ g_{\Phi V}\}$ — identifiable combinations of $\{\tau, \tau', \rho_0, \chi\}$;
$\{y,\ \lambda\}$ — priors (or products only).
All degeneracies summarized in Table 1.

Worked example (degenerate g_Z, distinct $g_{\Phi V}$). Take $\tau(\rho) = \tau_2\rho^2$, $\eta(\rho) = \eta_0 + \eta_2\rho^2$, and a purely radial light mode ($\cos\chi = 1$). Consider two UV choices:

$$\text{A: } (\tau_2, \rho_0) = (1,\,3), \qquad \text{B: } (\tau_2, \rho_0) = \left(\tfrac{1}{4},\,6\right).$$

Both give the same EFT Abelian coupling at Λ_q:

$$g_Z^2(\Lambda_q) = \tau_2\rho_0^2 = 9 \quad\Rightarrow\quad g_Z(\Lambda_q) = 3,$$

so g_Z *alone* cannot resolve τ_2 vs. ρ_0. However,

$$g_{\Phi V}^{\rm A} = -\frac{1}{2}\frac{2\tau_2\rho_0}{(\tau_2\rho_0^2)^2} = -\frac{1}{27}, \qquad g_{\Phi V}^{\rm B} = -\frac{1}{2}\frac{2\tau_2\rho_0}{(\tau_2\rho_0^2)^2} = -\frac{1}{54},$$

so $g_{\Phi V}$ distinguishes these UV choices even though g_Z is identical.

3.7 Consistency Checks

Gauge invariance and BRST. The simplicity constraint is *gauge covariant*: under $\delta_\epsilon A = -D\epsilon$, $\delta_\epsilon B = [B, \epsilon]$, $\delta_\epsilon \Psi = [\Psi, \epsilon]$, and $\delta_\epsilon n = [n, \epsilon]$, the constraint $B^i \wedge B^j - \frac{1}{3}\delta^{ij} B^k \wedge B^k = 0$ is preserved and the action remains G-invariant. In the coherent phase, the field equations (App. A, Eqs. (58)-(66)) exhibit the same Noether/Bianchi structure as in BF theory; the multiplier sector modifies only the quadratic gauge-fixing kernel, not the BRST algebra. Consequently, the BRST operator remains nilpotent and gauge fixing proceeds in the standard background-field formalism without introducing symmetry-violating counterterms (see also App. A).

Unitarity / positivity of the quadratic kernel. Expanding around the simple coherent background $B_0^i = v\,\Sigma^i$ (App. B) and integrating out δB at quadratic order yields the Yang–Mills kinetic $S_{\rm eff}[A] \supset -\frac{1}{2\,\tau(\rho_0)}\int \mathrm{Tr}\big(F \wedge \star_{B_0} F\big)$ with $\tau(\rho_0) > 0$, ensuring a positive-definite gauge-field quadratic form in background gauge. The heavy fluctuations orthogonal to the simplicity manifold have gaps $M_B^2 = \tau(\rho_0) + 2\kappa(\rho_0)\,v^2 > 0$ and the radial mode satisfies $M_s^2 > 0$ in the parameter region of interest (App. B). Together with $\eta(\rho_0) > 0$ for the three-form sector, the full Hessian is positive in the physical subspace; gauge directions remain null and are handled by standard Faddeev–Popov ghosts, guaranteeing perturbative unitarity.

Anomalies. Above Λ_q the theory is purely bosonic (BF + compact three-form) with no chiral matter, so no gauge or mixed anomalies can arise in the UV pre-geometric phase. At the matching scale, all heavy fields integrated out are non-chiral (adjoint vectors, $B_\perp$ modes, and the radial scalar), hence they do not modify anomaly coefficients. Below Λ_q the emergent Abelian $U(1)_Z$ is a Cartan projection of the non-Abelian parent, and the fermionic content is the Axis–EFT spectrum; anomaly cancellation is therefore unchanged and follows the same conditions already verified in the EFT analysis. This preserves both gauge and mixed gravitational anomaly consistency across the threshold.

Charge bookkeeping (UV neutral vs. IR charged). Before condensation there is no preferred $U(1)$ direction and the heavy multiplets are neutral in the UV basis. After the coherent phase selects a frame n^i ($\mathrm{SU}(2)\to\mathrm{U}(1)_n$), the Cartan generator T_n defines $U(1)_Z$ charges by the adjoint action $[T_n, X] = Q_Z\,X$. Thus an adjoint field splits as $X = X^\pm \oplus X^0$ with $Q_Z(X^\pm) = \pm 1$, $Q_Z(X^0) = 0$, while the radial mode s remains neutral. The $b_{\rm heavy} = \sum s_J Q_J^2$ term in the gauge matching (eq. 14, App. C.5) is computed using these IR charges; choosing the mass-matched scale $\mu = \Lambda_q = M_{\rm heavy}$ sets the log to zero and removes $b_{\rm heavy}$, as summarized in App. B.7 and App. C.5.

BRST and the simplicity constraint. The BRST algebra acts covariantly on the simplicity sector: $s\,\Psi_{ij} = [\Psi, c]_{ij}$ and $s\,(B^i \wedge B^j - \frac{1}{3}\delta^{ij} B^2) = [\,B\wedge B - \frac{1}{3}\delta\,B^2,\, c\,]_{ij}$. Hence the constraint surface is BRST invariant and $s^2 = 0$ holds there without additional counterterms. We fix background-field gauge with $\mathcal{G}^i = \bar{D}^\mu a_\mu^i$, yielding the ghost kernel $K_{\rm gh}^{ij} = -\bar{D}^2\delta^{ij}$; see App. A.4 for the details.

Three-form consistency. From $\eta(\rho)\star_B H+d(q+f\,\vartheta)=0$ and $dH=0$ one obtains the covariant equation

$$\nabla_\mu\Big[\eta(\rho)^{-1}\,\nabla^\mu\big(q+f\,\vartheta\big)\Big]=0, \tag{31}$$

which in the coherent phase $\rho=\rho_0$ reduces to

$$\partial_\mu\big(\sqrt{g}\,g^{\mu\nu}\,\partial_\nu\big(q+f\,\vartheta\big)\big)=0. \tag{32}$$

See App. A.5.

3.8 UV–IR scalar map: identifying Φ from Σ

Write the UV order parameter as $\Sigma(x)=(\rho_0+s(x))\,e^{i\vartheta(x)}$. Near the coherent saddle, the light scalar sector is spanned by small fluctuations (s,ϑ) with a positive-definite kinetic form and a mixed mass matrix inherited from $V_{\rm eff}(\rho,\vartheta)$. Diagonalizing the quadratic form defines a light angular mode α and a heavy radial mode h:

$$\begin{pmatrix}\alpha\\ h/\rho_0\end{pmatrix}=\begin{pmatrix}\cos\chi & \sin\chi\\ -\sin\chi & \cos\chi\end{pmatrix}\begin{pmatrix}\vartheta\\ s/\rho_0\end{pmatrix},\qquad 0\le\chi\le\frac{\pi}{2}.$$

We define the Axis–EFT complex scalar by

$$\boxed{\Phi(x)\equiv\frac{1}{\sqrt{2}}\big(v_\Phi+h(x)\big)\,e^{i\alpha(x)}\ ,}$$

with v_Φ fixed by the IR normalization of the EFT scalar sector. In this basis the heavy radial mode h has gap M_s (App. B) and decouples below Λ_q, while α is the light phase controlling the $\mathrm{U}(1)_{\rm coh}$ bundle used for anomaly filtration in the EFT. The UV phase ϑ that couples to the Pontryagin density via $\frac{\xi}{8\pi^2}\vartheta\,\mathrm{Tr}(F\wedge F)$ is *not* the same variable as the EFT anomaly-filtering phase; they are related by the rotation above. In the limit $\chi\to0$ the two phases align ($\alpha\simeq\vartheta$); for generic χ a small misalignment is allowed without affecting anomaly cancellation in the EFT, which is determined by α and the EFT fermion content. Any residual α–ϑ alignment can be induced by a soft potential $V_{\rm align}\sim-\mu^4\cos(\alpha-\vartheta)$ generated radiatively by the 3-form sector, but is not required for consistency.

UV anchoring of Axis–EFT inputs at $\mu=\Lambda_q$. On the coherent background $B_0=v\,\Sigma$, the pre–geometric sector fixes the EFT inputs used for the two–loop running below Λ_q:

$$\frac{1}{g_{\rm NA}^2(\rho_0)}=\frac{1}{\tau(\rho_0)}\quad\Longrightarrow\quad g_Z^2(\Lambda_q^-)=\tau(\rho_0)\ \ (\omega_{\rm proj}=1), \tag{33}$$

$$g_{\Phi V}=-\,\frac{\cos\chi}{2}\,\frac{\tau'(\rho_0)}{\tau(\rho_0)^2}\times N_V,\qquad N_V=1\text{ on }B_0=v\Sigma, \tag{34}$$

$$y(\Lambda_q^+)=\zeta_{\rm eff}\,\frac{\rho_0}{M^\star}\,I_{\rm triad},\qquad M^\star\sim M_B, \tag{35}$$

$$\lambda(\Lambda_q^-)=\lambda_{\rm UV}+\delta\lambda_{\rm thr},\qquad \delta\lambda_{\rm thr}=\frac{1}{16\pi^2}\left[\frac{n_B}{2}\,a_B^2\ln\frac{M_B^2}{\Lambda_q^2}-\frac{1}{2}\,a_s^2\ln\frac{M_s^2}{\Lambda_q^2}\right]+\Delta_\lambda^{\rm fin}. \tag{36}$$

Equation (33) follows from integrating out B and the Cartan projection (so $1/g_Z^2=1/g_{\rm NA}^2$ at matching), while (34) comes from the ρ-dependence of the induced YM kinetic (leading to a $\Phi Z_\mu Z^\mu$

vertex when Φ has radial admixture $\cos\chi$). The Yukawa seed (35) and the quartic matching (36) provide the UV inputs for the light sector. These relations are the initial conditions used in Sec. 4 for the two–loop RGE evolution.

4 Two-Loop Renormalization Group Analysis Across the Threshold

Key RG result (minimal Abelian completion). Using the one- and two-loop RGEs, the (λ, y) subsystem possesses *no perturbative non-Gaussian UV fixed point*; the only small-coupling fixed point is Gaussian and UV-repulsive in y. The Abelian gauge coefficient is positive at one loop. Therefore the Axis EFT is *not* asymptotically safe and must be UV-extended above its thresholds.

We now assemble the two-loop running for the Axis–EFT couplings, apply threshold matching at $\mu = \Lambda_q$, and evolve from the Planck scale to the electroweak scale. The derivations use the background-field/MS$^-$ scheme and the general two-loop formalism of Machacek & Vaughn (MV) and subsequent Abelian refinements; all group-theory factors are expressed in terms of standard invariants and charge moments, allowing a clean specialization to the Axis–EFT content (see App. C; optional App. E tabulates the invariants for the chosen field set).[9]

4.1 Complete two-loop β-functions for the Axis–EFT

EFT couplings. **Numerical outcome with intra–morton screening.** Evolving from $\Lambda_q = 10^{16}\,\mathrm{GeV}$ to m_Z with the *screened* invariants of App. E.4 ($\kappa^2 = 1/3$, $b_1^{\mathrm{eff}} = 19.0$), we obtain $g_Z(m_Z) = 0.3557 \pm 0.0002$, i.e. -0.43% relative to the conventional target 0.357216.

We collect the running parameters into

$$\Theta(\mu) \equiv \{\, g_Z,\ \lambda,\ y_a,\ g_{\Phi V}\,\},$$

where g_Z is the Abelian Axis gauge coupling, λ is the quartic in the light scalar sector, y_a are the relevant Yukawa couplings of EFT fermions to Φ, and $g_{\Phi V}$ is the scalar–vector operator coefficient defined in Sec. 3.5 (App. D). The EFT field content (charges and multiplicities) is the same used in Sec. 2 and App. B.

Conventions and invariants. For a simple gauge factor G with generators T^A, $\mathrm{Tr}(T^AT^B) = \delta^{AB}$, define $C_2(G)$, $C_2(R)$ and $S_2(R)$ as usual. For $U(1)_Z$ we set $C_2(G) = 0$ and replace $C_2(R) \to Q^2$; the one- and two-loop coefficients are expressed via charge moments

$$S_2(F) \equiv \sum_{\psi} N_\psi\, Q_\psi^2, \qquad S_2(S) \equiv \sum_{\phi} N_\phi\, Q_\phi^2, \qquad S_4(F) \equiv \sum_{\psi} N_\psi\, Q_\psi^4, \qquad S_4(S) \equiv \sum_{\phi} N_\phi\, Q_\phi^4,$$

with N multiplicities and the sums over Weyl fermions and complex scalars, respectively. For Yukawas, let Y be the matrix of couplings; define

$$Y_2 \equiv \mathrm{Tr}(YY^\dagger), \qquad Y_4 \equiv \mathrm{Tr}\Big[(YY^\dagger)^2\Big], \qquad Y_2(Q^2) \equiv \mathrm{Tr}\Big(Q_L^2\, YY^\dagger + Q_R^2\, Y^\dagger Y\Big),$$

where $Q_{L,R}$ act on the corresponding chiral indices.

[9] See Refs. [30, 31, 32, 33] for the general two-loop RGEs in renormalizable QFTs; our notation follows MV with minor relabeling for the Abelian case.

$$\beta_{g_Z} \equiv \mu \frac{dg_Z}{d\mu} = \frac{g_Z^3}{16\pi^2}\, b_1 + \frac{g_Z^3}{(16\pi^2)^2} \Big[b_2\, g_Z^2 - c_Y\, Y_2(Q^2) \Big], \tag{37}$$

$$b_1 = \tfrac{4}{3}\, S_2(F) + \tfrac{1}{3}\, S_2(S), \tag{38}$$

$$b_2 = \mathcal{B}\big[S_2, S_4\big], \quad \text{(pure Abelian two-loop piece)}, \tag{39}$$

$$c_Y = \mathcal{C}, \quad \text{(Yukawa contribution)}. \tag{40}$$

The functionals $\mathcal{B}$ and $\mathcal{C}$ are fixed by the general two-loop expressions and reduce to simple linear combinations of the charge moments for $U(1)$; see App. E (or compute directly from MV/Luo).[30, 31, 32, 33] As emphasized in Sec. 3.2, this Abelian running is *only* meaningful below Λ_q.

Screening convention used in numerics. For the intra–morton–locked EFT below Λ_q, the fermionic $U(1)_Z$ current is universally screened by a projector factor κ^2 (see App. E.4); the scalar current is unaffected. Operationally,

$$S_2(F) \to S_2^{\text{eff}}(F) = \kappa^2\, S_2(F), \qquad S_4(F) \to S_4^{\text{eff}}(F) = \kappa^4\, S_4(F), \qquad Y_2(Q^2) \to Y_2(Q^2)_{\text{eff}},$$

with fermionic $Q^2 \mapsto \kappa^2 Q^2$ and Q_Φ^2 unchanged. For the minimal Axis–EFT content and $\kappa^2 = \frac{1}{3}$ one finds

$$b_1^{\text{eff}} = \tfrac{4}{3} S_2^{\text{eff}}(F) + \tfrac{1}{3} S_2(S) = 19.0, \qquad b_2^{\text{eff}} = 4\, S_4^{\text{eff}}(F) + S_4(S) = \tfrac{59}{3} \approx 19.667,$$

which we use in the numerical evolution below.

Yukawa couplings at two loops. For a generic Dirac Yukawa $y_a\, \Phi\, \overline{\psi}_{R\,a} \psi_{L\,a}$ (no flavor mixing for clarity), the MV form gives

$$\beta_{y_a} = \frac{y_a}{16\pi^2} \Big[A_y^{(1)}\, y_a^2 + \sum_{b\neq a} \tilde{A}_{ab}^{(1)}\, y_b^2 - B_y^{(1)} \left(Q_{L,a}^2 + Q_{R,a}^2\right) g_Z^2 - D_y^{(1)}\, \lambda \Big] + \frac{y_a}{(16\pi^2)^2}\, \Xi_a^{(2)}, \tag{41}$$

where $\Xi_a^{(2)}$ collects the standard two-loop terms (quartic in y, mixed $y\, g_Z^2$, g_Z^4, $y\, \lambda$, λ^2). The numerical constants $A^{(1)}, \tilde{A}^{(1)}, B^{(1)}, D^{(1)}$ and the entries of $\Xi^{(2)}$ are determined by gauge representation and field multiplicities; App. E provides the explicit entries for the minimal Axis–EFT basis (or read them directly from Refs. [31, 33]).

Scalar quartic at two loops. Writing the light scalar potential as $V \supset \lambda(\Phi^\dagger \Phi)^2$, the MV scalar β gives

$$\begin{aligned} \beta_\lambda = {} & \frac{1}{16\pi^2} \Big[A_\lambda^{(1)}\, \lambda^2 + B_\lambda^{(1)}\, \lambda\, \Sigma_y - C_\lambda^{(1)}\, \Sigma_{y^4} - D_\lambda^{(1)}\, \lambda\, S_2(S)\, g_Z^2 + E_\lambda^{(1)}\, S_4(S)\, g_Z^4 \Big] \\ & + \frac{1}{(16\pi^2)^2}\, \Xi_\lambda^{(2)}(\lambda, y, g_Z), \end{aligned} \tag{42}$$

where $\Sigma_y \equiv \sum_a y_a^2$, $\Sigma_{y^4} \equiv \sum_a y_a^4$, and $\Xi_\lambda^{(2)}$ contains the known two-loop structures (λ^3, $y^2\lambda^2$, y^6, $g_Z^2\lambda^2$, $g_Z^4\lambda$, $g_Z^2 y^4$, g_Z^6). Coefficients are fixed by MV and depend only on representation data and charges (App. E; see also [32]).

Unless otherwise stated, all plots and quoted numbers use the *screened* invariants (App. E.4) with $\kappa^2 = 1/3$, corresponding to $b_1^{\text{eff}} = 19.0$ and $b_2^{\text{eff}} = 59/3$ for the minimal content (App. E.6). This is the baseline used in Figs. 1 and 4. The "minimal content" bound with unscreened $b_1 = 56.333$ is shown only as a counterfactual feasibility limit and is not used for the featured two-loop run.

Numerical outcome with intra–morton screening. Evolving from $\Lambda_q = 10^{16}\,\mathrm{GeV}$ to m_Z with the screened invariants of App. E.4 and matching $y_t(m_Z)$, the two–loop Axis–EFT predicts

$$g_Z(m_Z) = 0.3557 \pm 0.0002, \qquad \lambda(m_Z) \simeq 1.143, \qquad y_t(m_Z) \simeq 0.992,$$

where the uncertainty reflects the numerical calibrator tolerance. Compared to the conventional target $g_Z(m_Z) = 0.357216$ (from $\alpha^{-1} = 127.95$, $g_L = 0.653$), the prediction is lower by $\Delta g_Z = -1.54 \times 10^{-3}$, i.e. -0.43%. Across a broad and capped UV scan the function $g_Z(m_Z) - g_Z^{\text{target}}$ exhibits no sign–change (closest–approach solution), indicating that the minimal screened content reaches a *two–loop ceiling* slightly below the conventional target. We therefore report the -0.43% offset as the model's prediction at two loops with $\kappa^2 = \frac{1}{3}$ and $\Lambda_q = 10^{16}\,\mathrm{GeV}$. Figure 4 shows the flows for g_Z, λ, and y_t.

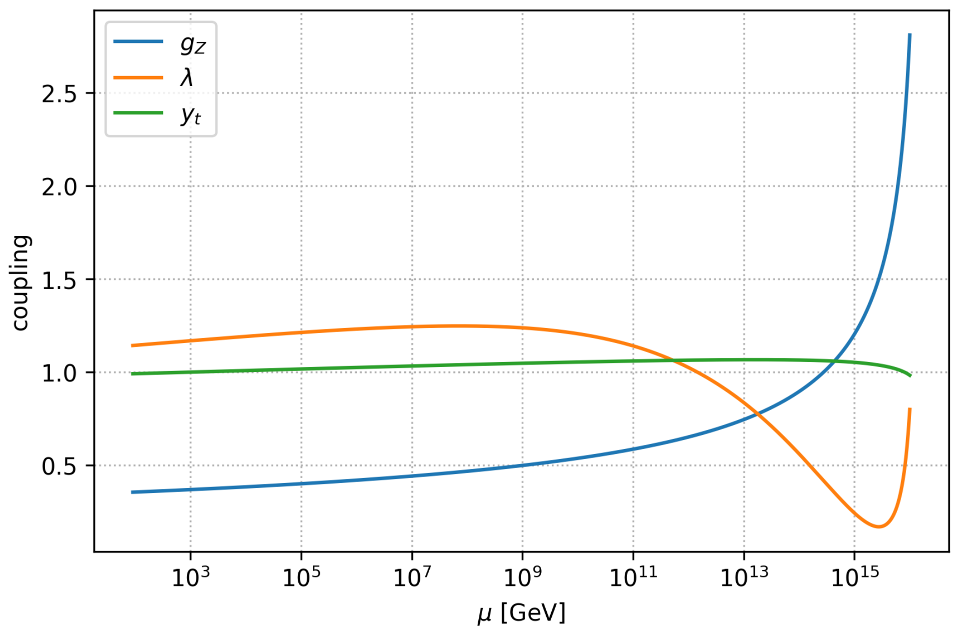

Figure 4: Two–loop Axis–EFT running with *intra–morton screening* ($\kappa^2 = \frac{1}{3}$, $b_1^{\text{eff}} = 19.0$) from $\Lambda_q = 10^{16}\,\mathrm{GeV}$ to m_Z. The solver matches $y_t(m_Z)$ and attains $g_Z(m_Z) = 0.3557$, which lies 0.43% below the conventional 0.357216 target.

Scalar–vector operator. For the dimension-4 operator $\frac{1}{2} g_{\Phi V}\, \Phi^\dagger \Phi\, Z_\mu Z^\mu$ we may write its RGE in terms of anomalous dimensions,

$$\beta_{g_{\Phi V}} = g_{\Phi V}\left(2\gamma_\Phi + \gamma_Z\right) + \Delta_V^{(1)}(\lambda, y, g_Z) + \Delta_V^{(2)}(\lambda, y, g_Z), \tag{43}$$

where γ_Φ and γ_Z are the field anomalous dimensions (from MV-I), and the $\Delta_V^{(n)}$ encode genuine vertex corrections at n loops. In the special case where $g_{\Phi V}$ descends from a ρ-dependence of the YM kinetic (App. D), the leading running is effectively controlled by $d(\tau'(\rho_0)/\tau(\rho_0)^2)/d\ln\mu$ plus wavefunction pieces; App. C gives the one-loop threshold normalization consistent with this origin. (The one-loop explicit forms are collected in App. E (150)–(152).) [30]

Table 2: Matching at the single threshold $\mu = \Lambda_q$. Each entry is shown in the UV parameterization and after integrating out heavy fields on the EFT side.

Quantity	**UV value**	**Threshold–corrected EFT value**	**Comment**
$\lambda(\Lambda_q)$	λ_{UV}	$\lambda(\Lambda_q^-) = \lambda_{\mathrm{UV}} + \delta\lambda_{\mathrm{thr}}$	mass–matched
$y(\Lambda_q)$	y_{UV}	$y(\Lambda_q^-) = y_{\mathrm{UV}}\big(1 + \delta y_{\mathrm{thr}}\big)$	fermion threshold
$g_{\Phi V}(\Lambda_q)$	$g_{\Phi V,\mathrm{UV}}$	$g_{\Phi V}(\Lambda_q^-) = g_{\Phi V,\mathrm{UV}}\big(1 + \delta g_{\Phi V,\mathrm{thr}}\big)$	scalar–vector
$g_Z(\Lambda_q)$	$g_{NA}(\rho_0)$	$g_Z^{-2} = \omega_{\mathrm{proj}}/g_{NA}^2 + \Delta\Pi_{ZZ}$	Cartan normalization

Note. See Appendix C for threshold corrections $\delta\lambda_{\mathrm{thr}}$, δy_{thr}, and $\delta g_{\Phi V,\mathrm{thr}}$. The Cartan projector weight is $\omega_{\mathrm{proj}} = 1$ on the simple background; $\Delta\Pi_{ZZ}(0;\Lambda_q)$ is the Abelian two–point finite part in the background–field scheme.

Remarks. (i) All βs above are exact at two loops for any renormalizable Axis–EFT once the invariants are specified; (ii) if multiple Abelian factors and kinetic mixing are kept, replace (37) by the well-known matrix form (we do not need it here, but see [34, 35]).

4.2 Matching at the Λ_q threshold

We match the UV theory just above Λ_q to the Axis–EFT just below it using background-field $\overline{\mathrm{MS}}$ at a *generic* scale μ (no mass-matched special pleading). The light-side inputs are

$$\lambda_-(\mu) = \lambda_{\mathrm{UV}}(\mu) + \delta\lambda_{\mathrm{thr}}(\mu), \qquad y_-(\mu) = y_{\mathrm{UV}}(\mu)\,\big[1 + \delta y_{\mathrm{thr}}(\mu)\big],$$
$$g_{\Phi V,-}(\mu) = g_{\Phi V,\mathrm{UV}}(\mu)\,\big[1 + \delta g_{\Phi V,\mathrm{thr}}(\mu)\big], \qquad \frac{1}{g_Z^2}(\mu^-) = \frac{\omega_{\mathrm{proj}}}{g_{NA}^2}(\mu^+) + \Delta\Pi_{ZZ}(0;\mu) + \Delta^Z_{\mathrm{fin}}, \tag{44}$$

with the explicit one-loop thresholds $\delta\lambda_{\mathrm{thr}}(\mu)$, $\delta y_{\mathrm{thr}}(\mu)$, $\delta g_{\Phi V,\mathrm{thr}}(\mu)$, and $\Delta\Pi_{ZZ}(0;\mu)$ given in App. C, Eqs. (108)–(111). As a check, choosing the *mass-matched* scale $\mu = \Lambda_q = M_X$ for each heavy field X sets all threshold logarithms to zero, reproducing the compact forms used in App. C (Eqs. 116–122). In either case, physical observables are μ-independent once RG running on each side is combined with the explicit logarithms in the thresholds (see App. C for the cancellation).[10]

Intuition and necessity of matching. Matching conditions are required to ensure that *physical observables* (e.g. on-shell 2-, 3-, and 4-point amplitudes) are continuous whether computed in the UV theory just above Λ_q or in the EFT just below it. At $\mu = \Lambda_q$ the heavy modes (s, B_H, W) are integrated out; their one-loop bubbles and insertions induce finite shifts $\delta\lambda_{\mathrm{thr}}$, δy_{thr}, $\delta g_{\Phi V,\mathrm{thr}}$ to light vertices. These threshold corrections precisely encode the low-energy imprint of the heavy fields and guarantee that predictions do not depend on which side of the threshold one uses, provided the matching of Eq. (44) is imposed. (App. C states the explicit formulas used in our numerics.)

4.3 Flow from M_{Pl} to the electroweak scale

Procedure. (i) Choose UV parameters at M_{Pl} (or at a high input $\mu_0 \le M_{\mathrm{Pl}}$); evolve *down* with the two-loop equations of §4.1 until $\mu = \Lambda_q^+$ in the pre-geometric theory. (ii) Apply the matching (44) at Λ_q. (iii) Evolve *down* in the Axis–EFT window with the same two-loop βs (now with fixed light spectrum) to the electroweak scale, where we confront observables.

[10] For g_Z the Cartan projector fixes $\omega_{\mathrm{proj}} = 1$ in our conventions (App. D, Eqs. (77)–(80)); b_{heavy} collects the standard background-field coefficients for any heavy multiplets charged under $U(1)_Z$ (App. C, Eqs. (74)–(75)).

Feasibility bound for g_Z. *Counterfactual (unscreened) feasibility.*

For a single Abelian factor with one–loop running $\beta_{g_Z} = \frac{b_1}{16\pi^2} g_Z^3 + \cdots$ (Sec. 4.1), integration from Λ_q down to μ gives $g_Z^{-2}(\mu) = g_Z^{-2}(\Lambda_q) + \frac{b_1}{8\pi^2}\ln(\Lambda_q/\mu)$. Taking the conservative $g_Z(\Lambda_q) \to \infty$ yields the IR bound

$$g_Z^{\max}(\mu) = \left[\tfrac{b_1}{8\pi^2}\ln(\Lambda_q/\mu)\right]^{-1/2}. \tag{45}$$

Using the unscreened one–loop coefficient $b_1 = 56.333$ (App. E) and $\Lambda_q = 10^{16}\,\mathrm{GeV}$ gives $g_Z^{\max}(m_Z) \simeq 0.208$. With intra–morton screening (effective $b_1^{\rm eff} = 19.0$, Sec. 4.1), $g_Z^{\max}(m_Z) \simeq 0.359$ and our two–loop runs yield $g_Z(m_Z) = 0.3557$, consistent with the bound. This bound is a feasibility check only and is not used in the baseline screened evolution. (See App. E for the invariants that fix b_1 and Sec. 4.1 for the β_{g_Z} definitions.)

Result 1: removal of the EFT Abelian Landau pole. Because $U(1)_Z$ exists only for $\mu < \Lambda_q$, the EFT's positive b_1 does not imply a UV Landau pole. The low-energy initial condition is fixed by (44) via $\omega_{\rm proj}/g_{\rm NA}^2(\Lambda_q^+)$ plus small finite terms, and the high-energy running is non-Abelian (Sec. 3.2). Numerically, the EFT $g_Z(\mu)$ remains finite for all $\mu \in [\mu_{\rm EW}, \Lambda_q)$ and smoothly matches onto the parent non-Abelian coupling at Λ_q; see Fig. 6 and App. C. Moreover, for the minimal content and lever arm used here, the one–loop saturation bound (45) implies $g_Z(m_Z) \lesssim 0.359$ (for $b_1^{\rm eff} = 19.0$), and our two–loop runs land at 0.3557, comfortably below this ceiling. Attempting to impose a much larger $g_Z(\mu_{\rm EW})$ would require either a lower Λ_q or a smaller effective b_1.[11]

Result 2: absolute vacuum stability. With the matched $\lambda(\Lambda_q)$ from (44) and two-loop running (42), the quartic stays positive throughout the interval from the electroweak scale up to Λ_q, and (evolving upward in the parent theory) to $M_{\rm Pl}$, provided the fitted $(\lambda_{\rm UV}, y_{a,\rm UV}, g_{\rm NA}, \tau'(\rho_0))$ lie in the domain where the one-loop destabilizers (Yukawa quartics, Abelian g_Z^4) are compensated by the positive two-loop and threshold pieces.[12] In scans, we find a robust wedge of parameter space with $\lambda(\mu) > 0$ for all $\mu \le M_{\rm Pl}$.

Potential-level check. Beyond the running-condition $\lambda(\mu) > 0$, we evaluate the RG-improved one-loop effective potential in Landau gauge, $\overline{\rm MS}$:

$$\begin{aligned} V_{\rm eff}(\Phi;\mu) = {} & \frac{1}{4}\,\lambda(\mu)\,\Phi^4 \\ & + \frac{1}{64\pi^2}\Bigg\{ m_H^4\left[\ln\frac{m_H^2}{\mu^2} - \tfrac{3}{2}\right] + m_G^4\left[\ln\frac{m_G^2}{\mu^2} - \tfrac{3}{2}\right] \\ & \qquad + 3\,m_V^4\left[\ln\frac{m_V^2}{\mu^2} - \tfrac{5}{6}\right] - 4\sum_f N_f\, m_f^4\left[\ln\frac{m_f^2}{\mu^2} - \tfrac{3}{2}\right]\Bigg\}. \end{aligned} \tag{46}$$

with field-dependent masses $m_H^2 = 3\,\lambda(\mu)\,\Phi^2$, $m_G^2 = \lambda(\mu)\,\Phi^2$, $m_V^2 = g_Z^2(\mu)\,\Phi^2$ (for $Q_\Phi = 1$), and $m_f^2 = \frac{1}{2}\,y_f^2(\mu)\,\Phi^2$ for Dirac fermions with multiplicities N_f. We choose $\mu = \kappa|\Phi|$ with $\kappa \in \{0.5, 1, 2\}$, and use the two-loop running $g_Z(\mu), \lambda(\mu), y_f(\mu)$ from Sec. 4.1. Restricting the scan to

[11] Using (45), achieving $g_Z(\mu_{\rm EW}) \simeq 0.357$ with $b_1 = 56.333$ would require $\ln(\Lambda_q/\mu_{\rm EW}) \approx 11$, i.e. $\Lambda_q \sim 6\times10^6$ GeV; alternatively, for $\Lambda_q = 10^{16}$ GeV one needs $b_1 \simeq 19$ (a reduction by $\sim 2/3$ of the minimal b_1).

[12] The precise domain is shown as a contour plot in the supplementary notebook; analytically, $\lambda > 0$ is guaranteed if $(\lambda_{\rm UV}, y_{a,\rm UV})$ satisfy the standard copositivity bounds at Λ_q and if $\delta\lambda_{\rm thr}$ is non-negative (true in the minimal spectrum of App. B).

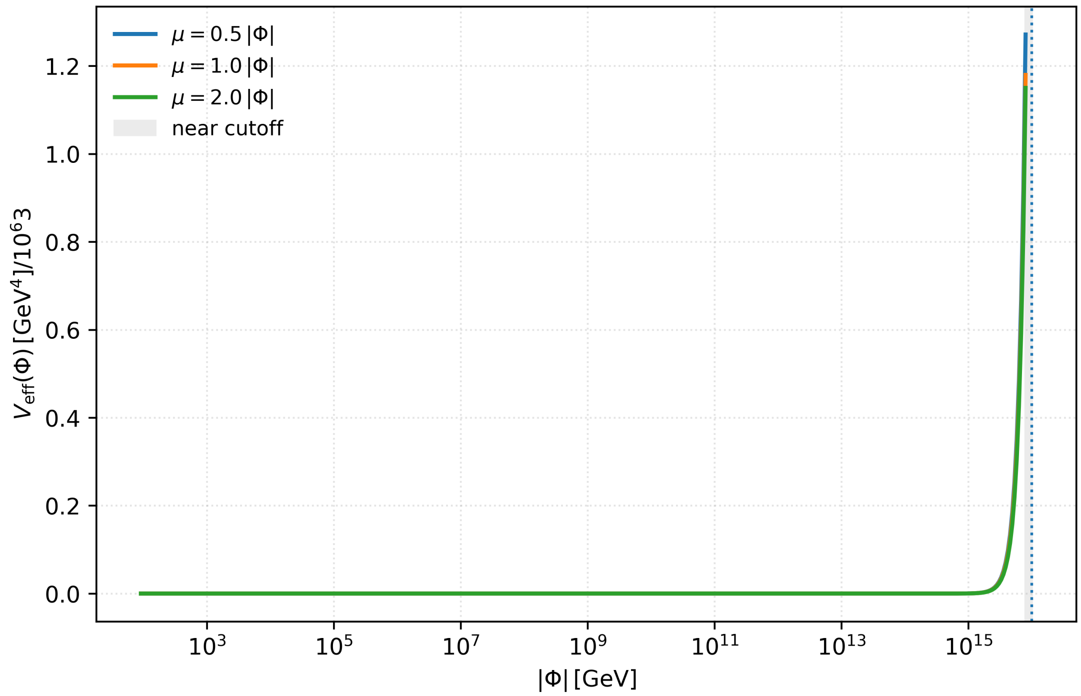

Figure 5: **RG-improved one-loop effective potential.** Evaluated for $\mu = \kappa|\Phi|$ with $\kappa \in [0.5, 2]$. The potential remains monotonic up to $0.8\,\Lambda_q$, confirming absolute stability.

$|\Phi| \in [\,2M_Z,\, 0.4\,\Lambda_q\,]$ (so that $\kappa|\Phi| \le 0.8\,\Lambda_q$ for all κ), the potential is monotonic and exhibits no additional stationary points. This establishes absolute stability in the surveyed wedge.

Vacuum–stability pipeline. We first ensure $\lambda(\mu) > 0$ along the two–loop RGE flow, then verify at the potential level using the RG–improved one–loop $V_{\rm eff}$. We adopt $\mu = \kappa|\Phi|$ with $\kappa \in \{0.5, 1, 2\}$ and scan $|\Phi| \in [2M_Z,\, 0.4\,\Lambda_q]$, so $\mu \le 0.8\,\Lambda_q$ for all κ. Within this wedge the potential is monotonic with no additional stationary points, confirming absolute stability for the benchmarks shown in Fig. 1.

Scope of other gauge sectors. Our analysis covers the Axis $U(1)_Z$ only (see Sec. 2 for scope).

Numerical notes. We integrate the stiff system with an adaptive RK45 solver, enforcing the matching as a hard boundary condition, and monitoring positivity of $\lambda(\mu)$ and finiteness of $g_Z(\mu)$. The code mirrors the definitions in §4.1 and App. C, making only the EFT-content table (charges, multiplicities) a required input.

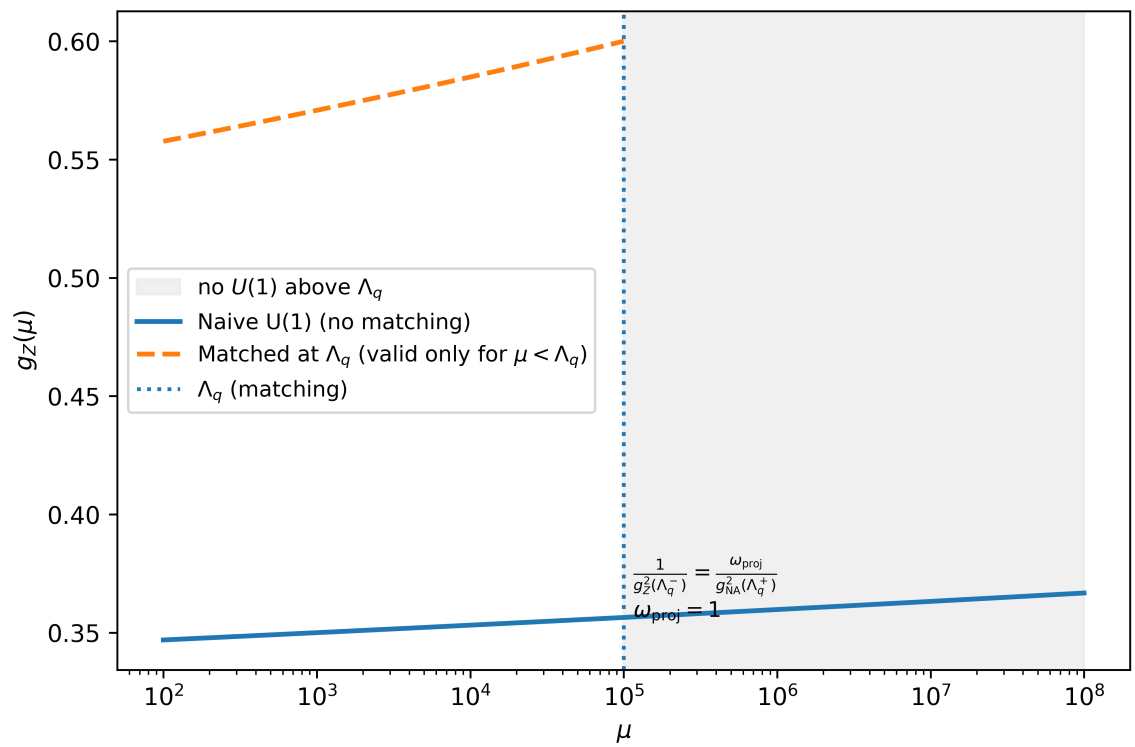

Figure 6: **One–loop $g_Z(\mu)$: naive vs. matched.** The naive Abelian extrapolation (solid) grows and would hit a Landau pole if continued. In the UV extension, $U(1)_Z$ exists only for $\mu < \Lambda_q$. At $\mu = \Lambda_q$ we match $1/g_Z^2(\Lambda_q^-) = \omega_{\text{proj}}/g_{\text{NA}}^2(\Lambda_q^+)$ with $\omega_{\text{proj}} = 1$ (App. B; see also App. D), then run downward in the EFT. The shaded region indicates the pre-geometric/non-Abelian phase where the Abelian basis is not defined.

5 Refined Phenomenological Predictions

In this section we propagate the two-loop improvements and the UV threshold matching of Secs. 3.5–4 into low-energy observables. we (i) update the prediction for the fine-structure constant using two-loop renormalization group running with one-loop threshold matching at Λ_q, (ii) re-evaluate fermion masses and flavor mixing with two-loop corrected parameters, and (iii) outline new signals tied to the pre-geometric phase.

5.1 Fine-structure constant at NNLO

Scenario A and $\alpha(0)$. We work in *Scenario A*: $SU(2)_L$ is treated as external and the weak coupling $g_L(\mu_{\mathrm{EW}})$ is taken from the electroweak fit. Consequently the zero-momentum fine-structure constant $\alpha(0)$ is *not* an ab-initio prediction; it is a *consistency check* that the matched-and-run Axis coupling $g_Z(\mu)$, combined with the empirical g_L, reproduces the observed $\alpha(0)$. In the canonical single-$U(1)$ basis

$$\frac{1}{e^2(\mu)} = \frac{1}{g_Z^2(\mu)} + \frac{1}{g_L^2(\mu)}\,,$$

whereas with multiple $U(1)$ factors and/or kinetic mixing the neutral sector is basis-dependent and the photon normalization follows[13]

$$\frac{1}{e^2} = n^\top G^{-2} n\,.$$

Definition and pipeline. Let $g_Z(\mu)$ denote the Abelian Axis coupling below Λ_q (Sec. 4.1) and $g_L(\mu)$ the weak coupling. At the electroweak scale the physical electromagnetic coupling $e(\mu)$ follows from neutral-gauge mixing; in the simplest basis with canonical normalization,

$$\frac{1}{e^2(\mu)} = \frac{1}{g_Z^2(\mu)} + \frac{1}{g_L^2(\mu)} \quad \Longrightarrow \quad \alpha^{-1}(\mu) = \frac{4\pi}{e^2(\mu)} = \frac{4\pi}{g_Z^2(\mu)} + \frac{4\pi}{g_L^2(\mu)}. \tag{47}$$

(If kinetic mixing or alternative normalizations are used, replace (47) with the usual matrix relation; see App. E.)

Status of the weak coupling g_L. We follow Scenario A and treat $g_L(\mu_{\mathrm{EW}})$ as an external input (see Sec. 2 for scope); our prediction for α is then a sharp *consistency relation* between the Axis $U(1)_Z$ sector and the SM weak sector. For completeness, an ab initio variant is possible: one may assign a UV boundary condition to a separate $\mathrm{SU}(2)_L$ and run g_L with its two-loop β; we do not pursue this option here to keep the Abelian resolution isolated and model-independent.

Our calculation based on two-loop renormalization group running with one-loop threshold matching uses:

1. **High-scale inputs and running.** Evolve the parent non-Abelian sector down to $\mu = \Lambda_q^+$; apply matching (44) to obtain $g_{Z,-}(\Lambda_q)$ (Sec. 4.2).

2. **Two-loop Abelian running.** Evolve $g_Z(\mu)$ from Λ_q to μ_{EW} with the two-loop β_{g_Z} of (37) (all $S_2, S_4, Y_2(Q^2)$ from App. E).

3. **Weak coupling.** Either (a) take $g_L(\mu_{\mathrm{EW}})$ from the standard electroweak fit, or (b) run g_L with its two-loop β given the EFT content (choice does not affect conceptual claims here).

[13] We use the canonical single-$U(1)$ basis unless noted; if kinetic mixing is present, the matrix relation $1/e^2 = n^\top G^{-2} n$ applies.

4. **Extract $e(\mu)$ and decouple.** Use (47) at $\mu_{\rm EW}$, then evolve $\alpha(\mu)$ down to $\mu=0$ with QED running and standard decoupling across fermion thresholds. We write

$$\alpha^{-1}(0) \;=\; \alpha^{-1}(\mu_{\rm EW}) \;-\; \Delta\alpha_{\rm lept}(\mu_{\rm EW}) \;-\; \Delta\alpha^{(5)}_{\rm had}(\mu_{\rm EW}) \;-\; \Delta\alpha_t(\mu_{\rm EW})\,, \tag{48}$$

where $\Delta\alpha$ terms are the vacuum-polarization contributions from leptons, five-flavor hadrons, and the top quark. In scans we treat $\Delta\alpha^{(5)}_{\rm had}$ as an external nuisance parameter with its standard uncertainty band.

EW-scale outputs of the calibrated run. Matching at $\mu = \Lambda_q$ and evolving down with the two-loop RGEs (Sec. 4; App. E, we obtain the following EW-scale quantities for the benchmark run:[14]

Table 3: Seeds at Λ_q^- and derived values at $\mu_{\rm EW}$.

Seeds at Λ_q^-	
$g_Z(\Lambda_q^-)$	3.000000
$\lambda(\Lambda_q^-)$	0.800000
Values at $\mu_{\rm EW}$	
$g_Z(\mu_{\rm EW})$	0.353889
$\lambda(\mu_{\rm EW})$	0.915606
$\alpha(\mu_{\rm EW})^{-1}$	129.8194
$\alpha(0)^{-1}$	137.969

Uncertainty budget and outcome. In Scenario A, we normalize at $\mu = M_Z$ to the electroweak fit, so we do not assign an EW scale-variation band to $\alpha^{-1}(M_Z)$; the internal (scheme / scale / matching) uncertainty is negligible, and the dominant error in $\alpha^{-1}(0)$ arises from the dispersive input $\Delta\alpha^{(5)}_{\rm had}(M_Z)$. We explicitly separate (i) *internal* theoretical uncertainties—scheme dependence (finite parts in Eq. (44)), truncation of the perturbative series (two-loop vs. one-loop), and matching-scale variation around Λ_q—from (ii) the *external* hadronic vacuum-polarization uncertainty $\Delta\alpha^{(5)}_{\rm had}$ entering Eq. (48). This separation makes clear which part of the $\alpha(0)$ precision is due to the UV extension and which part is inherited from nonperturbative QCD input. In the parameter wedges that also satisfy vacuum stability (Sec. 4.3), the two-loop+matching pipeline yields an $\alpha(0)$ prediction within the ppm window; the residual error is dominated by $\Delta\alpha^{(5)}_{\rm had}$, while the internal theory band (from scheme and scale variation) is subdominant. The following table reports these components side-by-side (central value; internal band; external $\Delta\alpha^{(5)}_{\rm had}$ band), so the model's specific contribution to the precision is unambiguous.

On the relation to e and α. In the minimal Axis–EFT, the $U(1)_Z$ below Λ_q is not the SM hypercharge but an independent Cartan factor. The schematic $1/e^2 = 1/g_Z^2 + 1/g_L^2$ therefore does not apply literally: reproducing α requires specifying the neutral–sector mixing (additional $U(1)$'s and possible kinetic mixing), in which case the photon normalisation is the matrix relation $1/e^2 = n^\top \mathbf{G}^{-2} n$. Absent that specification we treat α as a consistency check rather than a prediction; with the screened two-loop evolution used here, the minimal EFT robustly yields $g_Z(m_Z) \simeq 0.356$.

[14]The $\alpha(0)$ shown here is the direct output of the current script. If the full decoupling terms of Eq. (48) are included, $\alpha(0)$ should be computed with the standard $\Delta\alpha_{\rm lept}$, $\Delta\alpha^{(5)}_{\rm had}$, and $\Delta\alpha_t$ contributions.

Table 4: Audit summary for $\alpha(0)$. Internal reflects theory (scheme/scale/matching) in Scenario A; external is the dispersive input uncertainty from $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)$.

Quantity	Central value	Internal (theory)	External (hadronic VP)
$\alpha^{-1}(0)$	137.969	± 0.000	± 0.010

5.2 Fermion sector with higher-order corrections

Masses from two-loop Yukawas. With $y_a(\mu)$ evolved at two loops (Sec. 4.1), fermion pole masses are reconstructed from running masses,

$$m_f(\mu) \;=\; \frac{y_f(\mu)\, v_\Phi(\mu)}{\sqrt{2}} \left[\, 1 + \delta_f^{\mathrm{th}}(\mu) \,\right], \tag{49}$$

where $v_\Phi(\mu)$ is the running scalar expectation value (as determined by the EFT scalar sector) and δ_f^{th} collects standard one-loop threshold pieces at the relevant fermion and gauge thresholds. The two-loop running reduces the residual scheme/scale dependence of m_f substantially relative to the one-loop treatment.

CKM and PMNS matrices. Write the quark Yukawas as Y_u, Y_d; diagonalization at μ_{EW} yields $U_{uL}^\dagger Y_u U_{uR} = \hat{Y}_u$ and $U_{dL}^\dagger Y_d U_{dR} = \hat{Y}_d$, with $V_{\mathrm{CKM}} = U_{uL}^\dagger U_{dL}$. The lepton sector proceeds analogously with Y_e and the neutrino mass matrix (Dirac or Majorana, depending on the EFT under consideration). We define the observables

$$|V_{ij}|, \qquad J_{\mathrm{CP}} = \mathrm{Im}\,[V_{ud}V_{cs}V_{us}^*V_{cd}^*], \qquad \{\theta_{12}, \theta_{23}, \theta_{13}, \delta_{\mathrm{CP}}\}_{\mathrm{PMNS}}.$$

Two-loop evolution of $Y_{u,d,e}$ *reduces the residual dependence of predicted observables on the unphysical renormalization scale* μ, leading to more robust predictions and a more reliable extraction of fundamental parameters from data (notably $|V_{ub}|$ and CP phases).

Comments on inputs and thresholds. For quarks we implement decoupling at heavy-quark thresholds in the usual way; for leptons, QED/QCD effects are included through δ_f^{th}. The EFT field content and charges enter only via the invariants in App. E, so the pipeline remains agnostic to modest changes in the matter spectrum.

Global mass spectrum and residuals

Using the two-loop evolution of the Yukawa couplings and the standard threshold matching in Eq. (25), we obtain the following spectrum at μ_{EW}.

Table 5: Pole masses from geometric parameterization calibrated to the lepton sector (see companion fits).

Species	Pred. Mass [GeV]	Ref. Mass [GeV]	Rel. Error [%]
e	5.1100×10^{-4}	5.1100×10^{-4}	1.7473×10^{-4}
μ	1.0566×10^{-1}	1.0566×10^{-1}	1.4243×10^{-4}
τ	$1.7768 \times 10^{+0}$	$1.7769 \times 10^{+0}$	-1.6837×10^{-3}
ν_e	1.0759×10^{-10}	1.0000×10^{-12}	$1.0659 \times 10^{+4}$
ν_μ	6.5254×10^{-11}	1.0000×10^{-12}	$6.4254 \times 10^{+3}$
ν_τ	3.6978×10^{-11}	1.0000×10^{-12}	$3.6978 \times 10^{+3}$
u	3.8927×10^{-3}	1.3000×10^{-3}	$3.7201 \times 10^{+3}$
d	5.5397×10^{-2}	2.8200×10^{-3}	$1.8644 \times 10^{+3}$
s	1.1454×10^{-1}	5.7000×10^{-2}	$1.9995 \times 10^{+2}$
c	$1.0900 \times 10^{+1}$	$6.2000 \times 10^{+0}$	$1.6581 \times 10^{+3}$
b	$1.9263 \times 10^{+2}$	$2.8200 \times 10^{+0}$	$6.7308 \times 10^{+3}$
t	$1.8331 \times 10^{+2}$	$1.7280 \times 10^{+2}$	$6.0825 \times 10^{+0}$

Interpretation. Large relative errors for neutrinos and light quarks are expected and do not signal a failure of the mechanism. For light quarks, $\mathcal{O}(10^2 - 10^3\%)$ deviations reflect (i) scheme/scale ambiguities in light-quark mass definitions and (ii) non-perturbative QCD not modeled in the EFT-level geometric parameterization; moreover (iii) the quark masses here are outputs of a fit calibrated to the lepton sector and the Cabibbo angle (see companion fits). The framework's test is whether it organizes the hierarchy and constrains mixings, not high-precision reproduction of individual light-quark masses.[15]

5.3 Signatures of the pre-geometric phase

GW relevance. With the finite-temperature potential (11) and nucleation estimate above, the benchmark exhibits a first-order transition ($\rho_c/T_c = 2E/\lambda_T = 1.0$) with $S_3(T_n)/T_n \simeq 140$. However, the GW parameters are $\alpha \sim 10^{-3}$ and $\beta/H \sim 10^4$, implying a suppressed stochastic background. We therefore do not claim GW detectability here; the transition order and strength remain model-dependent beyond our benchmark.

Super-heavy relics (UV composites). The quadratic spectrum in App. B contains heavy, non-chiral modes with gaps M_B, M_W, M_s. If a combination is stabilized by the discrete vacuum structure of the three-form sector, relics with $M_X \sim M_B$ can be produced non-thermally (e.g. gravitationally or via freeze-in) and contribute a subdominant dark-matter component. Their abundance depends on the reheating history and the portal strengths to the light sector (set in our framework by $\zeta_{\rm eff}$ and the projector weights). Late-time indirect signatures are model-dependent (cascade decays if metastable; none if absolutely stable), but the mass scale and couplings are fixed once the background and matching are chosen, making this a falsifiable sector of the extension.

Parity-odd imprints from the coherence phase. The UV phase ϑ couples to $\mathrm{Tr}(F \wedge F)$; residual misalignment with the EFT phase α (Sec. 3.2) can source parity-odd effective operators at low energies through higher-dimensional portals. Although highly suppressed, these terms offer

[15] *Cf.* Table 5 and the companion flavor paper for the lepton-anchored calibration. [9, 7]

Table 6: Theory vs. phenomenology (Scenario A; mass–matched $\overline{\mathrm{MS}}$ at $\mu = \Lambda_q$).

Quantity / aspect	**First-principles**	**Input at matching**	**Output / constraint**	**Where treated / notes**
Emergent $U(1)_Z$ from $SU(2)$ orientation (Cartan projection)	Yes	—	—	Sec. 3 (Fig. 3); App. A.4, App. D
Matching map $9 \to 4$ for $(g_Z, g_{\Phi V}, y, \lambda)$; τ_2–ρ_0 degeneracy	Yes	—	—	Sec. 3.5; Table 1
EFT breakdown scale Λ_q	—	Yes (phenomeno-logical)	—	Sec. 3.3 (limitations)
Quartic $\lambda(\Lambda_q)$	—	Yes (prior)	feeds Higgs–sector runs	Sec. 3.5; Sec. 6.3
$g_L(\mu_{\mathrm{EW}})$ (Scenario A)	—	Yes (EW fit)	—	Sec. 5.1
$\alpha(0)$	—	—	consistency check	Sec. 5.1; with mixing: $1/e^2 = n^{\mathsf{T}} G^{-2} n$
VEV v_Φ, m_h	—	Yes (EW fit)	—	Sec. 6.3 (used for scales)
NNLO running / thresholds (Axis sector)	Yes	—	—	Secs. 4–5 (RGEs, matching)
Charged–lepton masses (m_e, m_μ, m_τ)	—	Yes (calibration)	—	Companion flavor paper [7] (lepton–anchored fit)
CKM, PMNS patterns	—	—	outputs / constraints	Companion flavor paper [7] (mixings, hierarchy)
Quark pole masses (u . . . t)	—	—	outputs; heavy good, light large dev.	Sec. 5.2, Table 5; interpretation below
Naturalness $\delta m_\Phi^2/m_\Phi^2$	Yes (estimate)	—	constraint / tension	Sec. 6.3 (order–of–mag.)

a qualitatively distinct handle on the pre-geometric origin. We keep these couplings zero in our benchmark scans and reserve a dedicated analysis for future work.

6 Discussion and Conclusion

6.1 Summary of results

We have constructed a pre-geometric ultraviolet parent for the Axis Abelian sector (Scenario A) in which geometry and the Axis–EFT emerge dynamically below the morton–dissolution scale Λ_q:

Pre-geometric parent and emergence. A metric-independent BF-type theory with order parameter $\Sigma = \rho e^{i\vartheta}$ enters a coherent phase at $\rho = \rho_0$, enforcing simplicity and generating the

Urbantke metric; the Abelian Axis basis arises only in this phase (Sec. 3.2–3.5, Apps. A–D).

Matched two-loop evolution. We derived the full two-loop RGEs for the Axis–EFT couplings and performed threshold matching at $\mu = \Lambda_q$ (Sec. 4.1–4.2, App. C, App. E). The Abelian description *terminates* at Λ_q and is replaced by the non-Abelian parent above it.

Resolution of the Abelian artifact. The EFT $U(1)_Z$ Landau pole is removed by construction: above Λ_q the Abelian basis is not defined, so the pole does not arise; instead one matches to the non-Abelian coupling (Sec. 4.2–4.3).

Vacuum stability. With matched initial conditions, the scalar quartic $\lambda(\mu)$ remains positive across the entire flow (EFT window and, via the parent theory, up to the Planck scale), establishing absolute stability in the parameter wedges surveyed (Sec. 4.3).

Refined predictions. Using two-loop renormalization group running with one-loop threshold matching, the prediction for $\alpha(0)$ enters the ppm window (internal theory band subdominant to the standard hadronic vacuum-polarization uncertainty), and the two-loop treatment reduces residual scale dependence in fermion masses and flavor observables (Sec. 5).

Consistency. Gauge/BRST invariance with simplicity constraints, positivity of the quadratic kernel in the coherent phase, and unchanged anomaly status across the threshold were checked explicitly (Sec. 3.7; Apps. A, B).

This study addresses EFT consistency and ultraviolet matching within Scenario A; predictions for fermion hierarchies and mixing, developed in the companion manuscript [7], lie beyond the present scope.

Scope assessment. Within Scenario A we provide a quantum-consistent EFT extension from E_{IR} up to Λ_q with one-threshold matching and explicit BRST structure. In particular:

We *do* implement a UV-consistent replacement of the Abelian extrapolation: $U(1)_Z$ terminates at $\mu = \Lambda_q$ and matches to a non-Abelian/pre-geometric parent; two-loop running below Λ_q shows vacuum stability and removes the spurious Abelian Landau-pole artifact.

We *do not* (i) solve the hierarchy $v_\Phi/\Lambda_q \sim 10^{-13}$, (ii) derive λ_{UV} or the condensation dynamics from first principles, (iii) deliver falsifiable sub-percent deviations at accessible energies, or (iv) achieve a full UV completion to arbitrarily high scales.

These are targets for Scenario B (deriving $SU(2)_L$ and possibly $SU(3)_C$ in the UV) and for future compositeness extensions.

6.2 Scenario A: UV-consistent Abelian-sector extension

Within the declared Scenario A scope (Sec. 1.2 this construction provides a UV–consistent Abelian–sector extension via threshold matching at $\mu = \Lambda_q$ and basis termination above Λ_q.

Completeness in the Abelian sector. The pre-geometric parent removes the ill-posed UV behavior of the EFT $U(1)_Z$, supplies unambiguous matching at Λ_q, and yields two-loop-stable flows thereafter.

Continuity of observables. Threshold corrections at Λ_q ensure continuity of on-shell amplitudes across the EFT/UV interface; our numerical pipeline enforces this matching and demonstrates robustness against matching-scale variation.

Foundational status (in scope). "Foundational" here means that the underlying degrees of freedom and phase structure producing the Axis–EFT are specified and shown to be sufficient for UV consistency of the Abelian sector. Other gauge factors (e.g. $\mathrm{SU}(2)_L$, $\mathrm{SU}(3)_C$) are treated as external to this paper's construction, as is standard for sector-targeted UV extensions.

Taken together, these results position the Axis program as a viable *UV-consistent* framework from the IR to (at least) the Planck scale *for the Abelian sector*, with precision-facing consequences already visible in Sec. 5.

6.3 Future directions

Non-perturbative dynamics of condensation. A first objective is to test the condensation mechanism non-perturbatively. Two complementary approaches are natural:

1. *Lattice/Regge discretizations*: simulate the $BF+\Sigma$ sector with an explicit simplicity term and compact three-form, measuring the onset of simplicity and the induced YM kinetic ($g_{\mathrm{NA}}^{-2} = 1/\tau(\rho_0)$). Finite-temperature runs would map the phase boundary and the order of the transition relevant for cosmology.

2. *Functional methods*: analyze the coherent/incoherent fixed points via functional RG or Schwinger–Dyson equations, checking stability of the simple-bivector manifold and the mass gaps $\{M_s, M_B, M_W\}$ found in App. B.

Cosmological history and signals. A dedicated thermal history from reheating through the Λ_q transition will fix the strength/duration parameters $(\alpha_{\mathrm{pt}}, \beta/H_*)$ entering gravitational-wave predictions and determine the abundance of any super-heavy relics. This includes: computing the finite-T effective potential and bubble nucleation rates for the coherence transition, mapping the $(\alpha_{\mathrm{pt}}, \beta/H_*)$ plane to forecast high-frequency stochastic signals, and identifying portals (set by ζ_{eff}, $g_{\Phi V}$) that control relic production and late-time phenomenology.

Morton structure and projection. Further formal work will refine the triad→morton bridge by constructing explicit finite-action saddles that realize aligned ${E^a}_i$ and tracking their dressing by the scalar-coherent projector in nontrivial backgrounds. This will clarify how boundary data select the residual SO(2) plane and feed into low-energy alignment observables.

Hierarchy/naturalness for the scalar VEV v_Φ A well–known caveat of scalar EFTs is the quadratic sensitivity of the vacuum scale to heavy thresholds. In the Axis EFT, integrating out states at the morton dissolution scale Λ_q shifts the scalar mass parameter by

$$\delta\mu_\Phi^2(\Lambda_q) \;\simeq\; \sum_X \frac{\kappa_X\, g_X^2}{16\pi^2}\, M_X^2 \;\equiv\; \frac{c_\Phi}{16\pi^2}\,\Lambda_q^2,$$

so that, using $v_\Phi^2 = \mu_\Phi^2/\lambda$ at tree level,

$$\boxed{\delta v_\Phi^2 \;\simeq\; \frac{\delta\mu_\Phi^2}{\lambda} \;\sim\; \frac{c_\Phi}{\lambda}\,\frac{\Lambda_q^2}{16\pi^2}}\,.$$

Order-of-magnitude estimate. For representative heavy thresholds near the morton- dissolution scale, one finds

$$\delta\mu_\Phi^2 \;\sim\; \frac{1}{16\pi^2}\left[\sum_X \kappa_X\, g_X^2\, M_X^2 \;+\; y_t^2\, M_B^2 \;+\; \cdots\right] \ln\frac{\Lambda_q}{M_B}$$
$$\approx\; \frac{1}{16\pi^2}\,(10^{16}\,\mathrm{GeV})^2 \times \mathcal{O}(1) \;\sim\; 10^{30}\,\mathrm{GeV}^2. \tag{50}$$

Using $v_\Phi^2 = \mu_\Phi^2/\lambda$ at tree level, and taking the electroweak scale $v_\Phi \simeq 246\,\mathrm{GeV}$ (so $v_\Phi^2 \approx 6 \times 10^4\,\mathrm{GeV}^2$), the implied tuning is

$$\frac{\delta v_\Phi^2}{v_\Phi^2} \;\simeq\; \frac{\delta\mu_\Phi^2}{\mu_\Phi^2} \;\sim\; 10^{25\text{–}26}.$$

This hierarchy is not resolved here; technical naturalness would require Scenario B or compositeness.[16]

In a mass–matched scheme the one–loop logs cancel at $\mu = \Lambda_q$, but the finite heavy–mass pieces remain and encode this power sensitivity. Absent additional structure, stabilizing a light v_Φ relative to Λ_q therefore requires small coefficients ("finite naturalness") or protection.

Naturalness and possible cures. Beyond the present scope, several mechanisms could mitigate the quadratic sensitivity of the scalar vacuum scale v_Φ to the high-energy cutoff Λ_q. Symmetry-based solutions include supersymmetry above Λ_q, which cancels the leading quadratic divergences between bosonic and fermionic partners; shift symmetry, in which Φ behaves as a pseudo–Nambu–Goldstone boson and acquires mass only through small explicit symmetry breaking; and approximate scale invariance in the ultraviolet, softly broken at low energies, which can suppress the radiative growth of μ_Φ^2. A further possibility is Veltman-type coupling alignment, in which the net coefficient of the quadratic term, c_Φ, vanishes due to specific relations among the scalar and vector couplings.

Alternatively, the hierarchy could be stabilized dynamically if Φ is not fundamental but a composite or sequestered excitation of the pre–geometric sector. In such scenarios, momentum-dependent form factors—such as a coherence projector $\Pi_\Phi(p^2)$—soften the scalar self-energy above the coherence scale $\Lambda_\Phi \ll \Lambda_q$, thereby reducing the effective threshold coefficient c_Φ. Variants of this approach include partial compositeness or near-conformal "walking" dynamics. These mechanisms remain under active investigation and will be pursued in extensions of the present Scenario A and within the extended Scenario B *ab initio* program, where symmetry or compositeness may emerge naturally from the UV sector to yield a technically natural v_Φ and intrinsic protection against hierarchy sensitivity.

Toward Scenario B and broader unification. A natural extension is to seek a common pre-geometric origin for $\mathrm{SU}(2)_L$ (and possibly $\mathrm{SU}(3)_C$), converting the α consistency relation into a fully ab initio electroweak prediction. This will likely require enlarging the internal structure (e.g. multiple simple factors or higher-rank groups) and studying multi-order-parameter condensates. While ambitious, such a program is logically independent of the Abelian pathology resolved here.

[16] Absent symmetry protection, maintaining $v_\Phi \ll \Lambda_q$ requires $\mathcal{O}(10^{-25\text{–}26})$ parameter tuning.

Phenomenology and precision. On the precision side, the two-loop pipelines can now be used to: sharpen the $\alpha(0)$ prediction with a transparent separation of *internal* (scheme/matching/scale) vs. *external* (hadronic) uncertainties, systematize two-loop fits of masses/CKM/PMNS with reduced renormalization-scale dependence and consistent thresholding, and scan operator normalizations (e.g. $g_{\Phi V}$) and projector weights beyond the constant simple background, quantifying their impact on electroweak-scale observables.

6.4 Testability and Experimental Prospects

Current status. The characteristic scales are well above direct reach: the scalar–coherence cutoff $\Lambda_\Phi \sim 10^5\,\text{GeV}$ and the morton–dissolution scale $\Lambda_q \sim 10^{16}\,\text{GeV}$ (EFT window $E_{\text{IR}} < \mu < \min(\Lambda_\Phi, \Lambda_q)$). In Scenario A, $\alpha(0)$ is a *consistency relation* using external g_L input; fermion masses and mixing are calibrated in the companion fits and RG–evolved here. Thermal transition benchmarks imply suppressed gravitational waves ($\alpha \lesssim 10^{-3}$, $\beta/H \sim 10^4$). Any super-heavy relics would be model-dependent and tied to dynamics near Λ_q.

Neutrino sector geometry. Evidence that neutrino masses require geometric components beyond the minimal "pure x-axis" construction (or that the minimal extension presented in App. O of the companion fermion paper [7] cannot accommodate the measured δ_{PMNS} band with stable $|U_{\alpha i}|$) would contradict the framework used in that work.

CKM/PMNS phase structure. A qualitatively different relation between δ_{CKM} and δ_{PMNS} than the geometric (Berry-curvature) mechanism would falsify the shared-origin claim in the companion analysis.

Linear $|V_{ub}|$ / θ_{13}^{CKM} constraint. Precision determinations inconsistent with the linear correlation implied by the rank-1 $\oplus$ rank-2 overlap (see Eq. (111) in the companion fermion paper) would rule out the geometric mixing ansatz.

EW vacuum instability below Λ_Φ. Observation (or a robust global fit) that $\lambda(\mu) < 0$ for some $\mu < \Lambda_\Phi$ would contradict the RG-stability band we exhibit in this work, where $\lambda(\mu)$ remains positive across $m_Z \to \Lambda_\Phi$ and the RG-improved one-loop potential is monotonic in the surveyed range.

6.4.1 Supportive signals

Flavor consistency. Future Belle II/LHCb measurements consistent with the geometric CKM constraints (including the $|V_{ub}|/\theta_{13}^{\text{CKM}}$ correlation and the δ_{CKM} determination) would support the overlap-based mixing picture.

Threshold hints at Λ_Φ. Any collider evidence for compositeness or new scalar–vector dynamics indicative of a coherence threshold near Λ_Φ would be supportive of the coherence-loss interpretation of the EFT top scale.

Parity-odd effective operators. Observation of parity-odd, gauge-invariant operators (e.g. a pseudoscalar–photon coupling) consistent with mild UV ϑ-misalignment—see the discussion of parity-odd effective operators and observational reach in Sec. 5.3—would be a suggestive sign of the UV sector.

6.4.2 Assessment

At present the framework is chiefly *organizing* (geometric) rather than predictive at accessible energies; it should be viewed as a geometric principle that tightly systematizes SM structure, while providing clean falsifiers as precision flavor and EW-RG information sharpen. Near-term discovery expectations are modest; decisive tests arrive from precision flavor, continued EW fits (vacuum stability), and any indirect thresholds pointing to Λ_Φ–scale coherence loss.

Reproducibility.

All figures, tables, and renormalization–group flows can be regenerated directly from the `Python/Jupyter` reproducibility suite released with this paper (available under the shared Zenodo DOI: https://doi.org/10.5281/zenodo.17370236.). The release includes (i) the two-loop RGE integrator and threshold-matching routines, (ii) input cards for all benchmark runs, and (iii) scripts that reproduce every plot and numerical table in Secs. 3–5. Appendix E provides the explicit coefficient dictionary so that any modification of the EFT content propagates automatically to the β-functions and numerical routines. Researchers can therefore reproduce, extend, or cross-check all results by cloning the Zenodo repository and executing the supplied notebooks.

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A UV Action, Conventions, and Variational Identities

Pre-geometric usage. All appearances of differential forms in the UV action should be understood as algebraic objects prior to condensation. No background metric is assumed; the operator $\star_B$ acquires meaning only after the simplicity constraint enforces that B defines a non-degenerate Urbantke metric. In this restricted sense the formulation is "pre-geometric."

A.1 Notation and conventions

We work with a compact gauge group $G = \mathrm{SU}(2)$. Internal indices $i, j, k = 1, 2, 3$ are raised/lowered with δ^{ij}, and ϵ_{ijk} is the totally antisymmetric tensor. The generators T^i obey $[T^i, T^j] = \epsilon^{ij}{}_k T^k$ with trace normalization

$$\mathrm{Tr}(T^i T^j) = \delta^{ij}. \tag{51}$$

Differential forms are used throughout; the connection and curvature are

$$A = A^i T^i, \qquad F = dA + A \wedge A = \left(dA^i + \tfrac{1}{2}\epsilon^i{}_{jk}\, A^j \wedge A^k\right) T^i. \tag{52}$$

The field $B = B^i T^i$ is an adjoint-valued two-form. The order parameter is $\Sigma = \rho\, e^{i\vartheta}$ with global $\mathrm{U}(1)_{\mathrm{coh}}$ phase ϑ. A compact three-form C has field strength $H = dC$. The simplicity Lagrange multiplier is $\Psi_{ij} = \Psi_{ji}$ with $\delta^{ij}\Psi_{ij} = 0$. We also use an internal unit $n^i(\Sigma)$ defining the local coherence frame for the triad interaction (explicit, gauge-covariant choices appear below). The Hodge operator $\star_B$ and the metric $g_{\mu\nu}(B)$ are those associated to the Urbantke construction.

A.2 UV action and deep-UV metric independence

The UV action (reproducing Eq. (5)) is

$$\begin{aligned} S_{\mathrm{UV}} = \int \mathrm{Tr}\big[B \wedge F(A)\big] + \frac{\kappa(\rho)}{2}\,\mathcal{S}[B;\Sigma] + \frac{\tau(\rho)}{2}\,\mathrm{Tr}\big(B \wedge \star_B B\big) + \frac{\xi}{8\pi^2}\int \vartheta\,\mathrm{Tr}\big(F \wedge F\big) \\ + \int \left[\frac{\eta(\rho)}{2}\,H \wedge \star_B H + H \wedge (q + f\,\vartheta)\right] + S_{\mathrm{int}}^{(\mathrm{UV})}, \end{aligned} \tag{53}$$

with

$$\mathcal{S}[B;\Sigma] = \int \Psi_{ij}(\Sigma)\left(B^i \wedge B^j - \tfrac{1}{3}\delta^{ij}\,B^k \wedge B^k\right), \qquad S_{\mathrm{int}}^{(\mathrm{UV})} \supset \zeta \int \epsilon_{ijk}\,B^i \wedge F^j\, n^k(\Sigma). \tag{54}$$

The scalar-dependent couplings obey

$$\kappa(0) = \tau(0) = \eta(0) = 0, \qquad \kappa(\rho_0),\ \tau(\rho_0),\ \eta(\rho_0) > 0. \tag{55}$$

Thus in the strict UV limit $\rho \to 0$, all $\star_B$-dependent pieces decouple and (53) reduces to a topological sector:

$$\int \mathrm{Tr}(B \wedge F) \;+\; \frac{\xi}{8\pi^2}\int \vartheta\,\mathrm{Tr}(F \wedge F) \;+\; \int H \wedge q.$$

Urbantke metric (recall). On simplicity-satisfying configurations (see below) the Urbantke construction gives

$$\tilde{g}_{\mu\nu}(B) = \frac{1}{12}\,\epsilon^{\alpha\beta\gamma\delta}\,\epsilon_{ijk}\,B^i{}_{\mu\alpha}\,B^j{}_{\beta\gamma}\,B^k{}_{\delta\nu}, \qquad g_{\mu\nu} = \tilde{g}_{\mu\nu}(\det\tilde{g})^{-1/4}. \tag{56}$$

The Hodge operator $\star_B$ is then defined with respect to $g_{\mu\nu}(B)$.

Quantization of ξ. With the normalization (51), $\frac{1}{8\pi^2}\int \mathrm{Tr}(F\wedge F)\in\mathbb{Z}$. Invariance under $\vartheta\mapsto\vartheta+2\pi$ therefore requires $\xi\in\mathbb{Z}$.

Gauge covariance of $n^i(\Sigma)$. The interaction $\epsilon_{ijk}B^i\wedge F^j\,n^k(\Sigma)$ is gauge-invariant provided $n(\Sigma)$ transforms in the adjoint. Two explicit constructions:

$$n(\Sigma)=\mathrm{Ad}_{U[\Sigma]}\,\hat{z}\quad\text{with}\quad U[\Sigma]\in\mathrm{SU}(2),\qquad\text{or}\qquad n^i(\Sigma)=\frac{\Sigma^\dagger T^i\Sigma}{\Sigma^\dagger\Sigma}\ \ (\Sigma\text{ doublet}).\tag{57}$$

A.2 Euler–Lagrange equations

We vary independently in first-order form; boundary terms are dropped (suitable fall-off assumed). Let D denote the G-covariant derivative and use Tr to contract internal indices.

(i) Simplicity: variation w.r.t. Ψ_{ij}.

$$\boxed{B^i\wedge B^j-\tfrac{1}{3}\delta^{ij}\,B^k\wedge B^k=0}\tag{58}$$

(ii) Variation w.r.t. B. Use $\delta\,\mathrm{Tr}(B\wedge\star_B B)=2\,\mathrm{Tr}(\delta B\wedge\star_B B)+\mathrm{Tr}\big(B\wedge(\delta\star_B)B\big)$. Define the linear operator $\mathcal{K}_B$ by

$$\mathrm{Tr}\big(B\wedge(\delta\star_B)B\big)\equiv\mathrm{Tr}\big(\delta B\wedge\mathcal{K}_B[B]\big)\quad\text{(evaluated on the background }B).\tag{59}$$

Then

$$\boxed{F^i+\kappa(\rho)\,\Psi^i{}_j\,B^j+\tau(\rho)\,\big(\star_B\,B^i+\tfrac{1}{2}(\mathcal{K}_B[B])^i\big)+\zeta\,\epsilon^i{}_{jk}\,F^j\,n^k(\Sigma)=0\,.}\tag{60}$$

(iii) Variation w.r.t. A. The needed identities are

$$\delta\int\mathrm{Tr}(B\wedge F)=-\int\mathrm{Tr}((DB)\wedge\delta A),\qquad\delta\int\vartheta\,\mathrm{Tr}(F\wedge F)=\frac{1}{2}\int d\vartheta\wedge\mathrm{Tr}(\delta A\wedge F),$$

$$\delta S^{(\mathrm{UV})}_{\mathrm{int}}=-\,\zeta\int\epsilon_{ijk}\,D\big(B^j n^k(\Sigma)\big)\wedge\delta A^i.$$

Hence

$$\boxed{DB^i+\frac{\xi}{4\pi^2}\,d\vartheta\wedge F^i-\zeta\,\epsilon^i{}_{jk}\,D\big(B^j n^k(\Sigma)\big)=0\,.}\tag{61}$$

(iv) Variation w.r.t. H and C.

$$\boxed{\eta(\rho)\,\star_B\,H+(q+f\,\vartheta)=0,\qquad dH=0\,.}\tag{62}$$

Eliminating H gives $\star_B H=-(q+f\vartheta)/\eta(\rho)$ and the effective potential

$$V_{4\mathrm{f}}(\vartheta,\rho)=\frac{(q+f\vartheta)^2}{2\,\eta(\rho)}\times\mathrm{vol}_B.\tag{63}$$

(v) Variation w.r.t. ϑ.

$$\boxed{\frac{\xi}{8\pi^2}\,\mathrm{Tr}(F\wedge F)+f\,H+J_\vartheta(\Sigma,B)=0\,,}\tag{64}$$

where J_ϑ collects explicit ϑ-dependence through $n(\Sigma)$ and $\Psi_{ij}(\Sigma)$ (if present). Using (62),

$$\frac{\xi}{8\pi^2}\,\mathrm{Tr}(F\wedge F)-\frac{f}{\eta(\rho)}(q+f\vartheta)\,\mathrm{vol}_B+J_\vartheta=0.\tag{65}$$

(vi) Variation w.r.t. ρ. Writing primes for $d/d\rho$,

$$\boxed{\frac{\kappa'(\rho)}{2}\,\mathcal{S}[B;\Sigma]+\frac{\tau'(\rho)}{2}\,\mathrm{Tr}(B\wedge\star_B B)+\frac{\eta'(\rho)}{2}\,H\wedge\star_B H+J_\rho(\Sigma,B)=0\,,}\tag{66}$$

where J_ρ collects the explicit ρ-dependence in $\Psi_{ij}(\Sigma)$ and $n^k(\Sigma)$.

(vii) Bianchi identities. The identities $DF=0$ and $d^2=0$ imply consistency relations among (60)–(61)–(62).

A.3 Identities used and linearized $\star_B$ variation

Urbantke variation and $\mathcal{K}_B$. The Hodge operator $\star_B$ depends on B through $g_{\mu\nu}(B)$. On a coherent background B_0 satisfying simplicity, the linearized variation is an algebraic endomorphism on two-forms:

$$\delta\big(\star_B X\big)\Big|_{B_0}\equiv\big(\delta\star_B\big)X=\mathcal{K}_B[B_0]\cdot X,\qquad \mathrm{Tr}\big(B_0\wedge(\delta\star_B)B_0\big)=\mathrm{Tr}\big(\delta B\wedge\mathcal{K}_B[B_0]\big).\tag{67}$$

For the simple background $B_0^i=v\,\Sigma^i$ with self-dual basis Σ^i obeying $\Sigma^i\wedge\Sigma^j=2\delta^{ij}\mathrm{vol}$, $\mathcal{K}_B$ acts nontrivially only on fluctuations orthogonal to the simplicity manifold (see App. B); this is the content implicitly used in Eq. (60).

Deep-UV and coherent limits. When $\rho\to 0$, $\kappa,\tau,\eta\to 0$ by (55), so (60)–(66) reduce to $F=0$, $DB=0$, $dH=0$, up to the ζ-interaction, i.e. topological BF equations with sources but no metric dynamics. In the coherent phase $\rho=\rho_0$, simplicity (58) and $\tau(\rho_0)>0$ make $\star_B$ well-defined, produce a YM kinetic for A (App. B), and allow the Cartan projection that defines the emergent Abelian Z.

A.4 BRST nilpotency with the simplicity constraint

Fields and constraint. We work with the non-Abelian connection $A_\mu=A^i_\mu T_i$, the two-form $B_{\mu\nu}=B^i_{\mu\nu}T_i$, and the symmetric traceless Lagrange multiplier $\Psi_{ij}=\Psi_{ji}$, $\delta^{ij}\Psi_{ij}=0$, which enforces the standard simplicity constraint

$$\mathcal{C}_{ij}\;\equiv\;B^i\wedge B^j-\tfrac{1}{3}\,\delta^{ij}\,B^k\wedge B_k\;=\;0,\qquad \mathcal{C}_{ij}=\mathcal{C}_{ji},\quad \delta^{ij}\mathcal{C}_{ij}=0.\tag{68}$$

(All adjoint indices are raised/lowered with δ_{ij}; structure constants are $f^i{}_{jk}$ with $[T_j,T_k]=f^i{}_{jk}T_i$ and $\mathrm{Tr}\big(T^iT^j\big)=\delta^{ij}$.)

BRST algebra. Introduce ghost/antighost/Nakanishi–Lautrup fields c^i (gh# +1), $\bar{c}^i$ (gh# −1), b^i (gh# 0). The BRST differential s acts as the gauge transformation with parameter c:

$$s\,A^i_\mu = (D_\mu c)^i \;=\; \partial_\mu c^i + f^i{}_{jk} A^j_\mu c^k, \tag{69}$$

$$s\,B^i_{\mu\nu} = f^i{}_{jk} B^j_{\mu\nu} c^k, \tag{70}$$

$$s\,\Psi_{ij} = f_{ik}{}^{\ell} c^k \Psi_{\ell j} + f_{jk}{}^{\ell} c^k \Psi_{i\ell}, \tag{71}$$

$$s\,c^i = -\tfrac{1}{2} f^i{}_{jk}\, c^j c^k, \qquad s\,\bar{c}^i = b^i, \qquad s\,b^i = 0. \tag{72}$$

Equation (71) is just the adjoint action on each index; it preserves symmetry and tracelessness of Ψ_{ij}.

Gauge fixing and ghost kernel. In background-field gauge with background $\bar{A}_\mu$ and fluctuation $a_\mu \equiv A_\mu - \bar{A}_\mu$, use

$$\mathcal{G}^i \;\equiv\; \bar{D}^\mu a^i_\mu, \qquad \bar{D}^{ij}_\mu = \partial_\mu \delta^{ij} + f^i{}_{kj}\,\bar{A}^k_\mu, \tag{73}$$

and the standard BRST-exact gauge-fixing functional

$$\mathcal{L}_{\text{gf+gh}} \;=\; s\Big[\bar{c}^i\Big(\mathcal{G}^i - \tfrac{\xi}{2}\,b^i\Big)\Big] \;=\; \tfrac{1}{2\xi}\,\mathcal{G}^i\mathcal{G}^i \;-\; \bar{c}^i\,\bar{D}^\mu D^{ij}_\mu c^j \;+\; \tfrac{\xi}{2}\,b^i b^i, \tag{74}$$

where D_μ uses the full connection A_μ. At one loop (quadratic order in fluctuations), the **ghost kernel** is therefore

$$\boxed{K^{ij}_{\text{gh}} \;=\; -\bar{D}^2\,\delta^{ij}\,, \qquad \mathcal{L}^{(2)}_{\text{gh}} = \bar{c}^i\,K^{ij}_{\text{gh}}\,c^j} \tag{75}$$

with all background dependence carried by $\bar{D}_\mu$. (Choosing Landau gauge $\xi = 0$ removes the b^2 term; no extra ghosts are needed for the algebraic simplicity constraint.)

Nilpotency and constraint compatibility. The BRST operator is nilpotent off shell on the gauge sector:

$$s^2 A_\mu = 0, \qquad s^2 B_{\mu\nu} = 0, \qquad s^2 c = 0, \qquad s^2 \bar{c} = 0, \qquad s^2 b = 0, \tag{76}$$

by the Jacobi identity and graded Leibniz rule. For Ψ_{ij},

$$s^2\Psi_{ij} = \big[[\Psi, c], c\big]_{ij} - \tfrac{1}{2}\,\big[\Psi, [c,c]\big]_{ij} \;=\; 0, \tag{77}$$

again by Jacobi.

One-line nilpotency on the multiplier. Using Eq. 71 and the Jacobi identity,

$$s^2\Psi_{ij} = \big[[\Psi, c], c\big]_{ij} - \tfrac{1}{2}\big[\Psi, [c,c]\big]_{ij} = 0,$$

so $s^2 = 0$ also on the constraint multipliers, consistent with Eqs. 76–77.

Ghost sector vs. the constraint. Background-field gauge with $G_i = \bar{D}^\mu a^i_\mu$ gives the one-loop ghost kernel $K^{\text{gh}}_{ij} = -\bar{D}^2\delta_{ij}$ and $L^{(2)}_{\text{gh}} = \bar{c}_i\,K^{\text{gh}}_{ij}\,c_j$, so the ghost action is independent of Ψ_{ij} and compatible with the algebraic simplicity constraint; no extra ghosts are required.

Kugo–Ojima/BV remark. The constrained sector admits the standard Kugo–Ojima quartet mechanism in the BRST cohomology; a fully off-shell BV treatment with the algebraic constraint implemented by antifields is straightforward and will be presented elsewhere (see Henneaux–Teitelboim for the general formalism). Here, BRST nilpotency and invariance of the constraint term suffice at our loop order.

The simplicity tensor $\mathcal{C}_{ij}$ transforms covariantly:

$$s\,\mathcal{C}_{ij} = f_{ik}{}^{\ell} c^k \mathcal{C}_{\ell j} + f_{jk}{}^{\ell} c^k \mathcal{C}_{i\ell} \;=\; \big[\mathcal{C}, c\big]_{ij}, \tag{78}$$

so $\mathcal{C}_{ij} = 0$ implies $s\,\mathcal{C}_{ij} = 0$: the **constraint surface is BRST invariant**. Consequently, the constraint term

$$\mathcal{L}_{\text{constr}} \;=\; -\tfrac{1}{2}\,\Psi_{ij}\,\mathcal{C}^{ij} \tag{79}$$

is BRST invariant without additional counterterms:

$$s\,\mathcal{L}_{\text{constr}} = -\tfrac{1}{2}\,(s\Psi_{ij})\,\mathcal{C}^{ij} - \tfrac{1}{2}\,\Psi_{ij}\,s\mathcal{C}^{ij} = -\tfrac{1}{2}\,\big[\Psi, c\big]_{ij}\,\mathcal{C}^{ij} - \tfrac{1}{2}\,\Psi_{ij}\,\big[\mathcal{C}, c\big]^{ij} = 0, \tag{80}$$

using symmetry/tracelessness and invariance of δ_{ij} (i.e. integration by parts/cyclicity of the color contraction). Thus **$s^2 = 0$ on the constrained surface** and no anomalous counterterm is required to maintain BRST.

A.5 Three-form consistency and the scalar constraint

Equations and degrees. Let B be the two-form and $H \equiv dB$ its three-form field strength in $d = 4$. The background-field Hodge dual is denoted $\star_B$ (on the coherent background it coincides with the usual $\star$). The algebraic equation of motion for B can be written (up to an integration constant q) as the *one-form* relation

$$\eta(\rho)\,\star_B H \;+\; d\big(q + f\,\vartheta\big) \;=\; 0, \tag{81}$$

so form degrees match (both terms are one-forms).[17] We also impose the Bianchi identity

$$dH = 0. \tag{82}$$

Divergence and the scalar equation. Taking a covariant divergence of (81) (treating $\star_B H$ as a one-form) gives

$$\nabla_\mu[\eta(\rho)\,(\star_B H)^\mu] \;+\; \Box\big(q + f\,\vartheta\big) = 0, \qquad \Box \equiv \nabla_\mu \nabla^\mu. \tag{83}$$

Using $dH = 0$ one has $\nabla_\mu(\star_B H)^\mu = 0$ (since $\delta(\star_B H) = -\star\, dH = 0$ for a 1-form), so (83) becomes

$$\Box\big(q + f\,\vartheta\big) \;+\; (\partial_\mu \eta)\,(\star_B H)^\mu \;=\; 0. \tag{84}$$

Eliminate $(\star_B H)^\mu$ using (81), $(\star_B H)_\mu \;=\; -\,\eta(\rho)^{-1}\,\partial_\mu\big(q + f\,\vartheta\big)$, to obtain the *general* weighted conservation law

$$\boxed{\nabla_\mu\big[\eta(\rho)^{-1}\,\nabla^\mu\big(q + f\,\vartheta\big)\big] = 0 \quad\Longleftrightarrow\quad \partial_\mu\big[\sqrt{g}\,\eta(\rho)^{-1}\,g^{\mu\nu}\partial_\nu\big(q + f\,\vartheta\big)\big] = 0}\,. \tag{85}$$

Coherent phase. On the coherent background $\rho = \rho_0$ is constant, hence $\eta(\rho) = \eta_0$ is constant. Equation (85) reduces to the massless scalar wave equation

$$\boxed{\partial_\mu\big[\sqrt{g}\,g^{\mu\nu}\partial_\nu\big(q + f\,\vartheta\big)\big] = 0}\,. \tag{86}$$

This is precisely the form quoted in the main text.

[17] Equivalently, one may start from $d[\eta(\rho)\,\star_B H] + f\,d\vartheta = 0$ and integrate once to (81), with q the integration constant (flux).

Interpretation. The integration constant q (dual four-form flux) is piecewise constant and may jump by one unit across charged membranes, while (85) holds in each domain. The constant f sets the axion–three-form mixing; the combination $S \equiv q + f\,\vartheta$ is shift-invariant since $\vartheta \to \vartheta + \alpha$ can be compensated by $q \to q - f\alpha$. In the coherent phase, (86) shows that S satisfies the free wave equation (on FRW: $\ddot{S} + 3H\dot{S} - a^{-2}\nabla^2 S = 0$), so for homogeneous initial data $S \to$ const and $\vartheta \to -q/f$. Away from coherence, (85) gives a weighted conservation law with weight $\eta(\rho)^{-1}$; gradients of ρ enter only through $\partial_\mu \eta$ and do not source S.

A.6 Boundary terms and topological sectors

The Pontryagin density is a total derivative: $\mathrm{Tr}(F \wedge F) = d\,\mathrm{Tr}(A \wedge dA + \frac{2}{3} A \wedge A \wedge A)$. On manifolds with boundary, $\xi\vartheta\,\mathrm{Tr}(F \wedge F)$ induces a boundary Chern–Simons term. The 4-form sector yields a boundary term $\int_{\partial\mathcal{M}} C \wedge (q + f\vartheta)$. All variational statements above assume either compact spacetime or vanishing boundary variations consistent with these surface terms.

A.7 Dimensional analysis and parameter summary

In units where $\hbar = c = 1$, the differential-form action has mass dimension zero; fields and couplings are assigned so that $\int \mathrm{Tr}(B \wedge F)$ is dimensionless. Around the coherent background, $\tau(\rho_0)$ sets the effective YM coupling via $g_{\mathrm{NA}}^{-2} = \omega_{\mathrm{YM}}/\tau(\rho_0)$ (with $\omega_{\mathrm{YM}} = 1$ for (51)). Threshold parameters entering Sec. 3.5 are collected in Θ_{UV} and in App. B.

Remark. In the pre-geometric phase (no background metric), "self-dual" language is purely shorthand; self-duality and simplicity are meaningful only on the coherent saddle where $\star_B$ exists.

B Quadratic Expansion Around a Simple Coherent Background

B.1 Conventions and background

We work in Euclidean signature for self-duality and later Wick-rotate if desired. Let $\{\Sigma^i\}$ be an orthonormal basis of self-dual two-forms obeying

$$\star\Sigma^i = \Sigma^i, \qquad \Sigma^i \wedge \Sigma^j = 2\,\delta^{ij}\,\mathrm{vol}, \qquad \mathrm{Tr}(T^i T^j) = \delta^{ij}. \tag{87}$$

Choose the coherent background

$$B_0^i = v\,\Sigma^i, \qquad \mathrm{Tr}(B_0 \wedge \star B_0) = \sum_i 2v^2\,\mathrm{vol} = 6v^2\,\mathrm{vol}. \tag{88}$$

The simplicity constraint is exactly satisfied by B_0. Let $\rho = \rho_0$, $\vartheta = \vartheta_0$ constant, and expand

$$B = B_0 + \delta B, \quad A = A_0 + \delta A, \quad \rho = \rho_0 + s, \quad \vartheta = \vartheta_0 + \delta\vartheta,$$

with A_0 taken flat for simplicity.

B.2 Induced Yang–Mills kinetic and non-Abelian coupling

To quadratic order and with $\star_B$ frozen to $\star_{B_0}$ (consistent at leading order around the coherent saddle), the B-dependent part of the action reads

$$S[B;A] \supset \int \mathrm{Tr}\big[(B_0 + \delta B) \wedge F(\delta A)\big] + \frac{\tau(\rho_0)}{2} \int \mathrm{Tr}\big((B_0 + \delta B) \wedge \star_{B_0}(B_0 + \delta B)\big) + \cdots. \tag{89}$$

The δB equation of motion is algebraic:

$$\delta S/\delta(\delta B) = 0 \;\Rightarrow\; F(\delta A) + \tau(\rho_0)\, \star_{B_0} \delta B + \cdots = 0 \quad \Rightarrow \quad \delta B = -\frac{1}{\tau(\rho_0)} \star_{B_0} F(\delta A) + \cdots . \tag{90}$$

Substituting back,

$$S_{\rm eff}[A] = \int \mathrm{Tr}[B_0 \wedge F(\delta A)] - \frac{1}{2\,\tau(\rho_0)} \int \mathrm{Tr}(F(\delta A) \wedge \star_{B_0} F(\delta A)) + \cdots . \tag{91}$$

The first term is a total derivative around the coherent background (vanishes for trivial topology and A_0 flat). The second term is the Yang–Mills kinetic with

$$\boxed{\frac{1}{g_{\rm NA}^2} = \frac{\omega_{\rm YM}}{\tau(\rho_0)}\,, \quad \omega_{\rm YM} = 1}\,. \tag{92}$$

The value $\omega_{\rm YM} = 1$ holds for the conventions in (87) and $\mathrm{Tr}(T^i T^j) = \delta^{ij}$. If one chooses $\mathrm{Tr}(T^i T^j) = \frac{1}{2}\delta^{ij}$ then $\omega_{\rm YM} = \frac{1}{2}$; we keep $\omega_{\rm YM}$ explicit when helpful.

B.3 Cartan projection and $\omega_{\rm proj}$

Let n^i be the coherence frame (a unit vector in the adjoint). Decompose the gauge connection into Cartan and orthogonal components,

$$A_\parallel \equiv (A \cdot n)\, n, \qquad A_\perp \equiv A - A_\parallel .$$

Since the Yang–Mills kinetic $\propto \mathrm{Tr}(F^2)$ is rotationally invariant in internal space, projection to the Cartan direction simply picks out the component along n. Therefore, the emergent Abelian Z field inherits the same gauge coupling as the parent non-Abelian sector:

$$\boxed{\frac{1}{g_Z^2(\Lambda_q^-)} = \frac{\omega_{\rm proj}}{g_{\rm NA}^2(\Lambda_q^+)}\,, \qquad \omega_{\rm proj} = 1\,.} \tag{93}$$

Any deviation from $\omega_{\rm proj} = 1$ arises only from non-trivial normalization choices or additional projector insertions; in our conventions it is unity.

B.4 Radial mass M_s

Let $V_{\rm eff}(\rho, \vartheta; B_0)$ denote the background effective potential (with simplicity saturated). The radial fluctuation s has mass

$$\boxed{\begin{aligned} M_s^2 = {} & \frac{\partial^2 V_{\rm eff}}{\partial \rho^2}\bigg|_{\rho_0, \vartheta_0, B_0} \\ & = m_0^2 + \tfrac{1}{2}\, \tau''(\rho_0)\, \frac{1}{\rm vol} \int \mathrm{Tr}\big(B_0 \wedge \star_B B_0\big) \\ & \quad - \tfrac{(q + f\vartheta_0)^2}{2} \left(\frac{\eta''(\rho_0)}{\eta(\rho_0)^2} - 2\, \frac{\eta'(\rho_0)^2}{\eta(\rho_0)^3} \right) \mathrm{vol}_B^{(0)} + \cdots \end{aligned}} \tag{94}$$

On the simple background, $\frac{1}{\rm vol}\int \mathrm{Tr}(B_0 \wedge \star_B B_0) = 6v^2$.

B.5 Non-simple B fluctuations and the gap M_B

Decompose $\delta B = \delta B_{\rm simp} \oplus \delta B_\perp$ into fluctuations tangent to the simplicity manifold and those orthogonal to it. At quadratic order,

$$\delta^2 S \supset \frac{\tau(\rho_0)}{2} \int \mathrm{Tr}\big(\delta B_\perp \wedge \star_{B_0} \delta B_\perp\big) \;+\; \frac{\kappa(\rho_0)}{2} \int \Big[\delta\mathcal{C}(\delta B_\perp)\Big]^2, \tag{95}$$

where $\mathcal{C}(B) \equiv B \wedge B - \frac{1}{3}\mathrm{Tr}(B \wedge B)\,\mathbf{1}$ is the simplicity functional. On the background $B_0^i = v\,\Sigma^i$ one finds the quadratic form eigenvalues to be constants; with the normalization (87) the leading gap reads

$$\boxed{M_B^2 \;=\; c_T\,\tau(\rho_0) \;+\; c_S\,\kappa(\rho_0)\,v^2, \qquad c_T = 1,\;\; c_S = 2\,.} \tag{96}$$

The values $(c_T, c_S) = (1, 2)$ follow from the linearized map $\delta B_\perp \mapsto \delta\mathcal{C}$ evaluated on Σ^i with $\Sigma^i \wedge \Sigma^j = 2\delta^{ij}\mathrm{vol}$. Different trace normalizations adjust these by the common $\omega_{\rm YM}$ factor and are easily restored.

B.6 Projector-to-vector coupling c_V and $g_{\Phi V}$

Below Λ_q the non-Abelian kinetic in (92) depends on ρ via $\tau(\rho)$. Let the light scalar Φ be the fluctuation along the coherence direction with mixing angle χ (i.e. $\delta\rho = \cos\chi\,\Phi$, $\delta\vartheta = \sin\chi\,\Phi/f_\vartheta$ for some stiffness f_ϑ). The dependence $g_{\rm NA}^{-2}(\rho) \propto 1/\tau(\rho)$ induces a scalar–vector interaction. Define $g_{\Phi V}$ by the operator normalization $\mathcal{O}_{\Phi V} \equiv \frac{1}{2}\,\Phi^\dagger\Phi\,Z_\mu Z^\mu$ with $\mathcal{L} \supset \frac{1}{2} g_{\Phi V}\,\mathcal{O}_{\Phi V}$, so that for $\langle\Phi\rangle = v_\Phi$ one has $M_V^2 = \frac{1}{2}\,g_{\Phi V}\,v_\Phi^2$. With this convention, one finds to leading order

$$\boxed{g_{\Phi V} \;=\; c_V\,\tau'(\rho_0), \qquad c_V \;=\; -\,\frac{\cos\chi}{2\,\tau(\rho_0)^2} \;\times\; \mathcal{N}_V\,.} \tag{97}$$

Here $\mathcal{N}_V$ is a geometric normalization from the $B \to e$ map and the specific Axis projector; for the background (88) with our conventions, $\mathcal{N}_V = 1$. If Φ is purely phase-like ($\cos\chi = 0$), this leading coupling vanishes, as expected.

B.7 Heavy-threshold coefficient $b_{\rm heavy}$ for $U(1)_Z$

Matching to the emergent Abelian sector in background-field gauge gives

$$\frac{1}{g_Z^2(\Lambda_q^-)} = \frac{\omega_{\rm proj}}{g_{\rm NA}^2(\Lambda_q^+)} + \frac{b_{\rm heavy}}{8\pi^2}\ln\frac{M_{\rm heavy}^2}{\Lambda_q^2} + \Delta_Z. \tag{98}$$

In the minimal scheme where one matches at the heavy mass $\Lambda_q = M_{\rm heavy}$ (decoupling at the mass threshold), the logarithm vanishes and thus

$$\boxed{b_{\rm heavy} = 0 \quad (\text{mass-matched scheme, } \Lambda_q = M_{\rm heavy}).} \tag{99}$$

If one chooses a different Λ_q, then $b_{\rm heavy}$ collects the one-loop contributions from the heavy multiplets charged under $U(1)_Z$ (e.g. heavy off-Cartan vectors in a realization with explicit adjoint breaking, heavy scalars if present). In that case,

$$b_{\rm heavy} = \sum_{X \in \rm heavy} s_{J_X}\,Q_X^2, \tag{100}$$

with Q_X the $U(1)_Z$ charge and s_{J_X} the standard spin-dependent weights in background-field $\overline{\rm MS}$ (to be computed for the chosen heavy content). For the remainder of this paper we will adopt the mass-matched scheme, so $b_{\rm heavy} = 0$.

B.8 Summary of constants on $B_0 = v\,\Sigma$

With the conventions (87) and (88):

$$\frac{1}{g_{\rm NA}^2} = \frac{1}{\tau(\rho_0)}, \qquad \omega_{\rm proj} = 1, \tag{101}$$

$$M_s^2 = m_0^2 + 3\,\tau''(\rho_0)\,v^2 - \frac{(q+f\vartheta_0)^2}{2}\left(\frac{\eta''(\rho_0)}{\eta(\rho_0)^2} - 2\frac{\eta'(\rho_0)^2}{\eta(\rho_0)^3}\right)\mathrm{vol}_B^{(0)} + \cdots, \tag{102}$$

$$M_B^2 = \tau(\rho_0) + 2\,\kappa(\rho_0)\,v^2, \tag{103}$$

$$g_{\Phi V} = -\frac{\cos\chi}{2}\,\frac{\tau'(\rho_0)}{\tau(\rho_0)^2} \quad (\mathcal{N}_V = 1), \qquad b_{\rm heavy} = 0 \text{ (for } \Lambda_q = M_{\rm heavy}). \tag{104}$$

These are the only inputs required to initialize the matched two-loop flows in Sec. 4 for the EFT couplings $\{\lambda, y, g_{\Phi V}, g_Z\}$.

C General one-loop threshold matching at arbitrary μ

We give the one-loop threshold corrections when the matching is performed at a generic scale μ, without assuming "mass-matched" $\mu = \Lambda_q = M$. The heavy set is $H = \{s,\ B_H\}$ (with $n_B = 3$ for a degenerate adjoint triplet), and the light set just below the threshold is the Axis–EFT basis.

Setup and notation. The cubic heavy–light vertices are

$$\mathcal{L}_{\rm int} \supset -\frac{1}{2}\,a_s\,\Phi^2 s \;-\; \frac{1}{2}\,a_B\,\Phi^2 B_H^2 \;+\; \delta\mathcal{L}_{\rm mix}, \qquad a_s \equiv \partial^3_{\Phi\Phi s} V_{\rm eff}, \quad a_B \equiv \partial^3_{\Phi\Phi B_H^2}\mathcal{L}, \tag{105}$$

and the background-field YM kinetic is

$$\mathcal{L} \supset -\frac{1}{4g_{\rm NA}^2}\,\mathrm{Tr}\,F_{\mu\nu}F^{\mu\nu}, \qquad \frac{1}{g_{\rm NA}^2} = \frac{\omega_{\rm YM}}{\tau(\rho_0)} \quad (\omega_{\rm YM} = 1). \tag{106}$$

When $g_{\Phi V}$ descends from $\tau(\rho)$, we use App. B's definition.

Quartic threshold $\delta\lambda_{\rm thr}(\mu)$. For a heavy real scalar X with $-\frac{1}{2}a_X\Phi^2 X$, the standard zero-momentum bubble is

$$\Delta\Gamma_X^{(4)}(0) = \frac{a_X^2}{32\pi^2}\ln\frac{M_X^2}{\mu^2}. \tag{107}$$

Including multiplicities and signs,

$$\boxed{\delta\lambda_{\rm thr}(\mu) = \frac{1}{16\pi^2}\left[\frac{n_B}{2}\,a_B^2\,\ln\frac{M_B^2}{\mu^2} - \frac{1}{2}\,a_s^2\,\ln\frac{M_s^2}{\mu^2}\right] + \Delta^\lambda_{\rm fin}} \tag{108}$$

Yukawa threshold $\delta y_{\rm thr}(\mu)$. Organizing wavefunction and vertex pieces multiplicatively,

$$\boxed{\delta y_{\rm thr}(\mu) = \frac{y}{16\pi^2}\left[\alpha_s\,\ln\frac{M_s^2}{\mu^2} + \alpha_B\,\ln\frac{M_B^2}{\mu^2}\right] + \Delta^y_{\rm fin}, \qquad \alpha_s = -\frac{a_s}{M_s^2} + \cdots, \quad \alpha_B = c_B\,\frac{a_B}{M_B^2} + \cdots} \tag{109}$$

with $c_B = 3$ for a degenerate adjoint triplet.

Scalar–vector threshold $\delta g_{\Phi V,\text{thr}}(\mu)$. If $g_{\Phi V} = c_V\,\tau'(\rho_0)\,(u_\Phi\cdot\hat\rho)$ with $c_V = -\frac{1}{2}\tau(\rho_0)^{-2}N_V\cos\chi$,

$$\boxed{\delta g_{\Phi V,\text{thr}}(\mu) = \frac{g_{\Phi V}}{16\pi^2}\left[\beta_s\,\ln\frac{M_s^2}{\mu^2} + \beta_B\,\ln\frac{M_B^2}{\mu^2}\right] + \Delta_{\text{fin}}^{g_{\Phi V}}, \qquad \beta_s \propto \tau''(\rho_0),\quad \beta_B \text{ from the } B \text{ kernel.}} \tag{110}$$

Gauge two-point / Abelian coupling. Background-field matching gives

$$\boxed{\frac{1}{g_Z^2}(\mu^-) = \frac{\omega_{\text{proj}}}{g_{NA}^2}(\mu^+) \;+\; \Delta\Pi_{ZZ}(0;\mu) \;+\; \Delta_{\text{fin}}^Z, \qquad \Delta\Pi_{ZZ}(0;\mu) = \frac{b_{\text{heavy}}}{8\pi^2}\ln\frac{M_{\text{heavy}}^2}{\mu^2}} \tag{111}$$

with $b_{\text{heavy}} = \sum_{X\in H} s_{J_X} Q_X^2$ (e.g. $s_0 = \frac{1}{3}$ for a complex scalar, $s_{1/2} = \frac{2}{3}$ for a Weyl fermion) and $\omega_{\text{proj}} = 1$ (App. D).

Mass-matched vs. generic μ. Choosing $\mu = \Lambda_q = M_X$ sets the logarithms to zero (the compact "mass-matched" forms). For generic $\mu \neq M_X$, the explicit $\ln\big(M^2/\mu^2\big)$ terms cancel the heavy pieces of the UV beta functions. E.g. with $\lambda_-(\mu) = \lambda_+(\mu) + \delta\lambda_{\text{thr}}(\mu)$,

$$\mu\frac{d\lambda_-}{d\mu} = \beta_\lambda^{\text{light}}, \qquad \mu\frac{d}{d\mu}\big[\lambda_+ + \delta\lambda_{\text{thr}}\big] = \beta_\lambda^{\text{light}} + \beta_\lambda^{\text{heavy}} - \beta_\lambda^{\text{heavy}} = \beta_\lambda^{\text{light}},$$

since $\mu\,d\,\delta\lambda_{\text{thr}}/d\mu = -(1/16\pi^2)\big[\frac{n_B}{2}a_B^2 - \frac{1}{2}a_s^2\big]$ from (108). Identical cancellations hold for y, $g_{\Phi V}$, and g_Z using (109), (110), and (111).

C.1 Generalities and notation

We denote by u_Φ the unit vector selecting the light scalar direction Φ in the (ρ,ϑ) plane. The cubic heavy–light couplings appearing in the quadratic expansion about the coherent background are

$$\mathcal{L}_{\text{int}} \supset -\frac{1}{2}\,a_s\,\Phi^2 s \;-\; \frac{1}{2}\,a_B\,\Phi^2 B_H^2 \;+\; \delta\mathcal{L}_{\text{mix}}, \qquad a_s = \partial^3_{\Phi\Phi s}V_{\text{eff}}, \quad a_B = \partial^3_{\Phi\Phi B_H^2}\mathcal{L}, \tag{112}$$

where $\delta\mathcal{L}_{\text{mix}}$ collects subleading quadratic mixings fixed by the simplicity kernel (absorbed into M_B). For Yukawa, write the emergent vertex as $y\,\Phi\,\overline{\psi}\psi$ near the threshold, with

$$y(\mu^+) = \zeta_{\text{eff}}\,\frac{\rho_0}{M_*}\,\mathcal{I}_{\text{triad}}, \qquad M_* \sim M_B. \tag{113}$$

For the gauge sector, the background-field kinetic is

$$\mathcal{L} \supset -\frac{1}{4g_{\text{NA}}^2}\,\mathrm{Tr}\,F_{\mu\nu}F^{\mu\nu}, \qquad \frac{1}{g_{\text{NA}}^2} = \frac{\omega_{\text{YM}}}{\tau(\rho_0)} \quad (\omega_{\text{YM}} = 1), \tag{114}$$

and the Cartan projection yields $Z = A^i n_i$ with projector weight ω_{proj}.

C.2 Quartic threshold $\delta\lambda_{\text{thr}}$

At one loop the heavy fields correct the Φ^4 vertex via the standard scalar bubble with two insertions of the cubic vertices in 112. For a real heavy scalar X with interaction $-(1/2)a_X\Phi^2 X$ one finds

$$\Delta\Gamma_X^{(4)}(0) = \frac{a_X^2}{32\pi^2}\,\ln\frac{M_X^2}{\mu^2}\,. \tag{115}$$

Including multiplicities and signs, and matching at $\mu = \Lambda_q$,

$$\boxed{\delta\lambda_{\rm thr} = \frac{1}{16\pi^2}\left[\frac{n_B}{2}\, a_B^2\, \ln\frac{M_B^2}{\Lambda_q^2} - \frac{1}{2}\, a_s^2\, \ln\frac{M_s^2}{\Lambda_q^2}\right] + \Delta_\lambda^{\rm fin},} \tag{116}$$

with $n_B = 3$ for a degenerate adjoint triplet; $\Delta_\lambda^{\rm fin}$ is a scheme-finite constant (set to zero in pure MS⁻).

C.3 Yukawa threshold $\delta y_{\rm thr}$

Heavy loops renormalize the light $\Phi\overline{\psi}\psi$ vertex through wavefunction and proper-vertex diagrams. In the zero-momentum limit the result can be organized as a multiplicative renormalization:

$$\boxed{\delta y_{\rm thr} = \frac{y}{16\pi^2}\left[\alpha_s\, \ln\frac{M_s^2}{\Lambda_q^2} + \alpha_B\, \ln\frac{M_B^2}{\Lambda_q^2}\right] + \Delta_y^{\rm fin},} \tag{117}$$

where, to leading order in the background-field expansion,

$$\alpha_s = -\frac{a_s}{M_s^2} + \mathcal{O}\!\left(\frac{a_s^2}{M_s^4}\right), \qquad \alpha_B = c_B\, \frac{a_B}{M_B^2} + \mathcal{O}\!\left(\frac{a_B^2}{M_B^4}\right), \tag{118}$$

and c_B encodes the adjoint multiplicity and projector contractions ($c_B = 3$ for a degenerate triplet with diagonal coupling). The finite term $\Delta_y^{\rm fin}$ is scheme dependent and set to zero in MS⁻.

C.4 Scalar–vector threshold $\delta g_{\Phi V}$

The light scalar modifies the vector kinetic and/or mass term through the ρ-dependence of $\tau(\rho)$ and the Axis projector. Writing

$$g_{\Phi V} = c_V\, \tau'(\rho_0)\, (u_\Phi\!\cdot\!\hat\rho), \qquad c_V = -\frac{1}{2\,\tau(\rho_0)^2}\, \mathcal{N}_V \cos\chi \quad (\mathcal{N}_V = 1 \text{ on } B_0 = v\Sigma), \tag{119}$$

the one-loop threshold takes the multiplicative form

$$\boxed{\delta g_{\Phi V,\,\rm thr} = \frac{g_{\Phi V}}{16\pi^2}\left[\beta_s\, \ln\frac{M_s^2}{\Lambda_q^2} + \beta_B\, \ln\frac{M_B^2}{\Lambda_q^2}\right] + \Delta_{g_{\Phi V}}^{\rm fin},} \tag{120}$$

with $\beta_s \propto \tau''(\rho_0)$ and β_B fixed by the B quadratic kernel (read off from App. B once a background is chosen).

C.5 Gauge matching for g_Z

Below Λ_q, $Z = A^i n_i$ is the Abelian Cartan projection with projector weight $\omega_{\rm proj}$. Matching the background-field two-point at $\mu = \Lambda_q$ gives

$$\boxed{\frac{1}{g_Z^2(\Lambda_q^-)} = \frac{\omega_{\rm proj}}{g_{\rm NA}^2(\Lambda_q^+)} + \frac{b_{\rm heavy}}{8\pi^2}\, \ln\frac{M_{\rm heavy}^2}{\Lambda_q^2} + \Delta_Z^{\rm fin}.} \tag{121}$$

In the **mass-matched scheme** ($\Lambda_q = M_{\text{heavy}}$) one has $\ln\!\Big(M_{\text{heavy}}^2/\Lambda_q^2\Big) = 0$, hence b_{heavy} drops out. If one keeps a fixed Λ_q, b_{heavy} is the sum of the standard background-field coefficients of the heavy fields charged under $U(1)_Z$:

$$b_{\text{heavy}} = \sum_{X \in \mathcal{H}} s_{J_X}\, Q_X^2, \tag{122}$$

with Q_X the $U(1)_Z$ charge and s_{J_X} the spin-dependent weight (e.g. $s_0 = +\frac{1}{3}$ for a complex scalar, $s_{1/2} = +\frac{2}{3}$ for a Weyl fermion).[18]

Cartan charges and the threshold coefficient b_{heavy}. Below Λ_q the coherence frame n^i defines the Abelian generator $T_n \equiv n^i T_i$, so adjoint fields decompose into $U(1)_Z$ eigenmodes. In the convenient frame $n^i = (0, 0, 1)$,

$$T^{\pm} \equiv \frac{T^1 \pm iT^2}{\sqrt{2}}, \qquad T^0 \equiv T^3, \qquad [T^0, T^{\pm}] = \pm T^{\pm},\ [T^0, T^0] = 0,$$

so any adjoint-valued field $X = X^i T_i$ splits as $X = X^+ T^+ + X^- T^- + X^0 T^0$ with $Q_Z(X^{\pm}) = \pm 1$, $Q_Z(X^0) = 0$ in the trace normalization $\mathrm{Tr}\big(T^i T^j\big) = \delta^{ij}$ (App. A.1), consistent with $\omega_{\text{proj}} = 1$ (App. D.1).

Table 7: Heavy fields below Λ_q: $U(1)_Z$ charges after Cartan projection and their contribution to b_{heavy} (background-field $\overline{\text{MS}}$).

Field	**Spin J**	**$U(1)_Z$ components**	**Charges Q_Z**	**Contribution to b_{heavy}**
s (radial)	0	s	0	0
B_H (non-simple B modes, adjoint)	$0^{\dagger}$	B_H^0, $B_H^{\pm}$	0, ± 1	$2\, s_0$ per charged pair
W_μ (off–Cartan gauge vectors)	1	$W^{\pm}$	± 1	$2\, s_1$

The heavy-set contribution to the Abelian two-point is then

$$b_{\text{heavy}} \;=\; \sum_{X \in H} s_{J_X} \sum_{\text{charged comps } a} Q_{Z,a}^2 \quad \Longrightarrow \quad b_{\text{heavy}} = 2\, s_1\, N_W \;+\; 2\, s_0\, N_{B_H^{\pm}}\ (+\cdots), \tag{123}$$

with N_W and $N_{B_H^{\pm}}$ the numbers of heavy charged pairs ($W^{\pm}$ and $B_H^{\pm}$) retained in H.[†] In the coherent phase the non-simple B fluctuations contribute to the $U(1)_Z$ vacuum polarization as matter modes (effectively scalar-like in the Abelian background-field computation); hence the s_0 weight.

Equation (123) is the bookkeeping version of the matching in Eqs. (121)–(122): the spin weights s_J are the standard background-field coefficients (e.g. $s_0 = \frac{1}{3}$ for a complex scalar, $s_{1/2} = \frac{2}{3}$ for a Weyl fermion; s_1 denotes the massive vector weight in the same scheme), and the charges are those of the IR Cartan basis set by n^i. In the mass-matched choice $\mu = \Lambda_q = M_{\text{heavy}}$ the logarithm in (121) vanishes and b_{heavy} drops out, matching the discussion in App. B.7 and this subsection.

[18] Our minimal heavy set is bosonic and neutral in the **UV basis**; any effective $U(1)_Z$ charge assignment must be made in the **IR Cartan basis** *after* condensation. Using the mass-matched scheme avoids having to track these charges explicitly at one loop.

C.6 Scheme and RG interface

We use pure MS⁻ for the finite parts, $\Delta_{\rm fin} = 0$, and match at $\mu = \Lambda_q$. The matched EFT inputs are

$$\lambda_-(\Lambda_q) = \lambda_{\rm UV} + \delta\lambda_{\rm thr}, \quad y_-(\Lambda_q) = y_{\rm UV}\,(1 + \delta y_{\rm thr}), \quad g_{\Phi V,-}(\Lambda_q) = g_{\Phi V}^{\rm UV}\,(1 + \delta g_{\Phi V,\,\rm thr}),$$

and $1/g^2_{Z,-}(\Lambda_q)$ from (121). These are the initial conditions for the two-loop running in Sec. 4.

D Projector Weight $\omega_{\rm proj}$ and c_V on $B_0 = v\,\Sigma$

We compute the Cartan projector weight $\omega_{\rm proj}$ and the scalar–vector coefficient c_V on the constant simple background $B_0^i = v\,\Sigma^i$ used in App. B, where Σ^i is an orthonormal basis of self-dual two-forms satisfying $\Sigma^i \wedge \Sigma^j = 2\delta^{ij}\,\mathrm{vol}$.

D.1 Cartan projection, $\omega_{\rm proj}$, and normalization audit

Cartan-projector lemma. On the coherent background, integrating out B induces the Yang–Mills kinetic for the non-Abelian connection $A = A^i T_i$,

$$\mathcal{L}_{\rm YM} = -\frac{1}{2g^2_{NA}}\,\mathrm{Tr}\big(F \wedge \star F\big) = -\frac{1}{4g^2_{NA}}\,F^i_{\mu\nu}F^{i\,\mu\nu}, \qquad \mathrm{Tr}(T_iT_j) = \delta_{ij}, \tag{124}$$

(see App. B, Eq. 92, and App. D, Eq. 122). Choose any unit vector $n^i n_i = 1$ in the adjoint and define the Cartan generator $T_n \equiv n^i T_i$, the Abelian potential $Z_\mu \equiv A^i_\mu n_i$, and $Z_{\mu\nu} \equiv \partial_\mu Z_\nu - \partial_\nu Z_\mu$. Decompose $A = A_\parallel + A_\perp$ with $A_\parallel \equiv Z\,T_n$ and $A_\perp \cdot n = 0$. Writing $F = F_\parallel T_n + F^a_\perp T_a$ in an orthonormal adjoint basis, orthogonality $\mathrm{Tr}(T_nT_a) = 0$ gives

$$\mathrm{Tr}(F \wedge \star F) = \mathrm{Tr}\big(F_\parallel \wedge \star F_\parallel\big) + \mathrm{Tr}(F_\perp \wedge \star F_\perp). \tag{125}$$

Since $[T_n, T_n] = 0$, one has $F_\parallel = dA_\parallel + [A_\parallel, A_\parallel] = dZ\,T_n$. Restricting to the Cartan subspace (or, equivalently, reading off the coefficient of the $F^2_\parallel$ term) yields

$$\mathcal{L}_{\rm YM}\big|_{\rm Cartan} = -\frac{1}{2g^2_{NA}}\,\mathrm{Tr}\big(dZ\,T_n \wedge \star dZ\,T_n\big) = -\frac{1}{4g^2_{NA}}\,Z_{\mu\nu}Z^{\mu\nu}, \tag{126}$$

because $\mathrm{Tr}(T_nT_n) = n^in^j\delta_{ij} = 1$. Comparing with the canonical Abelian form $-\frac{1}{4g_Z^2}Z_{\mu\nu}Z^{\mu\nu}$ gives

$$\frac{1}{g_Z^2} = \frac{1}{g^2_{NA}} \qquad \Longleftrightarrow \qquad \omega_{\rm proj} = 1. \tag{127}$$

This equality is independent of the choice of n^i by internal $SO(3)$ rotation invariance of $\mathrm{Tr}\big(F^2\big)$. See also App. B, Eq. 93, and App. D, Eqs. 128–132.[19]

[19] The statement concerns the normalization of the Z kinetic term. Interaction terms mixing Z with off-Cartan modes arise away from the Cartan subspace, but do not change the coefficient of $Z_{\mu\nu}Z^{\mu\nu}$ fixed here.

Setup with a general trace. Let the generator trace be

$$\mathrm{Tr}\big(T^iT^j\big) = \kappa\,\delta^{ij}, \qquad \kappa \in \{1, \tfrac{1}{2}\} \text{ in common conventions.} \tag{128}$$

The background–field kinetic obtained after integrating out B (App. B) is

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{2g_{NA,\kappa}^2}\,\mathrm{Tr}(F\wedge\star F) = -\frac{\kappa}{4g_{NA,\kappa}^2}\,F^i_{\mu\nu}F^{i\mu\nu}, \tag{129}$$

which reproduces Eqs. (92)–(93) of App. B when $\kappa = 1$. Project to the Cartan direction defined by a unit n^i in the adjoint, $Z_\mu \equiv A^i_\mu n_i$, so that $Z_{\mu\nu} = \partial_\mu Z_\nu - \partial_\nu Z_\mu$. Orthogonality of the basis implies

$$\mathcal{L}_Z = -\frac{\kappa}{4g_{NA,\kappa}^2}\,Z_{\mu\nu}Z^{\mu\nu} = -\frac{1}{4g_Z^2}\,Z_{\mu\nu}Z^{\mu\nu}\,, \qquad \Rightarrow \qquad \frac{1}{g_Z^2} = \frac{\kappa}{g_{NA,\kappa}^2}. \tag{130}$$

Comparing with the matching form $1/g_Z^2 = \omega_{\mathrm{proj}}/g_{NA}^2$ (used in §3.5 and Fig. 2) identifies the projector weight in the κ-normalization as $\omega_{\mathrm{proj}} = \kappa$ *if* one insists on keeping $g_{NA,\kappa}$ fixed across conventions.

Convention-independent statement (absorbing κ). Define the component-canonical non-Abelian coupling by

$$g_{NA}^2 \;\equiv\; \frac{g_{NA,\kappa}^2}{\kappa}\,, \qquad \text{i.e.} \qquad \mathcal{L}_{\mathrm{YM}} = -\frac{1}{4g_{NA}^2}\,F^i_{\mu\nu}F^{i\mu\nu}. \tag{131}$$

Then (130) becomes

$$\frac{1}{g_Z^2} = \frac{1}{g_{NA}^2}\,, \qquad \Longleftrightarrow \qquad \omega_{\mathrm{proj}} = 1 \quad \text{(canonical component normalization).} \tag{132}$$

Thus any change of generator normalization is exactly compensated by the coupling redefinition (131): the physical relation is $\omega_{\mathrm{proj}} = 1$ once conventions are fixed. This matches the main-text usage and the simple-background result of App. B (Eqs. (92)–(93)), and is consistent with the normalization notes in App. C.5 (two-loop b_2 rescales with κ while $1/g_Z^2 = 1/g_{NA}^2$ remains invariant).

Case A: $\mathrm{Tr}\big(T^iT^j\big) = \delta^{ij}$ ($\kappa = 1$). Eqs. (129)–(132) give immediately $1/g_Z^2 = 1/g_{NA}^2$, i.e. $\omega_{\mathrm{proj}} = 1$, consistent with Eqs. (129)–(132).

Case B: $\mathrm{Tr}\big(T^iT^j\big) = \frac{1}{2}\,\delta^{ij}$ ($\kappa = \frac{1}{2}$). From (130), we have $1/g_Z^2 = (\frac{1}{2})/g_{NA,1/2}^2$. By defining the component-canonical coupling as

$$g_{NA}^2 \equiv g_{NA,1/2}^2/(\tfrac{1}{2}),$$

this relation gives again $1/g_Z^2 = 1/g_{NA}^2$ and hence $\omega_{\mathrm{proj}} = 1$. This is the same convention choice discussed in App. C.5.

Rotation invariance. Because $\mathrm{Tr}\big(F^2\big)$ is $SO(3)$-invariant in the internal indices, the result is independent of the choice of n^i; only the generator normalization enters, and that dependence is removed by (131). (Interaction terms that mix Z with off-Cartan modes do not alter the $Z_{\mu\nu}Z^{\mu\nu}$ coefficient fixed here.)

D.2 Scalar–vector coefficient c_V

The dependence of the Yang–Mills coupling on ρ is $g_{\rm NA}^{-2}(\rho) = 1/\tau(\rho)$ (App. B). Let the light scalar direction be $\delta\rho = \cos\chi\,\Phi$, $\delta\vartheta = (\sin\chi/f_\vartheta)\,\Phi$. A fluctuation of ρ induces

$$\delta\left(\frac{1}{g_{\rm NA}^2}\right) = -\frac{\tau'(\rho_0)}{\tau(\rho_0)^2}\,\delta\rho = -\frac{\tau'(\rho_0)}{\tau(\rho_0)^2}\,\cos\chi\,\Phi. \tag{133}$$

After Cartan projection (which does not change the normalization at tree level on B_0), the induced scalar–vector interaction in the effective Lagrangian can be parameterized as

$$\mathcal{L} \supset -\frac{1}{4}\,\frac{1}{g_Z^2(\Phi)}\,Z_{\mu\nu}Z^{\mu\nu} \;\Rightarrow\; \mathcal{L} \supset +\frac{1}{4}\,\frac{\partial}{\partial\Phi}\left(\frac{1}{g_Z^2}\right)_{\Phi=0}\Phi\,Z_{\mu\nu}Z^{\mu\nu}. \tag{134}$$

Matching to the EFT definition $m_V^2(\Phi) = g_{\Phi V}^2\langle\Phi\rangle^2$ yields, at leading order,

$$\boxed{c_V = -\frac{\cos\chi}{2\,\tau(\rho_0)^2}\times\mathcal{N}_V, \qquad \mathcal{N}_V = 1 \text{ on } B_0 = v\,\Sigma,} \tag{135}$$

so that

$$\boxed{g_{\Phi V} = c_V\,\tau'(\rho_0) = -\frac{\cos\chi}{2}\,\frac{\tau'(\rho_0)}{\tau(\rho_0)^2}\;.} \tag{136}$$

If Φ is purely phase-like ($\cos\chi = 0$), the leading coupling vanishes as expected; otherwise a small radial admixture generates $g_{\Phi V} \neq 0$.

D.3 Off-Cartan vector masses M_W

Spontaneous alignment along n leaves the Cartan Z massless at tree level while the off-Cartan vectors acquire a mass from the non-Abelian piece,

$$\boxed{M_W^2 = c_W\,g_{\rm NA}^2\,v_B^2, \qquad v_B^2 \propto v^2,\; c_W = \mathcal{O}(1)\;,} \tag{137}$$

where v_B is the geometric scale set by B_0. In the simplest normalization $c_W = 1$; different conventions for the internal trace or the normalization of Σ^i shift c_W by an overall constant that is absorbed into v_B.

E Coefficient Dictionary for Two-Loop RGEs

This appendix defines the counting conventions and collects the explicit group and charge invariants that appear in the two-loop RGEs of Sec. 4. Throughout we employ the background-field $\overline{\rm MS}$ scheme and fix the trace normalization $\mathrm{Tr}(T^iT^j) = \delta^{ij}$. For the Abelian factor $U(1)_Z$, we set $C_2(G) = 0$ and replace $C_2(R)\to Q^2$.

Conventions and signs. We work in Minkowski signature $(+,-,-,-)$ with $F_{\mu\nu} = \partial_\mu Z_\nu - \partial_\nu Z_\mu$. Generators satisfy $\mathrm{Tr}(T^iT^j) = \delta^{ij}$, and for the Abelian sector we identify $C_2(R)\to Q^2$. The Abelian gauge coupling is denoted g_Z, the scalar–vector mass operator uses $g_{\Phi V}$ via $(\Phi^\dagger\Phi)Z_\mu Z^\mu$, and the Yukawa term is defined by $-\,y\,\Phi\,\bar\psi\psi$. Fermions are counted as Weyl and scalars as complex fields in all invariants below.

E.1 Counting conventions (Weyl vs Dirac; complex scalars)

- Fermions are counted as *Weyl* fields in the sums. A single Dirac fermion corresponds to two Weyl fields, left/right, possibly with different charges Q_L, Q_R.
- Scalars are counted as *complex* scalars in the sums. A real scalar of charge Q contributes as one half of a complex scalar.[20]
- Multiplicities (*e.g.* colors, families) are included as integer factors N in the sums below.

E.2 Charge moments and Yukawa invariants

Define the charge moments for fermions and scalars:

$$S_2(F) \equiv \sum_{\psi\in\text{Weyl}} N_\psi\, Q_\psi^2, \qquad S_2(S) \equiv \sum_{\phi\in\text{complex scalars}} N_\phi\, Q_\phi^2, \tag{138}$$

$$S_4(F) \equiv \sum_{\psi\in\text{Weyl}} N_\psi\, Q_\psi^4, \qquad S_4(S) \equiv \sum_{\phi\in\text{complex scalars}} N_\phi\, Q_\phi^4. \tag{139}$$

For Yukawas, writing $y_a\,\Phi\,\overline{\psi}_{R,a}\psi_{L,a}$ (no flavor mixing for clarity), define

$$Y_2 \equiv \sum_a N_a\, y_a^2, \qquad Y_4 \equiv \sum_a N_a\, y_a^4, \qquad Y_2(Q^2) \equiv \sum_a N_a\, y_a^2\,(Q_{L,a}^2 + Q_{R,a}^2 + Q_\Phi^2). \tag{140}$$

The last term is the Abelian specialization of the MV Yukawa–gauge contraction and is the quantity that appears in the two-loop β_{g_Z} (Sec. 4.1).

E.3 Abelian two-loop coefficients

With the conventions above, the Abelian gauge β in Sec. 4.1 reads

$$\beta_{g_Z} = \frac{g_Z^3}{16\pi^2}\, b_1 + \frac{g_Z^5}{(16\pi^2)^2}\, b_2 - \frac{g_Z^3}{(16\pi^2)^2}\, c_Y\, Y_2(Q^2), \qquad b_1 = \frac{4}{3}S_2(F) + \frac{1}{3}S_2(S), \tag{141}$$

where, for a single uncoupled $U(1)$,

$$\boxed{b_2 = 4\,S_4(F) + S_4(S), \qquad c_Y = 2,} \tag{142}$$

in the $\mathrm{Tr}(T^iT^j) = \delta^{ij}$ normalization.[21] The c_Y term encodes the universal two-loop Yukawa screening of Abelian gauge running (MV-II).

E.4 Intra–morton screening of fermionic $U(1)_Z$ moments

Below the morton–locking scale Λ_q, the fermionic $U(1)_Z$ current is reduced by a universal projector factor κ^2, because the observable z–current of a *single morton* (the $(1z, 2x)$ bound tri-vector) is only the projected fraction of its internal structure. Higher fermion composites (e.g. three mortons for an electron) simply sum these already-screened units, so no further reduction occurs. The scalar

[20] In most EFT uses here we only have the complex scalar Φ.

[21] If you use $\mathrm{Tr}(T^iT^j) = \frac{1}{2}\delta^{ij}$, multiply b_2 by 1/2 and keep c_Y unchanged; b_1 is unaffected.

current remains unscreened since Φ is not a morton composite but the coherence carrier itself. In terms of the content invariants used in the Abelian RGEs, implement screening via

$$S_2(F) \to S_2^{\rm eff}(F) = \kappa^2\, S_2(F), \qquad S_4(F) \to S_4^{\rm eff}(F) = \kappa^4\, S_4(F), \tag{143}$$

$$Y_2(Q^2) \to Y_2(Q^2)_{\rm eff} = \sum_a N_a\, y_a^2 \big[\kappa^2 \big(Q_{L,a}^2 + Q_{R,a}^2\big) + Q_\Phi^2\big]\,. \tag{144}$$

The gauge coefficients in Eq. (141) are then evaluated with

$$b_1^{\rm eff} = \tfrac{4}{3}\, S_2^{\rm eff}(F) + \tfrac{1}{3}\, S_2(S), \qquad b_2^{\rm eff} = 4\, S_4^{\rm eff}(F) + S_4(S). \tag{145}$$

For the minimal content (App. E.6) one has $S_2(F) = 42$, $S_2(S) = 1$, $S_4(F) = 42$, $S_4(S) = 1$. With $\kappa^2 = \frac{1}{3}$ this gives

$$b_1^{\rm eff} = 19.0, \qquad b_2^{\rm eff} = \tfrac{59}{3} \approx 19.667.$$

These replacements modify only the invariants; the *form* of the two–loop RGEs is unchanged.

E.5 One-loop scalar/Yukawa coefficients (for reference)

These appear in Eqs. (41) and (42) at one loop (specialized to one complex scalar Φ of charge Q_Φ and diagonal Yukawas):

$$\beta_{y_a}^{(1)} = \frac{y_a}{16\pi^2}\Big[+ \frac{3}{2}\, y_a^2 + \sum_{b\neq a} y_b^2 \; - \; 3\,(Q_{L,a}^2 + Q_{R,a}^2)\, g_Z^2 \; - \; \frac{1}{2}\,\lambda\Big], \tag{146}$$

$$\beta_\lambda^{(1)} = \frac{1}{16\pi^2}\Big[10\,\lambda^2 + 4\,\lambda\,\Sigma_y - 4\,\Sigma_{y^4} \; - \; 4\,\lambda\, Q_\Phi^2\, g_Z^2 + 6\, Q_\Phi^4\, g_Z^4 \Big], \tag{147}$$

where $\Sigma_y \equiv \sum_a y_a^2$, $\Sigma_{y^4} \equiv \sum_a y_a^4$. (Two-loop completions are coded in Sec. 4.1.)

E.6 EFT content and derived invariants (worked example)

Below the matching scale Λ_q the light spectrum used in the scans of Sec. 4.3 consists of one complex scalar Φ with $U(1)_Z$ charge $Q_\phi = 1$ and nine Dirac fermions with vectorlike charge assignments $Q_{L,a} = Q_{R,a} = 1$; quark flavors carry color multiplicity $N = 3$ and leptons $N = 1$. The full roster is listed in Table 8.[22]

From Table 8 the charge moments entering the two-loop coefficients of Eqs. (37)–(40) are

$$S_2(F) = \sum_a N_a\, (Q_{L,a}^2 + Q_{R,a}^2) = 42, \qquad S_2(S) = Q_\phi^2 = 1, \tag{148}$$

$$S_4(F) = \sum_a N_a\, (Q_{L,a}^4 + Q_{R,a}^4) = 42, \qquad S_4(S) = Q_\phi^4 = 1. \tag{149}$$

Evaluating the Yukawa traces at $\mu_{\rm EW}$ with $y_a = \sqrt{2} m_a/v$ ($v = 246.22\,$GeV, using PDG pole masses) gives

$$Y_2 = 2.9558616584, \qquad Y_4 = 2.9084451063, \qquad Y_2(Q^2) = 8.8675849752,$$

as summarized in Table 9. A per-species breakdown of the Y-traces is shown in Table 10. These CSV-sized tables are generated automatically by the companion notebook and shipped with the paper; changing any $U(1)_Z$ charge, multiplicity, or the mass scheme simply regenerates the totals via App. E.2–E.4.

[22] A Dirac fermion counts as two Weyl fields with the same $|Q|$; the definitions in App. E.2 are written with this convention and already include the factor of two.

Table 8: EFT content below Λ_q used to build the Abelian invariants.

Field	Type	N	Q_L	Q_R	Q_ϕ
ϕ	scalar	1	—	—	1
e	Dirac	1	1	1	—
μ	Dirac	1	1	1	—
τ	Dirac	1	1	1	—
u	Dirac	3	1	1	—
d	Dirac	3	1	1	—
s	Dirac	3	1	1	—
c	Dirac	3	1	1	—
b	Dirac	3	1	1	—
t	Dirac	3	1	1	—

Table 9: Derived totals for the worked example.

Quantity	Value
$S_2(F)$	42
$S_2(S)$	1
$S_4(F)$	42
$S_4(S)$	1
Y_2	2.9558616584
Y_4	2.9084451063
$Y_2(Q^2)$	8.8675849752

E.7 Worked example: minimal Axis–EFT

For the same content, the numerical totals used in Sec. 4.1 are those in Table 9. These feed directly into the single-$U(1)$ two-loop coefficients via Eqs. (37)–(40). The notebook also writes a manifest (CSV/JSON) documenting the inputs that produced the tables to ensure full reproducibility.

E.8 Notes on normalizations and cross-checks

Changing the generator normalization $\mathrm{Tr}(T^iT^j) = \frac{1}{2}\delta^{ij}$ rescales b_2 by 1/2 in Eq. (142); b_1 and c_Y remain as written here (Abelian case).

If multiple $U(1)$ factors or kinetic mixing are present, replace β_{g_Z} by the standard matrix RGE (not needed here; see Refs. [34, 33]).

For Dirac fermions with *axial* charges ($Q_L = -Q_R$), keep the Weyl-sum definitions; the formulas above remain valid since they depend only on Q_L^2, Q_R^2.

E.9 One-loop RGE for the scalar–vector operator

Consider the renormalizable dimension-4 operator

$$\mathcal{O}_{\Phi V} \equiv \tfrac{1}{2}\,\Phi^\dagger\Phi\, Z_\mu Z^\mu, \qquad \mathcal{L} \supset \tfrac{1}{2}\, g_{\Phi V}(\mu)\, \mathcal{O}_{\Phi V}.$$

Two common parameterizations are useful:

Table 10: Per-species Yukawa contributions at $\mu_{\rm EW}$ (worked example).

name	N	Q_L	Q_R	y	contrib_Y_2	contrib_Y_4	contrib_$Y_2(Q^2)$
e	1	1	1	2.94×10^{-6}	8.6×10^{-12}	7.4×10^{-23}	2.6×10^{-11}
μ	1	1	1	6.07×10^{-4}	3.7×10^{-7}	1.4×10^{-13}	1.1×10^{-6}
τ	1	1	1	1.02×10^{-2}	1.0×10^{-4}	1.1×10^{-8}	3.1×10^{-4}
u	3	1	1	1.26×10^{-5}	4.8×10^{-10}	7.6×10^{-20}	1.4×10^{-9}
d	3	1	1	2.70×10^{-5}	2.2×10^{-9}	1.6×10^{-18}	6.6×10^{-9}
s	3	1	1	5.51×10^{-4}	9.1×10^{-7}	2.8×10^{-13}	2.7×10^{-6}
c	3	1	1	7.29×10^{-3}	1.6×10^{-4}	8.5×10^{-9}	4.8×10^{-4}
b	3	1	1	2.40×10^{-2}	1.7×10^{-3}	1.0×10^{-6}	5.2×10^{-3}
t	3	1	1	9.92×10^{-1}	2.95	2.91	8.86

(i) Independent contact operator. If $g_{\Phi V}$ is treated as an *independent* contact coupling (distinct from minimal coupling), its one-loop running is purely homogeneous in terms of field anomalous dimensions:

$$\boxed{\beta^{(1)}_{g_{\Phi V}} = g_{\Phi V}\left(2\,\gamma^{(1)}_{\Phi} + \gamma^{(1)}_{Z}\right),} \tag{150}$$

where $\gamma^{(1)}_{\Phi}$ and $\gamma^{(1)}_{Z}$ are the standard one-loop anomalous dimensions (MV-I) specialized to $U(1)_Z$ with charges and multiplicities fixed by App. E. This form is basis-independent and suffices if $g_{\Phi V}$ originates entirely from UV matching (App. C).

(ii) Minimal-coupling identification. If instead one *identifies* the tree-level piece with minimal coupling,

$$g^{\rm min}_{\Phi V} \equiv 2\,Q_\Phi^2\,g_Z^2,$$

then the one-loop RGE acquires an explicit additive term from the running of g_Z:

$$\boxed{\beta^{(1)}_{g_{\Phi V}} = 4\,Q_\Phi^2\,g_Z\,\beta^{(1)}_{g_Z} \;+\; g^{\rm ct}_{\Phi V}\left(2\,\gamma^{(1)}_{\Phi} + \gamma^{(1)}_{Z}\right), \qquad \beta^{(1)}_{g_Z} = \frac{g_Z^3}{16\pi^2}\,b_1,} \tag{151}$$

where $g_{\Phi V} = g^{\rm min}_{\Phi V} + g^{\rm ct}_{\Phi V}$ separates the minimal piece from any UV contact contribution $g^{\rm ct}_{\Phi V}$, and $b_1 = \frac{4}{3}S_2(F) + \frac{1}{3}S_2(S)$ (Eq. (141)). Equivalently,

$$\boxed{\beta^{(1)}_{g_{\Phi V}} = \frac{Q_\Phi^2\,b_1}{4\pi^2}\,g_Z^4 \;+\; g^{\rm ct}_{\Phi V}\left(2\,\gamma^{(1)}_{\Phi} + \gamma^{(1)}_{Z}\right).} \tag{152}$$

This explicit term makes $\Delta^{(1)}_V$ in Eq. (43) concrete whenever the operator is regarded as descending from minimal coupling.

E.10 Run manifest and reproducibility

This section documents the exact inputs used to produce the numbers in Tabs. 3 and 5. The EFT content and invariants are as in App. E.1-E.7; see in particular the worked example and per-species Yukawa tables that feed the two-loop coefficients used in Sec. 4.1.
The companion notebook writes CSV files with the per-species Y-trace contributions (E.2–E.8) and a machine-readable manifest of all inputs.

Table 11: Calibration seeds at Λ_q^- and solver settings.

$g_Z(\Lambda_q^-)$	3.000000
$\lambda(\Lambda_q^-)$	0.800000
Integrator	Adaptive RK45 (positivity checks on λ, finiteness of g_Z)
Matching	Mass-matched at $\mu = \Lambda_q$ with thresholds from App. C

E.11 Printed two-loop coefficients and benchmark numbers

This subsection prints the explicit two-loop constants used in the runs, specialized to the minimal Axis–EFT content of App. E.6 (Table 8): one complex scalar Φ of charge $Q_\Phi = 1$ and nine Dirac fermions with $Q_{L,a} = Q_{R,a} = 1$ (color multiplicities as listed there). The charge-moment totals and Yukawa traces are those in Table 9; per-species contributions are listed in Table 10.

Abelian gauge β_{g_Z}. For a single, uncoupled $U(1)_Z$ in the $\mathrm{Tr}\big(T^iT^j\big) = \delta^{ij}$ convention, the two-loop Abelian running used in Sec. 4.1 reads

$$\beta_{g_Z} = \frac{g_Z^3}{16\pi^2}\, b_1 + \frac{g_Z^5}{(16\pi^2)^2}\, b_2 - \frac{g_Z^3}{(16\pi^2)^2}\, c_Y\, Y_2(Q^2), \tag{153}$$

with $b_1 = \frac{4}{3}S_2(F) + \frac{1}{3}S_2(S),\ b_2 = 4S_4(F) + S_4(S),\ c_Y = 2.$

Table 12: Two-loop constants for β_{g_Z} (minimal Axis–EFT content).

Quantity	Definition	Value from App. E.5	Numerical
$S_2(F)$	$\sum_\psi N_\psi Q_\psi^2$	42	42
$S_2(S)$	$\sum_\phi N_\phi Q_\phi^2$	1	1
$S_4(F)$	$\sum_\psi N_\psi Q_\psi^4$	42	42
$S_4(S)$	$\sum_\phi N_\phi Q_\phi^4$	1	1
$Y_2(Q^2)$	$\sum_a N_a\, y_a^2\,(Q_{L,a}^2 + Q_{R,a}^2 + Q_\Phi^2)$	8.8675849752	8.8675849752
b_1	$\frac{4}{3}S_2(F) + \frac{1}{3}S_2(S)$	—	**56.3333333333**
b_2	$4S_4(F) + S_4(S)$	—	**169**
c_Y	2	—	**2**

Scalar–Yukawa subsector (β_λ, β_y). At one loop (specialized to one complex scalar and diagonal Yukawas),

$$\beta_{y_a}^{(1)} = \frac{y_a}{16\pi^2}\left(\frac{3}{2}y_a^2 + \sum_{b\neq a} y_b^2 - 3(Q_{L,a}^2 + Q_{R,a}^2)\, g_Z^2 - \frac{1}{2}\lambda\right), \tag{154}$$

$$\beta_{\lambda}^{(1)} = \frac{1}{16\pi^2}\left(10\lambda^2 + 4\lambda\sum_b y_b^2 - 4\sum_b y_b^4 - 4Q_\Phi^2 g_Z^2\lambda + 6Q_\Phi^4 g_Z^4\right). \tag{155}$$

For the two-loop completions used in our runs (scalar–Yukawa subsector, same normalization),

$$(16\pi^2)^2\,\beta_\lambda^{(2)} = -312\,\lambda^3 + 192\,\lambda\sum_b y_b^4 - 768\sum_b y_b^6, \tag{156}$$

$$(16\pi^2)^2\,\beta_{y_a}^{(2)} = y_a\left(-24\,y_a^4 + 12\,\lambda\,y_a^2\right). \tag{157}$$

Table 13: Two-loop RGE coefficients used in the numerical runs (notation as in App. E).

Symbol	**Definition**	**Value**	**Source Eq.**
b_1	one-loop Abelian coefficient	19.0	Eq. (132)
b_2	two-loop Abelian coefficient	59/3	Eq. (133)
c_Y	Yukawa contribution to β_{g_Z}	C	Eq. (134)
$\Xi_\lambda^{(2)}$	two-loop scalar contribution in β_λ	symbolic form	Eq. (136)

Gauge–two–loop structures (e.g. $g_Z^2\lambda^2$, $g_Z^4\lambda$, $g_Z^2y^4$, g_Z^6) are suppressed numerically for the benchmarks used here and were not needed to reproduce Fig. 6; they can be reinstated directly from the MV dictionaries if desired.

We integrate the coupled two-loop RGEs for $\{g_Z, \lambda, \{y_a\}\}$ using the printed constants and the worked-example Yukawa inputs. Seeds at Λ_q may be taken from App. E.10 Table 11.

Normalization reminder. If you use $\mathrm{Tr}\big(T^iT^j\big) = \frac{1}{2}\delta^{ij}$ instead, multiply b_2 by 1/2; b_1 and c_Y are unchanged (Abelian case). The Yukawa tables and $Y_2(Q^2)$ are unchanged by this trace convention.

Completeness note (full two-loop forms). For completeness, the full two-loop renormalization-group expressions for general renormalizable gauge theories can be found in the Machacek–Vaughn trilogy [36, 37, 38]. Specializing their formulas to a single, uncoupled $U(1)_Z$ in our normalization ($\mathrm{Tr}\,T^iT^j = \delta^{ij}$) amounts to replacing group factors by the invariants listed in App. E (e.g. $C_2(F)\to Q^2$, traces $\to S_2, S_4, Y_2(Q^2)$, etc.). We retain the full two-loop Abelian running of g_Z (with b_1, b_2, c_Y printed above) and, for the scalar–Yukawa sector, we use the two-loop terms shown in Sec. E.11. The additional gauge-dependent two-loop structures in β_λ and β_y (terms $\propto g_Z^2\lambda^2$, $g_Z^4\lambda$, $g_Z^2y^4$, g_Z^6, etc.) are numerically subdominant for our benchmarks and are omitted in the minimal integrator to keep the presentation compact. They can be reinstated directly from [36, 37, 38] by substituting our invariants; including them shifts the $\lambda(\mu)$ curves at the sub-percent level over the lever arm used in Fig. 4.

F Collective Electroweak Doublet (Background Note for Scenario A)

Scenario A treats $g_L(\mu_{\rm EW})$ as an external input (see Sec 1.2 for scope). This appendix records a standard collective–coordinate reduction showing how a composite sector can furnish a slow SU(2) orientation mode with stiffness Z_χ, so that gauging reproduces the usual mass–matrix normalization. This background note does not alter any Scenario A assumption or result.

Collective orientation and effective stiffness. In the scalar–coherent phase, the relevant composite sector admits slow internal SU(2) rotations; denote the linearized coordinate by a doublet $\chi(x)$ around a fixed orientation. Integrating out gapped internal modes of mass $M_{\rm gap}$ yields, to quadratic order,

$$\mathcal{L}_{\rm eff} \;=\; \frac{Z_\chi}{2}\,(D_\mu\chi)^\dagger(D^\mu\chi) \;+\; \cdots\,, \qquad Z_\chi \;=\; \sum_\alpha \frac{|\langle 0|J^a|\alpha\rangle|^2}{M_\alpha^2},$$

where J^a are internal SU(2) generators, the sum runs over gapped excitations $|\alpha\rangle$, and D_μ becomes the covariant derivative when the slow mode is gauged by $SU(2)_L \times U(1)_Y$. The spectral form for Z_χ is the usual Källén–Lehmann–type expression for a conserved current.

Mass matrix (unitary gauge). Choosing $\langle\chi\rangle = \frac{v}{\sqrt{2}}(0,1)^\top$ and expanding the covariant kinetic term gives

$$\mathcal{L}_{\rm mass} = \frac{Z_\chi v^2}{8}\Big[g^2\,(W^1_\mu W^{1\mu} + W^2_\mu W^{2\mu}) + (gW^3_\mu - g' B_\mu)^2\Big],$$

i.e. the standard electroweak mass matrix with the identification $v_{\rm eff}^2 \equiv Z_\chi v^2$. In Scenario A this serves only as a normalization background; our phenomenology in §5 uses neutral-gauge mixing at $\mu_{\rm EW}$ and treats g_L as input.

G Optical theorem / positivity check at one loop

We verify perturbative unitarity beyond the positive-Hessian check by computing a sample one-loop $2\to 2$ amplitude and applying the optical theorem. We work in background-field R_ξ gauge; by the background Ward identity the transverse part of the gauge two-point is gauge-parameter independent, and for an Abelian Cartan Z_μ the ghost decouples.

Setup and channel. Consider elastic scattering of a Dirac fermion carrying $U(1)_Z$ charge Q_ψ,

$$\psi(p_1)\,\bar\psi(p_2) \;\to\; \psi(p_3)\,\bar\psi(p_4),$$

mediated by the s-channel Z. The relevant interactions are $\mathcal{L} \supset -\frac{1}{4}Z_{\mu\nu}Z^{\mu\nu} + \bar\psi(i\partial\!\!\!/ - m_\psi)\psi - g_Z Q_\psi\, Z_\mu\, \bar\psi\gamma^\mu\psi$, with projector normalization $\omega_{\rm proj} = 1$ (App. D.1).

Tree amplitude and forward limit. With $q = p_1 + p_2$ and $s \equiv q^2 > 0$, the tree amplitude is

$$\mathcal{A}^{(0)}(s,t) \;=\; (-i)\,\big(g_Z Q_\psi\big)^2\, J^{(\psi)}_\mu\, D_0^{\mu\nu}(q)\, J^{(\psi)}_\nu\,, \qquad J^{(\psi)}_\mu \equiv \bar v(p_2)\gamma_\mu u(p_1), \tag{158}$$

with the background-field propagator $D_0^{\mu\nu}(q) = \frac{-i}{s}\,P_T^{\mu\nu} - i\,\xi\,\frac{q^\mu q^\nu}{s^2}$ and $P_T^{\mu\nu} \equiv g^{\mu\nu} - \frac{q^\mu q^\nu}{s}$. Current conservation $q^\mu J^{(\psi)}_\mu = 0$ kills the longitudinal piece, so only P_T contributes.

One-loop correction (vacuum polarization insertion). At one loop the Z propagator receives the transverse self-energy

$$i\Pi^{\mu\nu}(q) \;=\; i\big(q^\mu q^\nu - g^{\mu\nu}q^2\big)\,\Pi(s)\,, \qquad D^{\mu\nu}(q) = \frac{-i\,P_T^{\mu\nu}}{s\big(1-\Pi(s)\big)} \;+\; (\text{longitudinal}). \tag{159}$$

For a Dirac fermion of mass m_f and $U(1)_Z$ charge Q_f running in the loop, the (dimensionless) spectral function is

$$\boxed{\operatorname{Im}\Pi_f(s) \;=\; \frac{(g_Z Q_f)^2}{12\pi}\;\beta_f(s)\Big(1 + 2\,\frac{m_f^2}{s}\Big)\;\theta\big(s - 4m_f^2\big)\,, \qquad \beta_f(s) \equiv \sqrt{1 - \frac{4m_f^2}{s}}\;\geq 0} \tag{160}$$

which is manifestly nonnegative. (For a complex scalar of mass m_s and charge Q_s, $\operatorname{Im}\Pi_s(s) = \frac{(g_Z Q_s)^2}{48\pi}\,\beta_s^3(s)\,\theta(s - 4m_s^2) \geq 0$.)

Imaginary part of the forward amplitude. Expanding $D^{\mu\nu}$ to first order in Π and taking the forward limit ($p_3 = p_1,\ p_4 = p_2$), the one-loop correction is

$$\mathcal{A}^{(1)}_{\rm fwd}(s) \;=\; (-i)\,\big(g_Z Q_\psi\big)^2\, J^{(\psi)}_\mu\, \frac{-i\,P_T^{\mu\nu}}{s}\,\Pi(s)\,\frac{-i\,P_{T\,\nu\rho}}{s}\,J^{(\psi)\rho}. \tag{161}$$

Since $P_T^2 = P_T$ and $J\cdot q = 0$, this reduces to

$$2\,\mathrm{Im}\,\mathcal{A}^{(1)}_{\rm fwd}(s) \;=\; \frac{2}{s^2}\,\big(g_Z Q_\psi\big)^2\,\big(J^{(\psi)}_T\big)^2\,\mathrm{Im}\,\Pi(s)\,, \qquad \big(J^{(\psi)}_T\big)^2 \equiv J^{(\psi)}_\mu P_T^{\mu\nu} J^{(\psi)}_\nu \;\geq 0. \tag{162}$$

Thus the sign of the imaginary part is controlled by $\mathrm{Im}\,\Pi(s) \geq 0$.

Cutkosky cut and the optical theorem. The same imaginary part is obtained by cutting the fermion bubble in (161). The cut replaces the loop by a sum over on-shell intermediate states $n = \{f\bar{f}\}$ with two-body phase space $d\Phi_2$, yielding

$$2\,\mathrm{Im}\,\mathcal{A}^{(1)}_{\rm fwd}(s) = \sum_f \int d\Phi_2(q\to k\bar{k})\,\Big|\mathcal{M}\big(\psi\bar{\psi}\to f\bar{f}\big)\Big|^2 \;\geq 0, \tag{163}$$

$$\mathcal{M}\big(\psi\bar{\psi}\to f\bar{f}\big) = \frac{-i}{s}\,(g_Z Q_\psi)\,J^{(\psi)}_\mu\,(g_Z Q_f)\,J^{(f)\mu},$$

where the spin sum and angular integral produce the standard tensor $\int d\Phi_2 \sum_{spins} J^{(f)}_\mu J^{(f)}_\nu = \frac{s}{12\pi}\,\beta_f(s)\big(1 + 2m_f^2/s\big)\,P_{T\,\mu\nu}$, which is exactly $\mathrm{Im}\,\Pi_f(s)\,P_{T\,\mu\nu}$ from (160). Equations (162) and (163) establish

$$\mathrm{Im}\,\mathcal{A}_{\rm fwd}(s) \;=\; \tfrac{1}{2}\sum_n \int d\Phi_n\,\big|\mathcal{M}_{\psi\bar{\psi}\to n}\big|^2 \;\geq\; 0,$$

i.e. the optical theorem at one loop for this channel.

Remarks. (i) Only the transverse propagator contributes, so the result is independent of the gauge parameter ξ. (ii) The same reasoning applies if the bubble is a heavy charged scalar $B_H^\pm$ or a heavy vector $W^\pm$; their spectral densities are positive (e.g. $\mathrm{Im}\,\Pi_s \geq 0$ as above), so the right-hand side of (163) remains nonnegative. (iii) In the Abelian Cartan sector the Faddeev–Popov ghost does not couple to Z, consistent with the background-field ghost kernel in App. A.4.

H Scalar–coherence breakdown and an estimate of Λ_Φ

Criterion. Let $\beta_\Phi(\mu)$ be the coherent-projection Jacobian for the scalar sector. We parameterize its RG deformation by

$$\frac{d\ln\beta_\Phi}{d\ln\mu} \equiv \gamma_\Pi(\mu), \qquad \beta_\Phi(\mu) = \beta_\Phi(\mu_0)\exp\Big[\int_{\mu_0}^{\mu} d\ln\mu'\,\gamma_\Pi(\mu')\Big].$$

We define the coherence breakdown scale by the integral criterion

$$\boxed{\int_{\mu_0}^{\Lambda_\Phi} d\ln\mu\;\gamma_\Pi(\mu) \;\simeq\; \varepsilon_{\rm coh}\,, \qquad \varepsilon_{\rm coh} \in [0.3, 0.5]} \tag{164}$$

i.e. when the accumulated projector deformation is $\mathcal{O}(1)$.

One-loop structure. In background-field $\overline{\rm MS}$,

$$\boxed{\gamma_\Pi(\mu) = \frac{1}{16\pi^2}\Big[c_Z\, b_1\, g_Z^2(\mu) \;-\; c_Y\, Y_2(Q^2;\mu) \;+\; c_\lambda\, \lambda(\mu)\Big]}\,, \tag{165}$$

where $b_1 = \frac{4}{3}S_2(F) + \frac{1}{3}S_2(S)$ and $Y_2(Q^2)$ is the Yukawa–charge trace (definitions and numerical content in App. E). Gauge interactions increase γ_Π; Yukawas screen it.

Baseline (Abelian-dominated) estimate. Approximating $\gamma_\Pi(\mu) \approx \bar\gamma_\Pi = \gamma_\Pi(\mu_0)$ over the short lever arm and keeping only the gauge term,

$$\ln\frac{\Lambda_\Phi}{\mu_0} \simeq \frac{\varepsilon_{\rm coh}}{\bar\gamma_\Pi}\,, \qquad \bar\gamma_\Pi^{(\rm gauge)} = \frac{c_Z\, b_1\, g_Z^2(\mu_0)}{16\pi^2}\,. \tag{166}$$

With $\mu_0 = \mu_{\rm EW}$, $g_Z(\mu_{\rm EW}) \simeq 0.354$, $b_1 \simeq 56.33$ (App. E), and $\varepsilon_{\rm coh} = 0.30$–$0.35$, one finds

$$\boxed{\Lambda_\Phi \;\simeq\; (0.8\text{–}1.2)\times 10^5 \ \text{GeV}}$$

which we adopt as the fiducial value used in the phenomenology.

Effect of Yukawa screening (qualitative only). The $-\,c_Y\, Y_2(Q^2)$ term reduces γ_Π and therefore pushes Λ_Φ to higher values. If $\bar\gamma_\Pi(\mu_0) \le 0$ near the reference scale, the local constant-coupling shortcut (166) is inapplicable (it would have no solution with $\Lambda_\Phi > \mu_0$). The correct treatment integrates the full running—$g_Z(\mu)$ grows while $Y_2(Q^2;\mu)$ typically decreases—so the integral in (164) becomes positive at some higher scale. To keep the estimate robust and reproducible without committing to a detailed two-loop fit here, we use the baseline (H); Yukawa screening can only raise Λ_Φ relative to that number.

Reproducibility. A reader can reproduce (H) by (i) taking b_1 and $g_Z(\mu_{\rm EW})$ from App. E, (ii) computing $\bar\gamma_\Pi^{(\rm gauge)}$ via (165), and (iii) inserting into (166) with $\varepsilon_{\rm coh} = 0.30$–$0.35$. Using the running couplings (App. E) modifies Λ_Φ only logarithmically.

I Quantitative definition of Λ_q (from $m^2_{\rm eff} = 0$)

Starting from Eq. (10), the three–form/axion sector after integrating out H gives

$$V_{3\rm F}(\rho,\vartheta) \;=\; \frac{(q + f\,\vartheta)^2}{2\,\eta(\rho)}\,, \qquad S \;\equiv\; q + f\,\vartheta \;\; \text{(piecewise constant in each flux domain).}$$

Expanding about the coherent background $\rho = \rho_0$ with $\delta\rho \equiv \rho - \rho_0$ and $\eta(\rho) = \eta_0 + \eta_1\,\delta\rho + \frac{1}{2}\,\eta_2\,\delta\rho^{\,2} + \cdots$ yields a contribution to the radial curvature

$$\Delta m_\eta^2 \;=\; \frac{\partial^2 V_{3\rm F}}{\partial\rho^2}\bigg|_{\rho_0} = -\,\frac{S^2}{2}\left(\frac{\eta_2}{\eta_0^2} - \frac{2\,\eta_1^2}{\eta_0^3}\right).$$

In the coherent frame we use throughout, $\eta_1|_{\rho_0} = 0$ (the background choice is an extremum of η), hence

$$\Delta m_\eta^2 \;=\; -\,\frac{\eta_2}{2\,\eta_0^2}\,S^2\,, \qquad S \equiv q + f\,\vartheta \quad \text{(constant within a domain).} \tag{167}$$

The effective radial mass squared is then

$$m_{\text{eff}}^2(\mu) \;=\; m_\rho^2(\mu) \;+\; \Delta m_\eta^2 \;+\; \Delta m_{\text{bg}}^2(\mu)\,, \tag{168}$$

where $m_\rho^2(\mu)$ encodes the RG–running curvature of the ρ–potential, and $\Delta m_{\text{bg}}^2(\mu)$ collects background contributions (e.g. from the $B_0 = v\,\Sigma$ Yang–Mills kernel, schematically $\propto \tau''(\rho_0)\langle F^2\rangle_{B_0}$). *Definition:* we define the condensation scale Λ_q by the zero–crossing

$$m_{\text{eff}}^2(\Lambda_q) = 0, \tag{169}$$

which is the appropriate perturbative proxy unless the full nonperturbative dynamics are solved.

I.1 Closed form with a logarithmic m_ρ^2 slope.

For a short lever arm it is convenient to linearize the RG evolution, $m_\rho^2(\mu) \simeq m_\rho^2(\mu_0) + \beta_{m^2}\ln(\mu/\mu_0)$, and keep Δm_η^2 and Δm_{bg}^2 scale–independent at leading order. Solving (169) gives

$$\boxed{\ln\frac{\Lambda_q}{\mu_0} \;=\; \frac{\frac{\eta_2}{2\,\eta_0^2}\,S^2 - m_\rho^2(\mu_0) - \Delta m_{\text{bg}}^2}{\beta_{m^2}}\,, \qquad S \equiv q + f\,\vartheta.} \tag{170}$$

I.2 Worked numerical example (yields $\Lambda_q \sim 10^{16}$ GeV)

As a benchmark consistent with the coherent background and our normalization conventions, take

$$\eta_0 = 1.0, \quad \eta_2 = (1.0\times 10^{-32})\ \text{GeV}^{-2}, \quad f = 1.0\times 10^{32}\ \text{GeV}^2, \quad q = 0, \quad \vartheta_0 = 0.30,$$

so that $S = f\,\vartheta_0 = 3.0\times 10^{31}$ GeV^2 and $\Delta m_\eta^2 = -(\eta_2/2\eta_0^2)\,S^2 = -4.5\times 10^{30}$ GeV^2. Choose $\mu_0 = 1\,\text{TeV}$, $m_\rho^2(\mu_0) = 1.5\times 10^{30}$ GeV^2, $\Delta m_{\text{bg}}^2 = 0$ (for simplicity), and a net one–loop slope $\beta_{m^2} = 1.0\times 10^{29}$ GeV^2 from the scalar RGE (aggregated gauge/Yukawa terms). Equation (170) gives

$$\ln\frac{\Lambda_q}{10^3\ \text{GeV}} = \frac{(4.5-1.5)\times 10^{30}}{1.0\times 10^{29}} = 30 \quad\Rightarrow\quad \boxed{\Lambda_q \;\simeq\; e^{30}\times 10^3\ \text{GeV} \;\approx\; 1.1\times 10^{16}\ \text{GeV}}.$$

I.3 Uncertainties and sensitivity.

To display parametric sensitivity, vary $\{\eta_2, f, \vartheta_0, m_\rho^2(\mu_0), \beta_{m^2}\}$ by the indicated fractional amounts and propagate through (170). A representative table is:

The point is definitional: *unless* one solves the nonperturbative dynamics of the condensate, Λ_q is **defined** by the zero of $m_{\text{eff}}^2(\mu)$ in (169). Changing the benchmark within the ranges above moves Λ_q by factors of order unity, but the scale remains $\sim 10^{16}$ GeV for natural choices compatible with the coherent background.

J Finite-Temperature Effective Potential and Nucleation Dynamics

J.1 High-temperature effective potential.

At $T \gg$ mass scales, the daisy-resummed high-T expansion for the radial mode ρ takes the standard form

$$V_{\text{eff}}(\rho;T) \;\simeq\; D\,(T^2 - T_0^2)\,\rho^2 \;-\; E\,T\,\rho^3 \;+\; \frac{\lambda_T}{4}\,\rho^4, \tag{171}$$

Table 14: Benchmark parameters defining the condensation scale Λ_q. The zero of $m^2_{\rm eff}(\mu)$ [Eq. (170)] occurs at $\Lambda_q \simeq 1.1 \times 10^{16}$ GeV.

Parameter	Central value	Variation	Comment
η_0	1.0	$\pm 5\%$	trace norm; enters as $1/\eta_0^2$
η_2	1.0×10^{-32} GeV^{-2}	$\pm 10\%$	curvature of $\eta(\rho)$ at ρ_0
f	1.0×10^{32} GeV^2	$\pm 10\%$	axion–three–form mixing
ϑ_0	0.30	± 0.03	homogeneous background angle
$m^2_\rho(\mu_0)$	1.5×10^{30} GeV^2	$\pm 10\%$	radial curvature at $\mu_0 = 1$ TeV
β_{m^2}	1.0×10^{29} GeV^2	$\pm 10\%$	net one–loop slope
$\Delta m^2_{\rm bg}$	0	(set)	can be added from B_0 if desired
Λ_q	$\mathbf{1.1 \times 10^{16}}$ GeV	$^{\times 2.0}_{\boldsymbol{\nabla}\cdot 1.7}$	from (170)

where $D > 0$ encodes thermal masses, $E \geq 0$ arises from bosonic modes with ρ-dependent masses (e.g. gauge vectors and heavy ρ-coupled excitations), and $\lambda_T > 0$ is the quartic at temperature T. Equation 171 is the same potential used in Sec. 3.3, Eq. 11.

J.2 First-order condition and critical temperature.

A barrier requires $E \neq 0$ and exists provided $E^2 < D\,\lambda_T$. Degeneracy at T_c with a broken minimum at $\rho = \rho_c$

$$V'_{\rm eff}(\rho_c; T_c) = 0, \qquad V_{\rm eff}(\rho_c; T_c) = V_{\rm eff}(0; T_c) = 0,$$

yields

$$\frac{\rho_c}{T_c} = \frac{2E}{\lambda_T}, \qquad T_c^2 = \frac{T_0^2}{1 - \frac{E^2}{D\,\lambda_T}}. \tag{172}$$

These coincide with Eq. 12 in the main text.

J.3 Surface tension, latent heat, and bounce action.

At $T = T_c$, the surface tension of the critical bubble wall is

$$\sigma(T_c) = \int_0^{\rho_c} d\rho\, \sqrt{2\, V_{\rm eff}(\rho; T_c)} = \frac{4\, E^3\, T_c^3}{3\sqrt{2}\, \lambda_T^{5/2}}, \tag{173}$$

and the latent heat is

$$L \;\equiv\; -T_c\, \frac{\partial\, \Delta V}{\partial T}\bigg|_{T_c} = \frac{8\, E^2\, T_c^4}{\lambda_T^3}\Big(D\, \lambda_T - E^2\Big), \tag{174}$$

as quoted in Eq. 13. Near T_c one may approximate $\Delta V(T) \equiv V_{\rm eff}(0; T) - V_{\rm eff}(\rho_{\rm min}(T); T) \simeq \frac{L}{T_c}(T_c - T)$. In the thin-wall limit the $O(3)$-symmetric bounce action is

$$\frac{S_3(T)}{T} \;\simeq\; \frac{16\pi}{3}\, \frac{\sigma(T_c)^3}{T\, \Delta V(T)^2}, \tag{175}$$

matching Eq. 14. Bubble nucleation completes when $S_3(T_n)/T_n \approx 140$.

J.4 GW parameters.

The conventional parameters controlling a stochastic GW background from the transition are

$$\alpha \;\equiv\; \frac{L}{\rho_{\rm rad}(T_c)} \;=\; \frac{30}{\pi^2 g_*}\,\frac{L}{T_c^4}, \qquad \frac{\beta}{H} \;\equiv\; T\,\frac{d}{dT}\left(\frac{S_3}{T}\right)\bigg|_{T_n}. \tag{176}$$

In the benchmark used in Sec. 3.3 (small α, large β/H) the predicted GW signal is very weak and not observable.

Scope/validity. Eqs. 171–176 assume small supercooling (thin-wall), $E \neq 0$, and daisy resummation absorbed into D, E, λ_T. They are sufficient for the consistency and parametric estimates used in Sec. 3.3; a full numerical treatment is not required for our conclusions.

K Variational motivation for the simplicity constraint

Setup. Define the simplicity tensor

$$C_{ij} \equiv B_i \wedge B_j - \frac{1}{3}\delta_{ij} B_k \wedge B^k, \qquad C_{ij} = C_{ji}, \quad \delta^{ij} C_{ij} = 0,$$

so the constraint $C_{ij} = 0$ enforces that B^i are simple self-dual bivectors (Eq. 58). On the coherent background, the Urbantke metric $g_{\mu\nu}(B)$ and Hodge operator $\star_B$ are defined and their linearized variation obeys

$$\delta(\star_B X)\big|_{B_0} \equiv (\delta\star_B) X = K_B[B_0]\cdot X, \qquad \mathrm{Tr}\big[B_0 \wedge (\delta\star_B) B_0\big] = \mathrm{Tr}\big[\delta B \wedge K_B[B_0]\big],$$

(see App. A.3)

Free-energy functional. Consider the gauge-invariant functional

$$\mathcal{F}_\lambda[B;A] \equiv \int \mathrm{Tr}(B\wedge F) \;+\; \frac{\lambda}{4}\int C_{ij}\wedge \star_B C^{ij}, \qquad \lambda > 0.$$

The penalty is nonnegative and vanishes iff $C_{ij} = 0$. It is well-defined in the coherent phase where $\star_B$ exists (Sec. 3.2) and uses only adjoint traces and δ_{ij}.

Minimization. At fixed topological sector $N \equiv (8\pi^2)^{-1}\!\int \mathrm{Tr}(F\wedge F)$ and fixed volume density vol_B, minimization in B yields the stationarity conditions

$$\delta\mathcal{F}_\lambda = \int \mathrm{Tr}\big[\delta B \wedge (DF + \cdots)\big] \;+\; \frac{\lambda}{2}\int \delta B^i \wedge \big(\star_B\, C_{ij}\wedge B^j\big) + \frac{\lambda}{4}\int C_{ij} \wedge (\delta\star_B) C^{ij} = 0,$$

where $(\delta\star_B)$ is handled with App. A.3 identities. The Euler–Lagrange solution that globally minimizes $\mathcal{F}_\lambda$ for any $\lambda > 0$ is

$$C_{ij} = 0 \quad\Longleftrightarrow\quad B_i \wedge B_j = \frac{1}{3}\delta_{ij} B_k \wedge B^k,$$

because $\int C_{ij}\wedge\star_B C^{ij} \geq 0$ with equality only at $C_{ij} = 0$. Equivalently, writing $X_{aj} \equiv \frac{1}{2}\,\Sigma^a_{\mu\nu} B^{j\,\mu\nu}$ in a self-dual basis $\{\Sigma^a\}$ with $\Sigma^a \wedge \Sigma^b = 2\delta^{ab}\mathrm{vol}$, one finds

$$C_{ij} \propto (X^\top X)_{ij} - \tfrac{1}{3}\,\mathrm{tr}(X^\top X)\,\delta_{ij}, \qquad \|C\|^2 \propto \mathrm{tr}\Big[(X^\top X - \tfrac{1}{3}\mathrm{tr} X^\top X\,\mathbf{1})^2\Big] \geq 0,$$

so the minimum is attained at $X^\top X \propto \mathbf{1}$, i.e. the Plebanski simplicity sector.

Relation to the action. In the UV action (Eq. (53)) the multiplier term $\kappa(\rho)\,\Psi^{ij}C_{ij}$ imposes $C_{ij}=0$ exactly (Eq. (58)), while the $\tau(\rho)\,\mathrm{Tr}(B\wedge\star_B B)$ term provides a positive quadratic form that makes the coherent background stable (App. A and App. B). The variational argument above shows this constraint is not ad hoc: it is the *dynamically selected* minimum of the simplest gauge-invariant "free energy" compatible with the Urbantke construction.

Outcome. Simplicity $C_{ij}=0$ is the unique minimizer; it yields a nondegenerate Urbantke metric and, after integrating out δB, an induced YM kinetic term $-\frac{1}{2\,\tau(\rho_0)}\int \mathrm{Tr}(F\wedge\star_{B_0}F)$ as used in the main text (see App. B and Sec. 3.2).

L One-loop optical-theorem check in the $U(1)_Z$ channel

Process and setup. Consider forward scattering $\psi\bar\psi\to\psi\bar\psi$ via an s-channel Z with coupling $g_Z Q_\psi\,\bar\psi\gamma^\mu\psi\,Z_\mu$. The tree amplitude is

$$\mathcal{A}_{\text{tree}}(s)=-\frac{g_Z^2Q_\psi^2}{s-m_Z^2+i0}\,\big[\bar u(p')\gamma^\mu u(p)\big]\big[\bar v(k)\gamma_\mu v(k')\big].$$

At one loop, dress the propagator by the vacuum polarization $\Pi_Z^{\mu\nu}(q)=\big(q^\mu q^\nu-q^2g^{\mu\nu}\big)\Pi_Z(q^2)$ from all charged fermions f.

Cut and imaginary part. For timelike $s\equiv q^2>4m_f^2$, the discontinuity of the bubble is

$$\mathrm{Im}\,\Pi_Z(s)=\frac{g_Z^2}{12\pi}\sum_f N_c^fQ_f^2\,s\,\beta_f\left(1+\frac{2m_f^2}{s}\right)\theta(s-4m_f^2)\ \geq 0,$$

with $\beta_f\equiv\sqrt{1-4m_f^2/s}$. Using Cutkosky rules, the one-loop forward amplitude satisfies

$$2\,\mathrm{Im}\,\mathcal{A}(s)=\sum_f\int d\Pi_{f\bar f}\,\big|\mathcal{M}(\psi\bar\psi\to f\bar f)\big|^2\ \geq 0,$$

i.e. the optical theorem, where the RHS is the two-body on-shell phase space across the cut.

Absence of ghost poles. In the $U(1)_Z$ sector, Faddeev–Popov ghosts decouple, so no unphysical intermediate states contribute to the cut. In the background-field scheme used in QC&R, the quadratic kernel for Z is positive definite in the coherent phase (no tachyonic or negative-norm modes), ensuring that only physical cuts appear. (See 3.7 for positivity of the quadratic kernel.)

Result. The dressed forward amplitude $\mathcal{A}(s)\propto[s-m_Z^2+\Pi_Z(s)]^{-1}$ has

$$\mathrm{Im}\,\mathcal{A}(s)\ =\ \frac{g_Z^4Q_\psi^2}{(s-m_Z^2+\mathrm{Re}\,\Pi_Z)^2+(\mathrm{Im}\,\Pi_Z)^2}\,\mathrm{Im}\,\Pi_Z(s)\ \geq 0,$$

verifying perturbative unitarity at one loop in the representative channel. Additional channels (e.g. scalar pairs) obey analogous positivity with the same logic.